

Modeling Armenia

Martin Cincibuch
Hynek Hrebicek
Guy Meredith
Jack Selody
David Vavra¹

May 2011

¹Corresponding author (david.vavra@ogresearch.com).

Contents

1	Armenia background	7
1.1	Macroeconomic Stylized Facts	7
1.2	Major Shocks	11
1.3	Disequilibrium Influences	15
1.4	Unique Model Features	19
1.5	Other Useful Modeling Information	20
2	Building an FPAS	24
2.1	Introduction	24
2.2	Monitoring Division	25
2.3	Monitoring	27
2.3.1	Role	27
2.3.2	Analysis	28
2.3.3	Meetings	30
2.4	Research	32
2.4.1	Role	32
2.4.2	Analysis	33
2.4.3	Meetings	33

3	Structural Model	35
3.1	Data used in the core model	45
3.2	Dynamic properties	47
3.3	Interpretation of history	63
4	Other Models in the FPAS	71
4.1	Near-term forecast models	72
4.2	The Global Projection Model (GPM)	76
4.3	A Small DGE Model for Armenia	78
5	Forecast performance	80
5.1	Model historical forecasts	80
5.2	Near term forecast performance	85
	Appendices	90
A	Structural model	90
B	Free model historical forecasts	114
C	A Small DGE model of Armenia	117
C.1	Introduction	117
C.2	Model Structure	118
C.3	Calibrating the Model for Armenia	131
C.4	Illustrative Shocks	139
C.5	Concluding Remarks	146

Preface

The work described in this document was undertaken in the context of a high-level workshop on “Improving Monetary Frameworks-Inflation Targeting (IT): Policy and Implementation.” The event, which was jointly sponsored by the IMF, the EBRD, and the Central Bank of Armenia, was held in Yerevan and Tshakhdzor, Armenia, during April 27 to May 6, 2011.

The 78 participants at the workshop included representatives from 14 countries in the eastern European region, as well as staff members from the IMF, EBRD, OGR Research, and private consultants. Its main purpose was to discuss issues related to the implementation of inflation targeting as an anchor of monetary and financial stability for countries in the region. In addition to providing a framework for maintaining inflation at a low and stable level, the benefits in terms of macroeconomic stability would help to deepen local financial markets, reduce “dollarization” of financial transactions, and promote healthy capital market intermediation between domestic savers and borrowers.

The first three days of the program consisted of a discussion of issues related to inflation targeting by senior policymakers from central banks in the region, based both on the international experience as well as specific examples from their countries. This forum allowed for a broad dialogue and transfer of experience from central banks with an established record of inflation targeting to those in the initial stages of putting an IT framework in place. The following week involved intensive work on economic modeling to support the implementation of IT, and on the communication process between technical specialists and policymakers in such an environment. Modeling activities included the development and use of: country-specific forecasting and policy simulation models; global models to provide a consistent projection of external developments; and smaller theoretical models to complement the results obtained from forecasting models. Work on the communications process featured presentations by several teams to a hypothetical central-bank policy board on Armenia’s economic prospects and their implications for monetary policy, focusing on the policy stance consistent with restoring low inflation.

This document describes the framework and tools presented at the workshop as building blocks in establishing a Forecasting and Policy Analysis System (FPAS) to underpin the successful implementation of IT. Taking Armenia as the representative country, it starts with a description of features of the economy that are relevant in understanding recent economic developments and prospects. The next chapter discusses general features of how the FPAS process works within a central bank, focusing both on organizational issues and the flow of information between the technical staff and policymakers. A forecasting and simulation model specific to Armenia is then presented and its properties described, including a decomposition of historical economic developments in terms of the model structure and shocks. This is followed by a discussion of other types of models that can be used in the FPAS framework to complement the core model, including a Bayesian VAR model to provide short-term forecasts, a global model to project the international environment, and a small structural model to provide insights on medium- and long-term responses to shocks. The final chapter analyzes the forecasting performance of both the core and Bayesian VAR models, noting areas of strengths and weaknesses, and underscoring the importance of using judgment in conjunction with models to arrive at realistic forecasts.

This document is accompanied by a compendium of model files, tutorials, hand-outs and presentations that the workshop participants used when preparing the forecasts for Armenia. This compendium can be obtained from the authors, the IMF or EBRD upon request.

Introduction

Benefits of Forward Looking Monetary Policy

Low-income emerging-market economies are likely to reap significant benefits from forward-looking monetary policy aimed at stabilizing inflation at a low level. There are a number of reasons for this.

First, forward -lookingness increases the timeliness of monetary policy actions, which builds confidence in the commitment to low inflation. Over time, this will result in lower inflation expectations, most importantly at longer horizons. As inflation expectations fall, so do long-term nominal interest rates since these rates must compensate borrowers for expected future inflation.

Second, forward-looking inflation control recommends policy action that moderates aggregate demand cycles since these cycles lead to future inflation in the standard monetary policy model used for inflation control. Applied consistently, this policy advice should reduce the volatility of output over time, thereby reducing the risk premium in long-term interest rates that compensates borrowers' for real economic volatility.

Third, the volatility of inflation should be reduced by forward-looking monetary policy actions by reducing the time between inflationary shock and policy response. The result will be inflation cycles with less amplitude, thereby reducing the inflation volatility risk premium which is also a component of long-term interest rates.

Fourth, the inflation target provides a stable nominal anchor for the economy which reduces the risk that inflation expectations will become unhinged periodically. The possibility of unhinged inflation expectations represents a risk that the economy will fall into periodic episodes of serious inflations where a unit root manifests itself in inflation expectations. This unit root is likely to be interpreted by central bank observers as a change in the objective of monetary policy. Long-term bond holders are likely to demand being

compensated for this risk, as so reducing the risk should result in a reduction in long-term interest rates.

Thus, successful forward-looking inflation targeting will reduce long-term interest rates over time, which will raise the long-term potential of the associated economies. The reduction in long-term rates should be greater than the reduction in trend inflation since inflation risk, policy risk, and macroeconomic volatility risk will all be lower.

Importance of FPAS for Forward-Looking Monetary Policy

The role of the Forecasting and Policy Analysis system (FPAS) is to create an analytical environment supporting robust policy decisions under uncertainty.

Forward-looking decision making is torn between two fundamental opposites. At one hand, all forecasts and decisions are wrong ex post, no matter how good the forecasting system and models are. Some of the assumptions always turn out incorrect. Yet, on the other hand, the policy makers have to feel comfortable to make a decision ex ante — today.

The objective of the FPAS is to assist in making these ex ante decisions under uncertainty. This implies that forecasting accuracy is not as important as the ability to consistently differentiate between various alternative future developments. The FPAS provides a disciplining tool for policy debates about these alternatives and a platform for analyzing risks and their policy implications.

The FPAS is therefore about creating a common language — a communication platform within a central bank. By using this language, the central bank creates an environment where policy discussion and advice is efficient and effective. The FPAS works best when all involved in monetary policy analysis are versant in the language, since this facilitates communication and reduces confusion. By facilitating a consistent and open internal communication of policy issues, the FPAS paves way for discussing policy externally.

Adopting a common language for policy analysis does not mean that all staff will conduct the same type of analysis, hold the same views, or work on the same set of issues. Specialization will remain an effective tool for increasing the depth of policy analysis. Neither it means that the policymakers identify themselves with the forecast all the time. The

interest rate trajectory in the forecast is not a prescription for policy actions. But the consistency of the forecast forces people to react either by agreeing or forming alternative views. And such an environment helps reaching good decisions.

Chapter 1

Armenia background

There is now overwhelming evidence that a framework of inflation-forecast targeting can effectively guide monetary policy in providing a stable nominal anchor while contributing to the stability of real economic activity. Central to the framework is an economic model that forecasts inflation, but also identifies and classifies economic shocks so the monetary authorities can determine how best to respond. The most successful models include features unique to the particular economy being forecast in addition to features that are standard to all economies. The purpose of this note is to provide guidance on how recent economic events in Armenia can be mapped into the type of economic model used for inflation targeting. The note begins with listing some of the stylized facts of recent Armenian macroeconomic experience that a model of the Armenian economy should strive to incorporate and explain. Then, the major shocks that have recently hit the economy are identified and interpreted in the context of explanations consistent with inflation-forecasting models. Third, the disequilibrium influences that affect the dynamics of macroeconomic variables are articulated. Fourth, this information is used to suggest some unique features that should be added to the standard model used for inflation targeting to better reflect the Armenian reality. Finally, a series of questions for Armenian experts are posed to help differentiate between competing explanations of the observed macroeconomic outcomes.

1.1 Macroeconomic Stylized Facts

1. The variance of inflation is high, in part because the agricultural sector is relatively large and the prices of domestically produced agricultural products are volatile

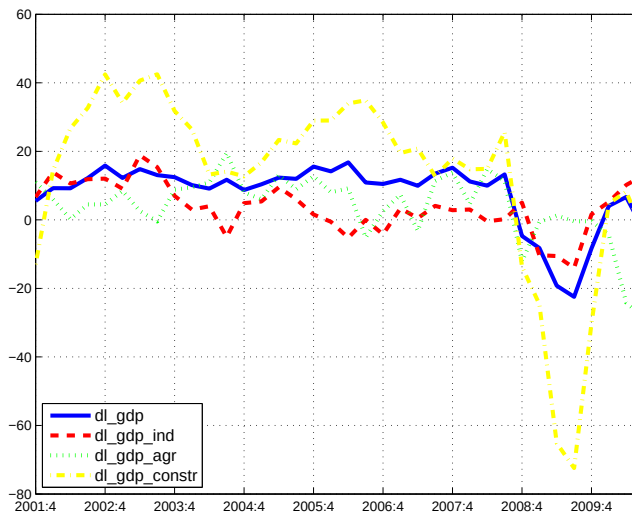


Figure 1.1: Sectoral Growth Decomposition in Armenia

owing to weather effects. Since 2000, 12-month CPI has fluctuated between -3 percent and +10 percent.

Implications for forecasting: (i) An independently-derived forward-looking indicator of inflation expectations would be useful since such measures often reflect the slow-moving components of inflation, which is where monetary policy has its greatest influence. (ii) Refined judgment will be needed to identify the transitory policy-independent factors that move inflation-these are the factors that will quickly vanish on their own, without help from monetary policy. (iii) Credibility will be an important conditioning variable for inflation forecasts-as monetary policy credibility builds over time, inflation volatility will fall and policy-independent factors will become less persistent.

2. The real economy is prone to boom-bust cycles. Real GDP growth was above 13 percent from 2005 through 2007 and below -14 percent in 2009. The composition of growth changed dramatically after the financial crisis with construction sector activity down significantly, being replaced by growth in the industry and service sectors.

Implications for modeling: (i) Foreign shocks are a key driver of economic growth-it will be important to have a well-understood methodology for determining which foreign shocks will have persistent inflation effects

on the Armenian economy (demand shocks) and which will have transitory inflation effects (supply shocks). (ii) The sectoral composition of growth will matter for the forecast dynamics of growth as construction, agriculture and services are all highly cyclical but likely have very different interest-rate sensitivities. In addition, some export industries have highly cyclical demand for their products, such as mining. (iii) Refined judgment will be needed to condition estimates of inflation dynamics as the response of inflation to excess demand in each of these sectors is likely to be very different.

3. Exchange rate pass through is very high and likely asymmetric between depreciations and appreciations.

Implications for modeling: (i) The exchange rate should be seen as a key financial variable in the transmission mechanism-special attention needs to be paid to the effect of the exchange rate on economic activity and inflation in both the short- and medium-run. (ii) The dynamics of the interest rate - exchange rate nexus should be well articulated in the model. (iii) Forecasters will need to decide the natural level of the exchange rate (i.e., the rate consistent with inflation at target on a sustained basis) in addition to the natural level of the interest rate and the natural level of output (i.e., potential) in the model-consistency should be maintained between movements in these three variables. (iv) The reasons for the asymmetric effect of exchange rate changes will need to be well understood and incorporated into the model.

4. The effect of interest rates on the real economy can be seemingly weak and unpredictable owing to an underdeveloped financial sector. The ratios of bank credit to GDP and bank assets to GDP are low compared to economies with more developed financial sectors. However, there are historical episodes where large changes in interest rates have had an obvious and significant effect on real economic activity.

Implications for modeling: (i) The policy interest rate will not be an accurate measure of the effect of policy on economic activity for the most part-a broader financial indicator, perhaps based on observable market-determined financial variables, needs to be developed or the parameters on the policy rate need to be adjusted to capture policy effects through other channels. (ii) Judgment will need to be applied to the link between the exchange rate and other financial variables since the exchange rate

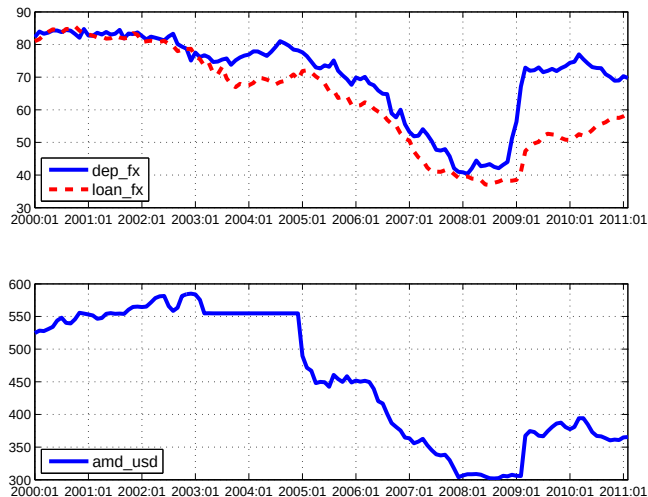


Figure 1.2: Dollarization and Exchange Rates in Armenia

is likely largely market determined whereas other financial variables are significantly less so. (iii) Judgment will be required to properly determine how a change in the financial sector affect will affect real economic activity.

5. The economy is highly dollarized and the degree of dollarization is highly sensitive to economic developments.

Implications for modeling: Changes in dollarization is driven by exchange rate expectations in the short run, and likely contains important information about perceived prospects for economic development. The effect of exchange rate movements on dollarization is asymmetric with depreciations having a more immediate and obvious impact than appreciations. It may be the case that domestic residents must be compensated in the form of a trend appreciation of the domestic currency for the risk of holding assets denominated in that currency. It will be important to have a well-understood methodology for determining whether a change in dollarization reflects a rational aversion to country risk that can be offset through a rise in country risk premiums, or an irrational reaction to uncertainty similar to what one sees in financial market panics and bubbles.

6. The inflation-targeting regime is relatively new and is gaining credibility, but could still experience credibility issues given that inflation has been high and volatile.

Implications for forecasting: (i) In a low-credibility regime the exchange rate path takes on added significance since it is one of the first variables to reflect a lack of commitment to the policy regime. (ii) Building credibility requires policy paths that are intuitive and that can be easily explained—complex, hard to understand features should be excluded from the model for the time being. (iii) Forecasters should be aware of how the different projected policy paths are likely to affect credibility. (iv) Forecasts should avoid wishful thinking in the sense that positive credibility effects should not be incorporated into the forecast before they are observed in the data. (v) Forecasts should support the current active communication strategy aimed at building credibility.

7. Fiscal policy is limited in its ability to raise revenue.

Implications for forecasting: (i) Monetary policy will bear a greater share of the burden of aggregate demand management—the projected policy path will likely be more sensitive to economic conditions than in similar countries where fiscal policy is less constrained. (ii) There may be spillover effects from fiscal deficits to monetary policy credibility—these effects will condition the judgment added to the model when generating alternative policy paths.

1.2 Major Shocks

1. Changes in remittances

Remittances were growing rapidly until 2008, declined precipitously in 2009, and recovered somewhat in 2010. This cycle had an impact on GDP growth, in part because the remittances were a source of finance for construction activity. Remittances are correlated with foreign economic activity and so exogenous to the model. However, the effect of remittances on domestic economic activity is likely endogenous to the state of the economy and to expected macroeconomic policy. The reason the effect of remittances is endogenous is that those who receive the remittance have a choice of what to do with the money, and this choice depends on current and expected economic conditions. Basically, the choice is:

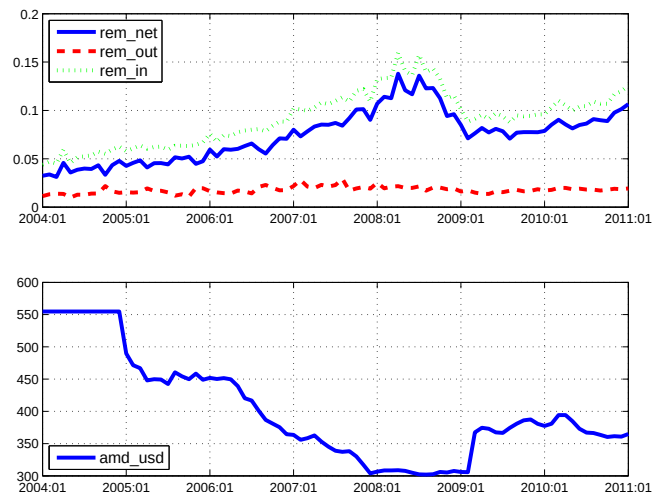


Figure 1.3: Remittances and Exchange Rate in Armenia

- i spend the money on consumption goods, in which case an increase in remittances is a positive demand shock that needs to be resisted by tighter monetary policy;
- ii spend the money on production goods, in which case the increase in demand is accompanied by a positive supply shock that does not need to be resisted by tighter monetary policy except to the extent of timing differences between the demand and supply dynamics;
- iii keep the money in the bank, in which case shock results in pent-up demand that will likely be unleashed once economic conditions warrant.

The model should recognize that the impact of a change in remittances on the economy will depend on how the money is used (for example, the import content of spending is likely to be different for consumption than it is for production) and the model should allow the forecaster to split remittance shocks into these three parts. The model should also recognize that changes in remittances significantly impact exchange rate developments.

2. An exogenous increase in the price of food

The Armenian economy experienced a sharp run up in food prices in 2010 that pushed headline inflation significantly above target. Since a once-off increase in prices will eventually drop out of the inflation mea-

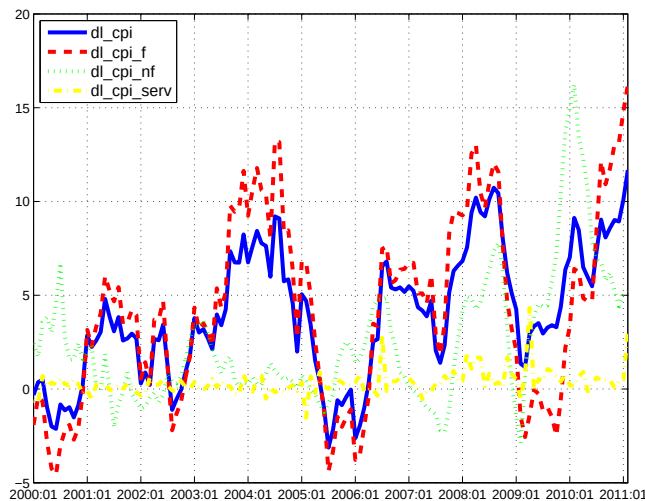


Figure 1.4: Inflation Decomposition in Armenia

sure, such an exogenous increase in a component of the CPI does not require a monetary response except insofar as it affects inflation expectations. Expectations are likely to be affected if the share of the exogenous component in the CPI basket is large, if the exogenous increase is widespread across components, large, persistent, and there is low credibility that future policy responses will be taken if needed. When expectations are affected, the objective of monetary policy is to keep inflation expectations from rising. This is less of a response than would be required to depress endogenous price components by an amount sufficient to offset the rise in the exogenous component of the price index. Generating such a policy path requires the forecaster to determine how much inflation expectations are being affected by the exogenous price shock. Current estimates suggest that a 1 percentage point rise in food prices leads to a 0.2 percentage point rise in nonfood prices. The model should aid the forecaster in illustrating the consequences of different assumptions about how inflation expectations will respond to the shock.

3. A change in dollarization

From 2004 to 2008 the Armenian financial sector experienced declining dollarization, which rapidly reversed in early 2009. The change in dollarization likely reflected improving prospects for the Armenian economy, which by this measure would have been seen as increasingly optimistic from 2004 to 2008 with a sudden loss of confidence in early 2009. The

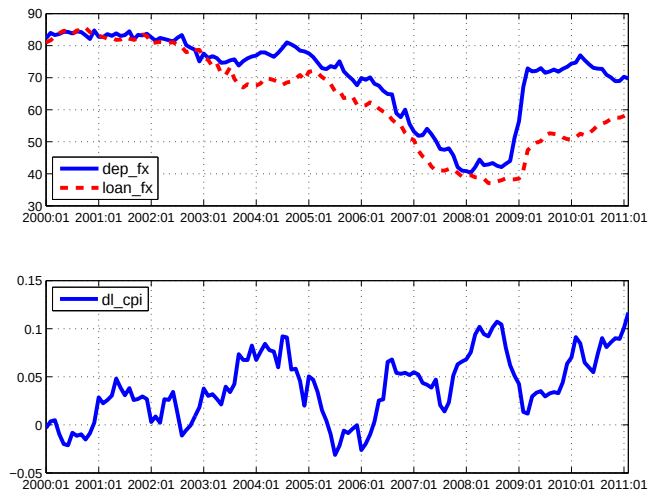


Figure 1.5: Dollarization and Inflation in Armenia

change in dollarization may have been rational response to country risk premiums that were too low to compensate for the risk of remaining in domestically-denominated assets, in which case a rise in domestic interest rates should stabilize the situation. Or, it may have been an irrational response to increasingly uncertain future prospects, in which case any reasonable rise in domestic interest rates would be inadequate to stabilize the situation. Whether behavior in such a shock is seen as rational (and thus responding to marginal changes in relative prices) or irrational (and thus unaffected by changes in relative prices) has implications for how such events are introduced into the model (i.e., endogenously or exogenously).

4. A change in monetary policy regime

In 2006, Armenia announced it was moving to an inflation-targeting framework. The key to securing the effectiveness of a new monetary policy framework is credibility. Credibility is being built by setting targets that are reasonable but that require a break from past behavior to achieve, using an active communication strategy to provide compelling explanations of how the targets were met, or if missed, compelling arguments as to why missing the targets was a better policy outcome, and drawing lessons from the experience that are seen to increase the probability of hitting targets in the future. Credibility is hard to build but easy to lose. Unfortunately, economic conditions were such that inflation continued its

upward trend after the introduction of the new policy regime, which, if not managed properly, may damage the credibility of the new regime. Further refinement of the policy regime will require that the active communication strategy be intensified. The communication strategy will need to include all variables that the public view as indicators of credibility, including headline inflation, core inflation, inflation expectations and the exchange rate.

1.3 Disequilibrium Influences

1. The output gap is close to being closed.

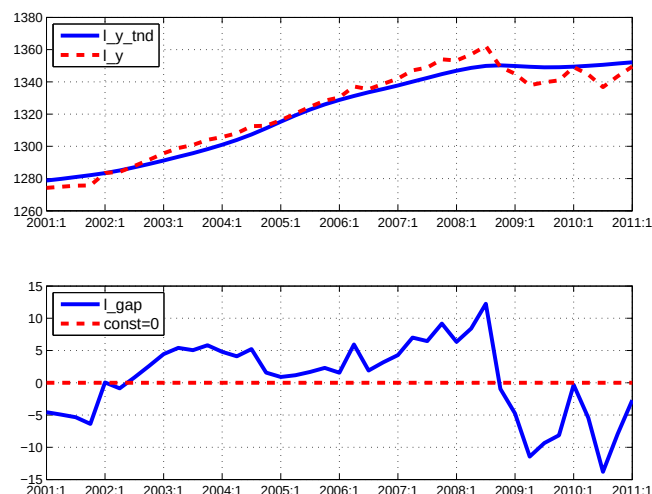


Figure 1.6: Output, Output Trend and Output Gap in Armenia

In a 2010 IMF Working Paper (10/197), El-Ganainy and Weber estimate that the output gap in Armenia would be closed at the start of 2010 if demand continued to strengthen rapidly. According to this analysis, the weakness in demand in 2009 was sufficient to offset the excess demand build up during the boom years of 2007-2008. Although the output gap is close to being closed, the composition of demand is not sustainable, with consumption being above what is sustainable, owing to government deficits and bank credit expansion, and investment and exports being below what is sustainable. The current account deficit is a reflection of

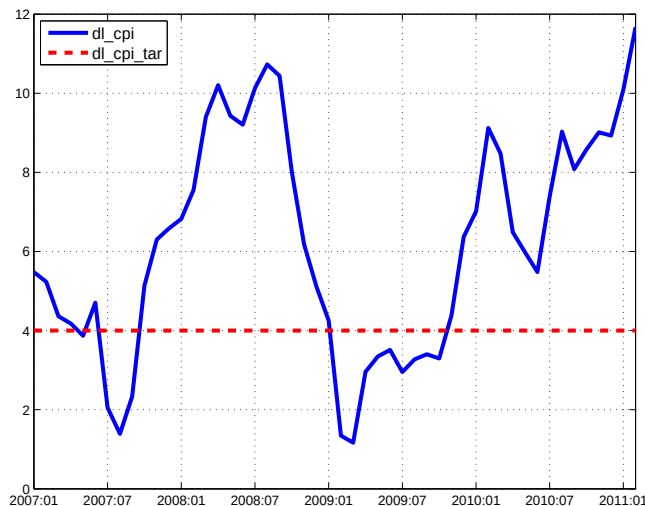


Figure 1.7: Inflation and Inflation Target in Armenia

these imbalances. This means that while the high inflation Armenia is currently experiencing is partly the result of exogenous transitory factors largely affecting food prices, there is also upward inflation pressure coming from the unsustainable strength in consumption spending.

2. It is difficult to determine whether or not inflation expectations are anchored near the inflation target.

There appear to be no direct measures of inflation expectations for the Armenian economy. Indirect measures indicate that inflation expectations could be near target. Non-food inflation has been stable for the last 6 months at a rate near target, which is consistent with the absence of an inflation expectations gap given the output gap is closed. Bank lending rates are not rising relative to the policy rate. If this spread was rising, it would likely indicate rising inflation expectations or a rising country risk premium. The degree of dollarization has stabilized and may be starting a renewed downward trend. If inflation expectations were rising above target, dollarization would likely be increasing. On the other hand, headline inflation has been rising and monetary policy is tightening. More analysis is needed to pin down this important disequilibrium influence.

3. No financial sector headwinds.

The Armenian financial sector is relatively sound and well regulated,

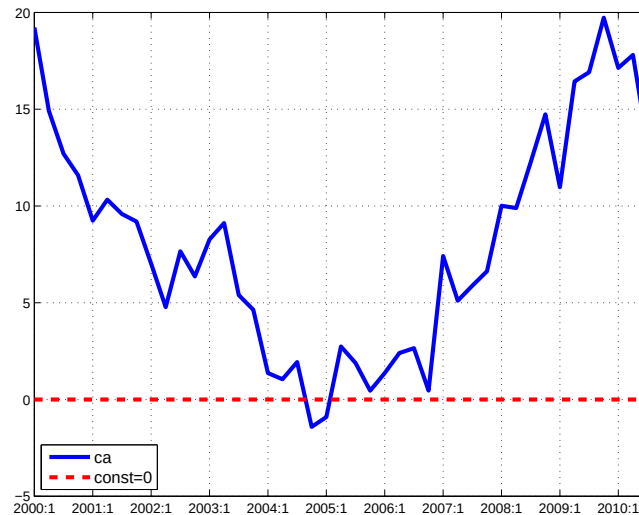


Figure 1.8: Current Account Deficit as % of GDP

which indicates that it should remain stable going forward. Dram loan-deposit spreads are stable, which is consistent with this view. Although these spreads are high by international standards, which indicates some tightness in credit, the spreads are not high by Armenian historical standards, which indicates no unusual tightness. This data suggest that real growth in Armenia is not being unusually constrained by disequilibrium in the financial sector.

4. A large and persistent current account deficit.

Armenia's large and persistent current account deficit indicates that consumption demand for import goods is unsustainable. It is likely that consumption demand will moderate given the current tightness of monetary policy, which suggests that the current account deficit will moderate without further adjustment to relative prices and without further impact on inflation expectations.

5. A large fiscal deficit.

The large fiscal deficit indicates a need to raise taxes or constrain public expenditures, either of which will have a dampening effect on private demand. To maintain a closed output gap, monetary policy will need to accommodate this adjustment to the extent that inflation expectations remain anchored at target.

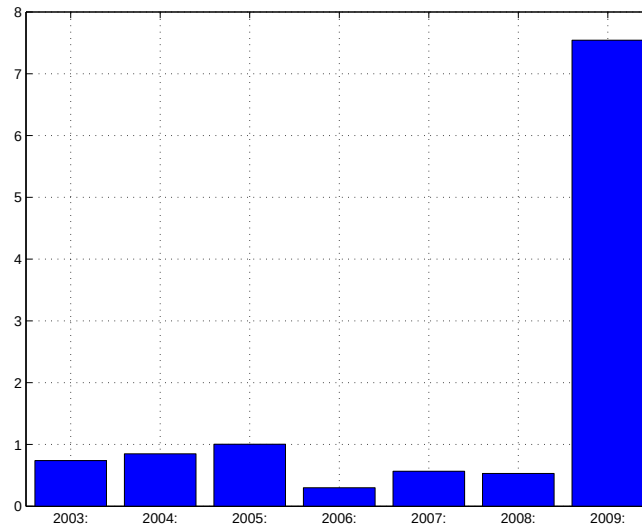


Figure 1.9: Fiscal Deficit as a Per Cent of GDP in Armenia

6. A policy rate slightly above its long-run equilibrium value.

In the long run, the policy rate should converge on its natural rate, equal to the foreign interest rate plus the justified expectation of exchange rate depreciation plus the country risk premium. The justified expectation of exchange rate depreciation is the difference in the inflation target in Armenia and abroad, the difference in potential growth rates in Armenia versus that of its trading partners, and the difference in the inflation credibility premium in Armenia versus that abroad (i.e., the differential in the lack-of-credibility premium). The current policy rate in Armenia appears to be slightly this value, assuming country risk premiums and inflation uncertainty premiums are not unusually large, which is appropriate given the view that consumption is higher than what is sustainable despite the output gap being closed, inflation is above target even though inflation expectations appear to be anchored, there are no financial sector headwinds, and that current account deficit adjustment will take place without further depreciation.

7. An overvalued real effective exchange rate.

Given that the current disequilibrium forces in the Armenian economy appear to be confined to current account and fiscal deficits, the real effective exchange rate is overvalued as it reflects the net disequilibrium

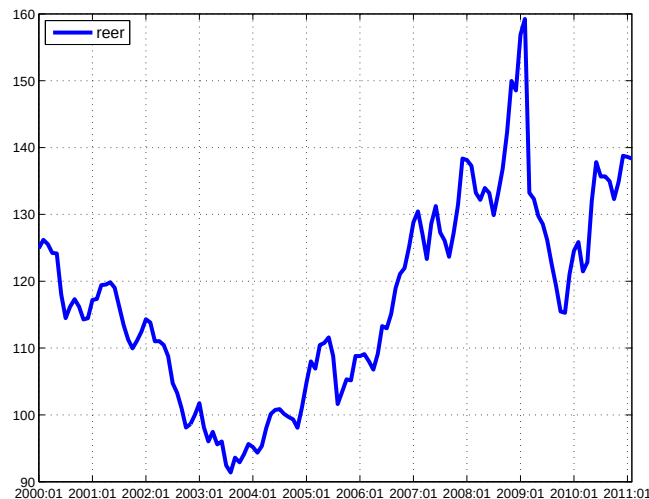


Figure 1.10: Real Effective Exchange Rate in Armenia

in the economy. The dynamic adjustment path of the exchange rate is most likely being heavily conditioned by the credibility of the monetary policy framework since depreciation is only appropriate if it does not result in an increase in inflation expectations.

1.4 Unique Model Features

The standard inflation-targeting macroeconomic model should work well for Armenia as the shocks and disequilibrium influences that affect the Armenian economy are similar to those that affect every small open economy. This said, there are three areas where the Armenian experience is richer, and potentially more variable, than allowed for in the standard model, and so the model could usefully be extended to better articulate the impact of shocks in these areas on the inflation forecast. First, the standard model assumes that all foreign shocks are demand shocks to the small open economy since potential output is largely a function of domestic variables. However, in the case of Armenia, remittances could be used to augment production as well as consumption. In other words, remittances are part supply shock, part demand shock. Potential output should therefore be linked to a portion of remittances to capture the prospective that remittances can be used to augment the capital stock of Armenia and/or the quality of Armenian labor by financing training. Second, the standard model assumes that the credibility of the monetary policy framework is either fixed or is a very slow moving process. Given the past

volatility of inflation in Armenia, credibility may adjust more rapidly, especially if inflation volatility or the exchange rate suddenly rises or falls. It would be useful to augment the standard model to link credibility to the characteristics of the inflation path that are likely to affect credibility, such as the volatility of inflation along the path, the volatility of forecast inflation and the policy rate path from one forecast to the next, or unexpected, large and precipitous changes in the exchange rate. Third, the standard model assumes that every movement in the exchange rate reflects the net state of macroeconomic disequilibrium in the model and therefore requires that the policy rate to adjust. However, in an economy as highly dollarized as Armenia, portfolio rebalancing effects can move the exchange rate independently of macroeconomic disequilibrium such that the resulting change in the exchange rate does not affect aggregate demand and hence requires no policy offset. It would be useful to augment the standard model to allow for such portfolio adjustment effects.

1.5 Other Useful Modeling Information

1. Measuring inflation expectations.

Current surveys of inflation expectations are derived from questionnaires sent to analysts at financial institutions. These surveys track CBA forecasts closely and so are probably not representative of expectations at the consumer or firm level. New surveys of consumer and firm inflation expectations are planned. There is no financial instrument that can be used to derive inflation expectations of financial market participants.

2. Factors driving policy credibility.

The factors that seem to erode policy credibility are persistently high levels of observed inflation and unusually high exchange rate volatility. Effective monetary policy communication has helped prevent a loss of monetary policy credibility resulting from the recent financial crisis and rising commodity prices.

3. Estimating how remittances are used.

Seasonal workers' income transfers are mainly used for consumption purposes, which is about 45% of transfers according to BOP statistics. Other transfers—the remaining 55%—is largely investment spending, mainly real estate. The Central Bank of Armenia did three surveys

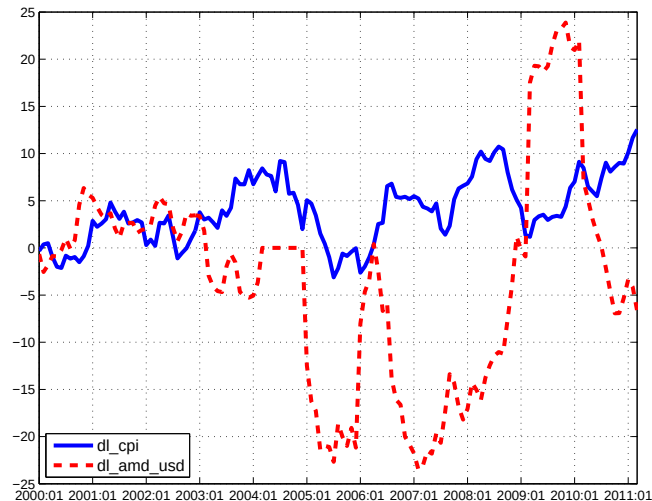


Figure 1.11: Exchange Rate and Inflation in Armenia

(in 2005, 2006, and 2010) which reported that 75% of remittances are spent on consumption goods. These survey results likely overestimate spending on consumption goods because the survey did not cover the wealthiest strata which are most likely engaged in real estate and other construction activity.

4. Estimating how policy affects construction, agriculture and services activity differentially.

Services and industry are significantly more dependent on domestic bank loans than are construction and agriculture. Industry exports significantly more than does agriculture or construction, and therefore is more exposed to changes in terms of trade.

5. Estimating the relative effect of interest rates and exchange rates on aggregate demand.

The estimate of the influence of interest rates relative to the exchange rate is 1:2.

6. Estimates relating to government activity.

Government revenues:

- seem exogenous to fluctuations in inflation and interest rates,

- tend to rise by 1
- are somewhat sensitive to the exchange rate through import taxes.

Government expenditures:

- seem exogenous to fluctuations in inflation and interest rates,
- seem exogenous to fluctuations in GDP,
- seem exogenous to fluctuations in the exchange rate.

Multipliers:

- a 1 percentage point increase in real government expenditures results in a 0.54 cumulative increase in real GDP,
- a 1 percentage point increase in real government revenues seems to have no impact on real GDP.

7. Exogenous drivers of the exchange rate.

Remittances, export prices, and expectations seem to be important.

8. Measures of credit conditions.

The Central Bank of Armenia has regular surveys of credit conditions but has not systematically studied the links between credit conditions and monetary policy.

9. Drivers of dollarization.

- Prices for real estate and motor vehicles tend to be denominated in dollars.
- Financial institutions tend to run matched books to avoid exchange rate risk.
- Dollarization is highly sensitive to exchange rate fluctuations.
- Residents are quicker to abandon the domestic currency than to adopt it.
- Reserve requirements and prudential regulation and supervision (micro and macro) can have an effect on dollarization.

10. Impact of dollarization on economic activity.

An increase in dollarization can contribute to exchange rate volatility which might have a negative impact on the volume of foreign trade.

11. Impact of inflation volatility on economic activity.

Inflation volatility mostly reflects seasonal factors. Good central bank communication has helped in preventing this volatility from having a negative impact on economic activity.

12. Estimates of the growth rate of potential output.

Current estimates are 3-4 percent over 2011-2012 and 5-6 percent in the long run.

13. What explains the asymmetric effect of exchange rate changes on inflation?

Appreciations are viewed as temporary and so rigidities prevent exchange rate pass through until it is persistent. Depreciations are viewed as persistent and so adjustment is relatively quick despite rigidities.

Chapter 2

Building an FPAS

2.1 Introduction

The Forecast and Policy Analysis System (FPAS) is about creating a common language for policy analysis within a central bank. By using this language, the central bank creates an environment where policy discussion and advice is efficient and effective. The FPAS works best when all involved in monetary policy analysis are versant in the language, since this facilitates communication and reduces confusion. Adopting a common language for policy analysis does not mean that all staff will conduct the same type of analysis, hold the same views, or work on the same set of issues. Specialization will remain an effective tool for increasing the depth of policy analysis.

In the FPAS, three teams work separately and together in a process that produces monetary policy advice (figure 2.1). Associated with this advice are three types of documents: the monetary policy report which articulates the views of the policy makers and provides supporting evidence and analysis compiled by the staff; the official forecast generated by the staff but that can incorporate the views of the policy makers; and research studies that provide in-depth background analysis of policy-relevant economic issues and ideas. These divisions in the work that contributes to FPAS provide a natural organization structure for the Monetary Policy department of a central bank (figure 2.2).

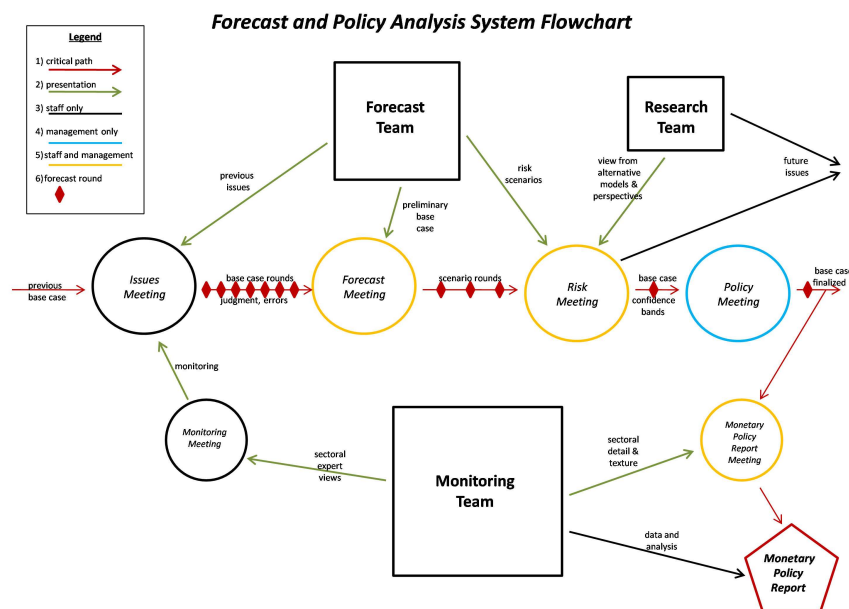


Figure 2.1: Three teams working for a common purpose

2.2 Monitoring Division

The monitoring team is in charge of compiling, interpreting, and analyzing historical data. The focus of analysis is on understanding the economic forces currently at play in the economy. This division monitors the data as it is released by statistical agencies and provides timely briefings to management about what the newly released data imply about the current state of the economy. Often, this division will gather its own data—such as a survey of business conditions or a survey of inflation expectations—to augment that from statistical agencies. Often, the monitoring division is itself divided into groups.

The Data group transforms and compiles the raw economic data received from statistical agencies, as well as managing the process of direct data collection if such processes exist. This group generates data reports that the expert economists in other groups use to analyze the economy and to brief management about current economic events. This group resides in the Monitoring division because the economists who interpret the data need intimate knowledge of how the data is constructed. This said, as information technology is increasingly used to automate the function of this group, and documentation of the statistical methodologies used to gather and construct the data improve, this group will shrink in size and can be moved to the group in the information technology

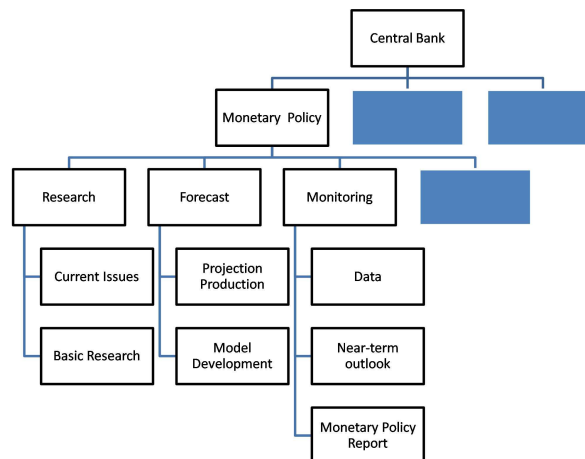


Figure 2.2: Organizational Chart

department that maintains and manages the corporate data bases.

The Near-term Outlook group consists of economists who are expert in interpreting recently released data. They maintain close contact with the statistical experts in the statistical agencies that compile and construct the data, and they conduct their own economic intelligence that helps them interpret the data from the statistical agency. These economists need to understand the models used in the Forecast and Research divisions, and they need to understand the monetary policy process, since economic data is best interpreted in reference to the context in which it will be used. The Near-term Outlook group provides the 'starting-point' data estimates for the Forecast. This starting point consists of an estimate of the data used in the forecast model for the current quarter, which will be constructed using all information that is available about the current quarter, usually the high-frequency indicators. And, the 'starting point' consists of a data-point estimate for the following quarter, that is, the second quarter of the forecast period. Although there will not be any partial data available for the second quarter, some of the non-model factors affecting the first-forecast-quarter data will persist to the second-forecast-quarter. In addition, having the data experts generate a future data point helps them focus on the identification of the economic forces present in the data that will be resolved over the forecast period.

2.3 Monitoring

2.3.1 Role

In the FPAS system, the role of monitoring is to inform the forecast by extracting information from recent data by means of analysis and interpretation. Forecasters look to the data for information about shocks, disequilibriums, frictions, and risks.

Monitoring starts with each expert analyst (members of the Monitoring Team) separating signal from noise. The signal consists of the economic forces that drive the dynamics the forecast (i.e., shocks, disequilibriums, and frictions), or that are used to question the validity of the forecast (i.e., risks). Noise consists of static (i.e., small unexplainable variations in the data that are unlikely to persist), anomalies (i.e., a large unexplainable variations in the data that is unlikely to persist), and transitory measurement error (i.e., explainable movements in the data unrelated to economic forces). In practice, noise is what is left over after the economic signal and transitory measurement error has been extracted from the data.

Shocks are large surprises in the data that are likely to have future economic consequences that cannot be attributed to forces already at play in the forecast story. Shocks are important to forecasters because they are new elements that must be taken into account in the forecast story. Typically, each forecast will one or two new shocks as a single shock can cause surprise movements in many variables. In addition, forecast exercises can become overly complicated with anything more than a few new shocks, which can confound the policy analysis. Thus, it is common practice to group large surprise movements in multiple data series into a single shock with a common origin and consistent analysis. For example, a demand shock might be identified as a simultaneous rise in consumption, rise in the GDP deflator, and rise in GDP, whereas a supply shock might be reflected as a simultaneous rise in mining production, fall in the GDP deflator, and rise in GDP. The forecaster looks to the monitoring for (i) the source of the shock, (ii) an explanation of what caused the shock, and the variables that are being affected by the shock along with an estimate of the magnitude by which the variables are being affected by the shock.

Disequilibriums are gaps between where variables are now and where they would be if there were no shocks or frictions. Disequilibriums represent the pressures on variables to move over time. A shock can open a new disequilibrium, widen an existing disequilibrium, or shrink an existing disequilibrium. Disequilibriums can only be identified in the context of an economic model. Thus, in looking to the monitoring for information

about starting-point disequilibriums, the forecaster is looking for an update of the forces that are currently acting on the variables being forecast. The forecaster is also looking for consistency in the form of an explanation of why the monitoring team's view of the starting-point forces in the current forecast are different from the one-period-out forces in the previous forecast. In practice, this amounts to explaining how the latest forecast errors changed the team's view about the disequilibriums that are thought to exist in the economy.

Frictions are factors or considerations that prevent gaps from closing quickly. Frictions, along with the magnitude of disequilibrium, determine the speed at which variables move over time. Since frictions are coded into the equations of motion in the model, forecasters look to the monitoring for special factors that might cause variables to move temporarily at a different speed from that coded into the model. In practice, frictions and disequilibriums (along with transitory measurement errors) are used to explain why the first forecast point (i.e., the second monitoring data point) is different from the last data-based point (i.e., the first monitoring data point).

Risks are alternative interpretations of the data that would lead to dramatically different forecast outcomes. Often, they are identified by the incongruencies between the economic story invoked to explain the monitoring and that invoked to explain the forecast. In practice, risks tend to be associated with large movements in the data that are treated as anomalies for purposes of the forecast, but in fact may reflect an economic shock. But risks can also be disequilibriums that are bigger or smaller than assumed, or frictions more or less powerful than incorporated into the forecast.

2.3.2 Analysis

Analysis is explaining “why” something happened, whereas description is describing “what” happened. Forecasters look to the monitoring team for analysis, not description.

Separating signal from noise requires judgment and knowledge of the economic model being used by the forecasters. Basically, signals are the large correlated movements in the data, whereas noise are the movements that are transitory and that have no predictive power for what will happen in the future. Signals can either be slow moving processes in which the data is correlated with itself over time (auto correlated data), or they can be more transitory components that are correlated with components in other data (cross correlated data), or both. In either case, the basics of analysis is providing an explanation of what is causing the correlation in the data.

As a rule, the explanation given for a correlation in the data is one that is consistent with the economic model being used by the forecasters with (i.e., the expert analyst uses the same economic paradigm to interpret the data as is embodied in the forecast model). In these cases, the monitoring team will be providing an economic explanation for the correlations in the data. There are, however, three exceptions to this rule.

The first exception relates to variables that are exogenous in the model. These are variables that are not driven by economics in the model, although they do drive the economics of the model. Movements in these variables can have explanations that do not appeal to economics (e.g., agricultural output fell because of a drought), or they can have explanations that appeal to economics not incorporated into the model (e.g., oil prices rose because OPEC is maximizing rents).

The second exception relates to predictable measurement error. Since data arrives to the statistical agency on a fixed schedule, and the statistical agency uses simple rules of thumb as placeholders for “yet to arrive” data (e.g., the data value from the previous period), predictable biases can occur in the data. For example, preliminary estimates of GDP may be consistently revised up when the economy is expanding, and be revised down when the economy is contracting. In such cases, the forecaster will accept the forecast error associated with such biases since it is inappropriate for policy to respond to movements in the data associated with measurement error.

The third exception relates to situations where the expert analyst believes (or has independent evidence) that the explanation for data movements provided by the model is wrong. For example, the model may attribute strengthening consumption to easing fiscal policy whereas the analyst believes the strength has more to do with a bubble in house prices, a phenomenon not in the model. Moreover, the difference in interpretation is important because the ease in fiscal policy is likely to be transitory whereas the rise in house prices is likely to be persistent. In this case, the analyst would provide a well-founded explanation for the movement in the data that was inconsistent with the model, and the analysts interpretation would be seen as a risk to the forecast.

Once the expert analyst identifies the signal in the data, it is most useful to provide starting-point update measures of model concepts. There are three concepts that might be useful to update and include in the monitoring story: shocks, disequilibriums, and frictions.

Updating the shock analysis requires the analyst to determine whether the movements they have identified in the data are better associated with currently identified shocks or whether they constitute a new shock. Clearly, there is a limit to the number of shocks that can be analyzed within a single projection exercise, so the burden of proof is fairly high

on the analyst to demonstrate that the data movements are best explained by introducing a new theme into the forecast story. In general, a single shock will hit multiple variables simultaneously.

Updating starting point disequilibrium estimates (i.e., gaps) requires the analyst to take a view as to whether the movement in the data represents a transitory disequilibrium phenomenon or a persistent equilibrium phenomenon as defined by the equilibrium structures in the forecast model. In a policy model, typically, movements that change the disequilibrium gap require a policy response whereas those that maintain the gap unchanged (i.e., equilibrium movements) do not. The analyst can split a movement between the two effects. For example, a movement in GDP may be judged to be 50% associated with a change in potential output and 50% a change in demand independently of potential output, in which case the output gap would move by 55% of the movement in GDP.

Updating frictions requires knowledge of the dynamics of the model in response to shocks and reference to past forecast values. If the new data is significantly different than the previous forecast predicted, and there are no new shocks, then it is highly likely that the frictions currently being observed in the economy are different from those coded into the model. The analysis should give an indication of which frictions they think might be causing the error so that the forecasters can decide if model calibration needs to be tweaked.

2.3.3 Meetings

Monitoring Meeting

The purpose of the Monitoring Meeting is to resolve inconsistencies between the stories being told by the different expert members of the monitoring team, garner a common understanding of the new shocks that have occurred since the last projection exercise, agree upon starting point disequilibrium estimates, and identify cases where the frictions in the model appear to be at odds with those currently being observed in the economy. The meeting is to understand the starting point of the projection so that issues can be communicated effectively and efficiently to the forecast team in the issues meeting.

The monitoring meeting is for the benefit of the monitoring team. Other forecast participants may find it useful to attend in order to garner a detailed understanding of some of the factors behind the monitoring team's collective view, but it is the monitoring teams view that is firmed up at the meeting.

Issues Meeting

The purpose of the Issues Meeting is to establish the initial conditions for the projection. This includes (i) establishing the first two data points for the projection, to which the forecast will be “tuned”, thereby providing initial values for the judgment that will be added to the projection, and (ii) identifying issues that the forecast team needs to pay special attention to when constructing the forecast. Issues are typically associated with the new data points (e.g., new shocks or revised estimates of initial disequilibrium conditions), reinterpretations of history (e.g., what was previously interpreted as a demand shock now appears to be a supply shock), or a previously identified and corrected model deficiency going live for the first time.

Time is always short in the issues meeting, since the starting-point analysis is presented by the monitoring team along with the modeling issues presented by the forecast team, both of which involve a lot of detailed background work. The “trick” to making the issues meeting effective is for presentations to focus on what the forecast team needs to know to produce the forecast. Details or discussions of other important aspects, whether they be conceptual or practical, are best left for seminars and workshops. Essentially, the forecast team needs to know how the macroeconomic story is changing, what signals in the data support the change, and what are the new risks. What they do not need is a lot of description about how the data changed or about the process that generated the change in view.

Monetary Policy Report Meeting

The purpose of the Monetary Policy Report Meeting is to establish the storyline for the Monetary Policy Report. Typically, the Monetary Policy Report is more descriptive and less analytic than that presented in the Issues Meeting because the audience for the report is less technically sophisticated. This said, the description in the report should be consistent with the results of the analysis used to identify the forecast issues. The purpose of the description in the report is to explain why policy path needed to change as it did.

Monitoring Presentation Checklist

- brief summary of previous story
 - current disequilibrium conditions

- economic pressures for adjustment
 - impediments to quick or smooth adjustment
- brief description of latest data
 - consistencies
 - surprises
 - anomalies
 - transitory measurement biases
- how latest data change the story
 - new shocks
 - reinterpretation of old shocks
 - new risks
- brief summary of updated story
 - current disequilibrium conditions
 - economic pressures for adjustment
 - impediments to quick or smooth adjustment

2.4 Research

2.4.1 Role

In the FPAS system, the role of Research is to inform management of the risk that the forecast will turn out differently than that in the base case. This is achieved by presenting thoughtful alternative perspectives on the economic outlook. Management and forecasters look to the Research Team for perspectives coming from alternative models, alternative perspectives, and academic studies.

Alternative models are most useful when they reflect a different economic paradigm, for example, the monetary view of inflation or the bank-credit view of economic cycles. These models are necessarily less elaborate and less data intensive than the forecast model since they are supported by significantly fewer resources. Alternative models are

not meant to provide competing forecasts, although in some cases it may be useful to an array of alternative forecasts to gain a sense of the importance of model uncertainty. Rather, alternative models are meant to condition judgment that can then be applied to the alternative scenarios produced routinely by the forecast team.

Alternative perspectives tend not to be embodied in mathematical models, but instead represent internally consistent ways of interpreting economic events that differ in important ways from the macroeconomic story embodied in the forecast model. The perspective is usually an alternative way of interpreting the economic data, which is a useful challenge to the logic embodied in the model. Often, these alternative perspectives are observationally equivalent to the perspective in the forecast model, and so a “horse race” between the competing views is neither possible nor productive. These alternative perspectives can be useful as a communication device where commentators are focused on phenomenon not embedded in the model. This is especially true if the policy advice from the different perspectives differs only in tactics not strategy.

Academic studies are useful for challenging the calibration of the frictions in the forecast model, for differentiating between competing hypotheses about the behavior driving the movements of the starting point data, and for determining the size of initial disequilibrium gaps. These studies are also useful references to provide elaboration when communicating the results of the forecast analysis because they are thoughtful and well founded in theory and data.

2.4.2 Analysis

The analysis underlying research should use accepted academic methods and be judged by academic standards (i.e., peer review) given their purpose is to challenge the forecast in order to identify risks. More specifically, the output should not be judged on the acceptability or reasonableness of the policy advice to policy makers.

2.4.3 Meetings

Risk Meeting

The purpose of the Risk Meeting is to inform policymakers about the seriousness of the uncertainties surrounding with the outlook. The meeting consists of a presentation of alternative scenarios by the forecast team that capture and quantify what they consider

to be the biggest risks to the projection, and a presentation by the Research Team on alternative views, interpretations, and perspectives that might provide better explanations of the forces at work in the economy. The presentation by the Research Team should be seen as a challenge function in that it is not intended to result in a change to the base case scenario, but rather to inform the judgment that management will apply to the base case scenario in their policy deliberations.

The risk meeting is mostly for the benefit of management, but also is useful for the forecast team in that it is likely to raise issues with the model or macroeconomic story that they will want to resolve in future projection exercises. All forecast participants may find it useful to attend in order to garner a sense of how the analysis behind the outlook may change going forward.

Research Presentation Checklist

- brief summary of alternative view
 - a missing shock?
 - a different interpretation for an existing shock?
 - different consequences from the shock?
- brief description of how the shock would change the outlook
 - disequilibriums
 - frictions
 - policy response
- support for the alternative view
 - inconsistencies in the base line view
 - evidence from studies or models external to the forecast exercise
 - external commentary
- methods of distinguishing between views
 - contrasting predictions from the two views
 - future economic research possibilities

Chapter 3

Structural Model

This section shows the FPAS core structural model for Armenia and its properties. The first subsection describes in detail various blocks of the model, their motivation and main model equations. The complete model code and variables description can be found in Appendix. Subsection 3.1 gives the list and definitions of the observed data used in the model estimation. Dynamic properties in terms of impulse response functions and variance decompositions are shown in subsection 3.2. The last subsection – 3.3 describes the model's interpretation of the observed historical data using the Kalman filter technique.

The core model is an extended version of a linear small open economy gap model of New-Keynesian business cycle fluctuations (see [3, 4] for a canonical reference). Such models have been used by many central banks as core projection models in the FPAS. Examples include Columbia, the Czech Republic, Serbia or Turkey.

Main features of the gap models are: endogenous monetary policy, rational expectations, distinction between supply and demand shocks, nominal and real rigidities, flexible exchange rate and partial “ad-hocness” of the model equations.

Endogenous monetary policy provides a nominal anchor by setting the interest rates to attain a nominal objective in a long run – the inflation target. In the short run, it may also pursue some other objectives like smoothing real output or nominal exchange rate.

In modeling monetary transmission, the model embodies the main principles of monetary policy-making, such as the long-term monetary neutrality and monetary control over inflation and other stationary nominal variables. The main mechanism driving inflation over the business cycle is the fluctuations (i.e. gaps) of real variables – such as output

and relative prices –around their long-term trends. The New-Keynesian Phillips curves translate these real fluctuations into inflation movements in the short-term, while ensuring no long-term trade-off between inflation and real macroeconomic variables.

The model is a loose short-cut for a full structural model derived from optimization. Although the model structure has good grounding in theory, its calibration is flexible in order to account for many empirical phenomena that theory has difficulty in capturing and are important for policy analysis in Armenia and elsewhere in the region. Partial “ad-hocness” of model equations allows for adding new features and dynamics. The advantage of this approach is that the resulting dynamics can be very close to what policy deliberations need in practice at much smaller costs than developing a full-fledged micro-founded model.

The flexible structure of a gap model permits us to extend the standard gap model in several dimensions:

- On the expenditure side we distinguish among domestic demand, import and export expenditures. This brings external balance issues explicitly into a model based monetary policy analysis.
- We allow for financial dollarization to affect the model in several ways. This acknowledges the complications many economies face when running independent monetary policy in a highly dollarized environment.
- We add both demand as well as supply side effects of remittances and terms of trade. Both remittances and terms of trade are an important external source of business cycle dynamics in Armenia and elsewhere in the region.
- We allow monetary policy to manage the economy through both interest and exchange rates. Although the exchange rate of Armenian dram is relatively flexible, the central bank intervenes frequently in the FX market, while at the same time using the key policy rate as its main instrument.
- We introduce several relative prices in the model and allow them to follow trends determined by expert judgment and other forecasting assumptions. Besides the real effective exchange rate, the model features the domestic relative price of food to non-food, terms-of-trade and world relative price of oil and food. This allows capturing many important phenomena, such as the Balassa-Samuelson effect or a rising relative price of food observed in many economies in the recent decade.

Although the model is calibrated to Armenian stylized facts, its structure is more general than strictly needed for Armenian purposes. This should facilitate the model's application to other economies in the region. For this reason coefficients in certain channels that are not too important in Armenia have small values rather than zeros.

Despite substantial extensions and additions to the standard gap model, our model continues to be only one of several FPAS elements making together a good policy analysis environment. As usual, the core model needs to interact with other sources of information to provide for forecast realism and accuracy, as well as robustness to policy decisions. While the model aims at providing a workable tool for medium-term forecasting, it has a relatively limited explicit coverage of variables and policy relevant issue. The trade-off between simplicity and coverage must be addressed by adding other tools and techniques (including expert knowledge) to support the core model in different phases of the forecast process.

Expenditure

On the expenditure side, the model distinguishes among private domestic demand, exports and imports, and government demand. The structure of these expenditure equations is similar, though. The expenditure equations are all expressed in gaps from trends that are tied together by a GDP growth identity. Besides, we make an assumption that a similar identity holds for the expenditure gaps too.

Domestic Demand. The key driver of domestic demand is an inter-temporal effect of the real exchange rate gap. Demand is also affected by current income (motivated by income constrained consumers) less interest rate payments on dollarized loans. Besides, remittances and terms of trade add income and wealth effects too:

(A.1a)

$$\begin{aligned}\hat{d}_t = & a_{10} \cdot \hat{d}_{t-1} + (1 - a_{10}) \cdot E_t[\hat{d}_{t+1}] - a_{11} \cdot \hat{r}_{t-1} \\ & + a_{12} \cdot (\hat{y}_t + a_{14} \cdot q\hat{r}\hat{e}m_t + a_{13} \cdot \hat{z}_t - \hat{d}_t) + a_{15} \cdot (\Delta at_t - at_{sst}) + \varepsilon_t^{\hat{d}}\end{aligned}$$

Remittances are set in real terms in foreign currency bringing real exchange rate effects into the equation. Real exchange rate depreciation thus increases the expenditure of income-constrained households.

The workings of these channels is complicated by dollarization of the financial sector that affects this equation in two major ways:

- The real interest rate gap contains real foreign rates adjusted by expected depreciation (see later).
- Interest rate payments on dollar denominated loans reduce expenditure of income-constrained household, causing real exchange rate depreciation to have a negative effect on expenditure.

As a result, the real exchange rate term in the equation has an ambiguous sign. Real depreciation can boost consumption by increasing the dram value of remittances or reduce it by increasing the dram value of interest rate payments on dollar denominated loans.

Imports and Exports. These expenditure equations have a similar structure with domestic and foreign income as the main drivers respectively and substitution effects through the real exchange rate:

(A.1c)

$$\hat{x}_t = a_{31} \cdot \hat{x}_{t-1} + (1 - a_{31}) \cdot E_t[\hat{x}_{t+1}] + a_{32} \cdot (\hat{y}_t^* - \hat{x}_t + a_{33} \cdot \hat{z}_t) + \varepsilon_t^{\hat{x}}$$

(A.1d)

$$\hat{m}_t = a_{41} \cdot \hat{m}_{t-1} + (1 - a_{41}) \cdot E_t[\hat{m}_{t+1}] + a_{42} \cdot (\hat{y}_t - 0 \cdot \hat{m}_t - a_{43} \cdot \hat{z}_t) + \varepsilon_t^{\hat{m}}$$

Government demand. Government demand is modeled as mildly pro-cyclical. The purpose of this equation is only to introduce a government expenditure variable into the model. Its forecast should be tuned according to expert judgment or government budget forecasts.

(A.1b)

$$\hat{g}_t = a_{20} \cdot \hat{g}_{t-1} + (1 - a_{20}) \cdot E_t[\hat{g}_{t+1}] + a_{20} \cdot (\hat{y}_t - \hat{g}_t) + \varepsilon_t^{\hat{g}}$$

Expenditure constraint. We postulate a relationship inspired by the statistical expenditure identity tying down the expenditure gaps. Unlike the statistical identity, this one is a behavioral assumption, though.

(A.1e)

$$\hat{y}_t = (1 - a_1 - a_2 + a_3) \cdot \hat{d}_t + a_1 \cdot \hat{g}_t + a_2 \cdot \hat{x}_t - a_3 \cdot \hat{m}_t + \varepsilon_t^{\hat{y}}$$

Prices

In the price block we distinguish between food and non-food inflation rates. They are driven by domestic as well as foreign elements: imported food, non-food and oil prices and inflation rates. Expectation terms provide for a direct pass-through between food and non-food prices. Because we allow for a non-stationary trend in the relative price of food, both inflation rates have to grow at potentially different rates in the steady state. This requires a number of algebraic adjustments to the inflation equations.

Non-food inflation. A standard New-Keynesian Phillips-curve has nominal frictions and impacts from imported oil and effective foreign inflation, and a gap in real marginal costs:

$$\pi_t^{xf} = j_5 \cdot {}^e\pi_t^{xf} + (1 - j_5 - j_6 - j_7) \cdot \pi_{t-1}^{4,xf} + j_6 \cdot \pi_t^{*oil,im} + j_7 \cdot \pi_t^{*im,xf} + j_8 \cdot rmc_t^{xf} + \varepsilon_t^{\hat{\pi}^{xf}} \quad (A.2d)$$

The real marginal costs reflect the economic scale (output gap) and relative cost components: the real exchange rate gap, relative domestic price of food to non-food gap, and relative world price of oil gap. Taken together the coefficient restrictions ensure that domestic non-food prices are cointegrated with world CPI and oil prices. These cointegration relationships are exogenous and can be subject to expert judgment in a forecast.

$$rmc_t^{xf} = (1 - k_3 - k_4) \cdot \hat{y}_t + (k_3 + k_4) \cdot (\hat{z}_t + \omega \cdot \hat{r}p_t) + k_4 \cdot \hat{r}p_t^{oil} \quad (A.2e)$$

Food inflation. A standard New-Keynesian Phillips-curve has nominal frictions and effects from imported food and effective foreign inflation and a gap in real marginal costs.

$$\pi_t^f = j_1 \cdot {}^e\pi_t^f + (1 - j_1 - j_2 - j_3) \cdot \pi_{t-1}^{4,f} + j_2 \cdot \pi_{t-1}^{*food,im} + j_3 \cdot \pi_t^{*im,f} + j_4 \cdot rmc_t^f + \varepsilon_t^{\hat{\pi}^f} \quad (A.2b)$$

The real marginal costs reflect the economic scale (output gap) and relative cost components: the real exchange rate gap, relative domestic price of food to non-food gap, and relative world price of food gap. Taken together the coefficient restrictions ensure that domestic non-food prices are cointegrated with world CPI and food prices. These cointegration relationships are exogenous and can be subject to expert judgment in a forecast.

$$rmc_t^{xf} = (1 - k_3 - k_4) \cdot \hat{y}_t + (k_3 + k_4) \cdot (\hat{z}_t + \omega \cdot \hat{r}p_t) + k_4 \cdot \hat{r}p_t^{oil} \quad (A.2e)$$

Effective imported inflation. Effective imported inflation enters both the food and non-food Phillips curves – however adjusted by trend movements in relative prices to allow for different steady state levels of food and non-food inflation. This adjustment is therefore different for each equation:

(A.10a)

$$\pi_t^{*im} = 0.5 \cdot \pi_{t-1}^{*im} + (1 - 0.5) \cdot (\pi_{t-1}^* + \Delta s_{t-1}^{AMD/USD} - \Delta \bar{z}_t)$$

(A.10b)

$$\pi_t^{*im,f} = \pi_t^{*im} + (1 - \omega) \cdot \Delta \bar{r} \bar{p}_t$$

(A.10c)

$$\pi_t^{*im,xf} = \pi_t^{*im} - \omega \cdot \Delta \bar{r} \bar{p}_t$$

Imported commodity inflation. Imported oil and food inflation enter the non-food and food Phillips curves respectively, adjusted by relative price movements:

(A.13a)

$$\pi_t^{*food,im} = 0.5 \cdot \pi_{t-1}^{*food,im} + (1 - 0.5) \cdot (\pi_t^{*food} + \Delta s_t^{AMD/USD} - \Delta \bar{z}_t - \Delta \bar{r} \bar{p}_t^{food} + (1 - \omega) \cdot \Delta \bar{r} \bar{p}_t)$$

(A.14a)

$$\pi_t^{*oil,im} = \pi_t^{*oil} + \Delta s_t^{AMD/USD} - \Delta \bar{z}_t - \Delta \bar{r} \bar{p}_t^{oil} - \omega \cdot \Delta \bar{r} \bar{p}_t$$

Policy

The policy block features an interest rate rule typical for flexible IT regimes and a target change in the nominal exchange rate. The latter represents short-term policy preferences for smoothing the exchange rate and enters both the interest rate rule as well as the UIP where it serves as a proxy for effects of FX interventions.

Interest rate rule. The interest rate rule features several elements: expected deviation of inflation from the target, output gap and difference between actual and target change in the exchange rate. We assume that the interest rate rule determines both the key

policy rate of the central bank as well as the prevailing short-term interbank market rate (CBA refinancing rate).

$$i_t = b_1 \cdot i_{t-1} + (1 - b_1) \cdot (\bar{i}_t + b_2 \cdot \omega \cdot \hat{\pi}_t^f + b_3 \cdot (1 - \omega) \cdot \hat{\pi}_t^{xf} + b_4 \cdot \hat{s}_t + b_5 \cdot \hat{y}_t) + \varepsilon_t^i \quad (\text{A.4f})$$

$$\bar{i}_t = \bar{r}_t + E_t[\pi_{t+1}^{tar}] \quad (\text{A.4a})$$

$$\hat{\pi}_t^f = E_t[\pi_{t+1}^{4,f}] - \pi_t^{tar,f} \quad (\text{A.4c})$$

$$\hat{\pi}_t^{xf} = E_t[\pi_{t+1}^{4,xf}] - \pi_t^{tar,xf} \quad (\text{A.4d})$$

$$\hat{s}_t = \Delta s_t^{AMD/USD} - \Delta s_t^{AMD/USD,tar} \quad (\text{A.4e})$$

The expected deviation of inflation from the target is worked out separately for the food and non-food components of CPI (that grow at different rates at the steady state) to allow for policy experiments with different reactions to food and non-food inflation rates.

$$\pi_t^{tar,f} = \pi_t^{tar} + (1 - \omega) \cdot \Delta \bar{r} \bar{p}_t \quad (\text{A.3a})$$

$$\pi_t^{tar,xf} = \pi_t^{tar} - \omega \cdot \Delta \bar{r} \bar{p}_t \quad (\text{A.3b})$$

Target exchange rate change. This policy variable captures the preferences for smoothing the exchange rate fluctuations either through interest rates or FX interventions. In the long-term it is made consistent with the inflation target and macroeconomic fundamentals. However, in the short-term we assume monetary authority may wish to exercise certain discretion over the exchange rate movements. The purpose of this equation is just to introduce the policy variable into the model. In practice, it will be subject of expert/policy tuning in a forecast process.

$$\Delta s_t^{AMD/USD,tar} = 0.3 \cdot \Delta s_{t-1}^{AMD/USD,tar} + (1 - 0.3) \cdot (\Delta z_t - \pi_t^* + \pi_t^{tar}) + \varepsilon_t^{star} \quad (\text{A.4b})$$

Policy transmission

Policy is transmitted via interest and exchange rate channels. Policy (and interbank) rates are transmitted to domestic lending and deposit rates. Their real equivalents in gap form drive the inter-temporal element in domestic demand, feeding through the output gap into real marginal costs of inflation equations.

The nominal exchange is also affected by expected policy rates, but additionally by FX interventions of the central bank. Nominal exchange rate movements feed directly into the imported elements of inflation equations. The real exchange rate as a gap enters the inflation equations directly - as part of real marginal costs - and indirectly, by affecting output through the substitution effects in export and import equations.

Deposit and lending rates. The rates in domestic currency are linked to the policy (interbank) rate using a simple forward looking mechanism and fixed term spreads:

(A.6d)

$$i_{dt} = term_t^d + h_{31} \cdot i_{dt-1} + (1 - h_{31}) \cdot (i_t + E_t[i_{t+1}] + E_t[i_{t+2}] + E_t[i_{t+3}])/4 + \varepsilon_t^{i_d}$$

(A.6e)

$$i_{lt} = term_t^l + h_{32} \cdot i_{lt-1} + (1 - h_{32}) \cdot i_{dt-1} + \varepsilon_t^{i_l}$$

Real interest rate gap. It is a composite measure of gaps in real lending and deposit rates in AMD and real foreign rate gap adjusted by expected exchange rate depreciation. The latter term captures the real interest rate cost of borrowing in foreign currency.

(A.7e)

$$r_{ft} = i_t^{*Fed-US} + ({}^e s_t^{AMD/USD} - s_t^{AMD/USD}) \cdot 4 - {}^e \pi_t$$

(A.7f)

$$\hat{r}_t = 0.25 \cdot \hat{r}_{dt} + 0.25 \cdot \hat{r}_{lt} + (1 - 0.5 - 0.25) \cdot \hat{r}_{ft}$$

Nominal exchange rate. A standard UIP equation is modified in two major ways. First, the exchange rate expectations are partly backward looking. Second, the sensitivity of the exchange rate to the interest rate differential is reduced to allow for the effect of FX interventions (proxied by the presence of the target exchange rate movement).

Thanks to these modifications the exchange rate movement can in the short-term violate pure UIP logic, however in the long-term (i.e. also for implied long nominal rates) these modifications disappear and pure UIP is restored.

(A.5a)

$$s_t^{AMD/USD} = \eta \cdot (s_t^{AMD/USD} - (i_t - i_t^{*Fed-US} - prem_t)/4) + (1 - \eta) \cdot (s_{t-1}^{AMD/USD} + \Delta s_t^{AMD/USD, tar}/4) + \varepsilon_t^{AMD/USD}$$

(A.5b)

$$s_t^{AMD/USD} = \gamma \cdot E_t[s_{t+1}^{AMD/USD}] + (1 - \gamma) \cdot (s_{t-1}^{AMD/USD} + 2/4 \cdot (\Delta z_t - \pi_t^* + \pi_t))$$

FX intervention proxy. The deviation of the exchange rate from a UIP prediction serves as a crude proxy for FX intervention activity. It cannot be directly compared to FX intervention volumes, because it is expressed in interest rate terms and is relative to other elements of the UIP.

(A.5c)

$$int_t = s_t^{AMD/USD} - (s_t^{AMD/USD} - (i_t - i_t^{*Fed-US} - prem_t)/4)$$

Effective exchange rate. The bi-lateral AMD/USD exchange rate is recast in effective terms using the cross rates of the Russian ruble and the euro so that the real exchange rate and imported inflation rates appear in effective terms:

(A.10d)

$$p_t^* = \mu_{US} \cdot p_t^{*US} + \mu_{EZ} \cdot (p_t^{*EZ} + s_t^{USD/EUR}) + (1 - \mu_{US} - \mu_{EZ}) \cdot (p_t^{*RU} + s_t^{USD/RUB})$$

(A.10e)

$$\pi_t^{*,ss} = \mu_{US} \cdot \pi_{ss}^{*US} + \mu_{EZ} \cdot (\pi_{ss}^{*EZ} + \Delta s_{ss}^{USD/EUR}) + (1 - \mu_{US} - \mu_{EZ}) \cdot (\pi_{ss}^{*RU} + \Delta s_{ss}^{USD/RUB})$$

(A.7a)

$$z_t = s_t^{AMD/USD} + p_t^* - p_t$$

Trends

The trend block includes several expenditure and relative price trends. These trends are held together using only two identities, hence are largely independent and can be chosen freely in a forecast.

Expenditure identity. GDP growth identity is an approximation of the actual GDP identity that holds in data in growth rates of real expenditure elements. This statistical construct helps to tie down the trends of expenditure elements.

(A.1f)

$$\Delta y_t = (1 - a_1 - a_2 + a_3) \cdot \Delta d_t + a_1 \cdot \Delta g_t + a_2 \cdot \Delta x_t - a_3 \cdot \Delta m_t$$

UIP in real trends. It puts together the trends in domestic and foreign real interest rates and a real exchange rate change. Because the real exchange rate is defined in effective terms (whereas the nominal exchange rate with respect to the dollar), the equation is adjusted by cross exchange rates and relative foreign inflation rates to make both nominal and real UIP consistent with each other.

(A.6a)

$$\begin{aligned} \bar{r}_t - \bar{r}_t^{*Fed-US} = & \Delta \bar{z}_t + prem_t - \mu_{EZ} \cdot (\pi_{ss}^{*EZ} - \pi_{ss}^{*US} + \Delta s_{ss}^{USD/EUR}) \\ & - (1 - \mu_{US} - \mu_{EZ}) \cdot (\pi_{ss}^{*RU} - \pi_{ss}^{*US} + \Delta s_{ss}^{USD/RUB}) \end{aligned}$$

External sector

The external environment is composed of the USA, Eurozone and Russia with equal weights. These weights define effective inflation rates, effective exchange rate and effective foreign output gap (demand). Besides, the Fed funds rate serves as the benchmark foreign rate. There are world relative prices of oil and food, and remittances defined in real terms in foreign currency. In addition, we have a relative price of exports to world CPI so that we can construct the terms of trade.

3.1 Data used in the core model

The data used in the structural model are the following (data are not seasonally adjusted if not stated otherwise):

Spot exchange rate AMD/USD The monthly average of the spot exchange rate of the Dram against the U.S. Dollar. The time series is obtained from the Bloomberg database.

Real gross domestic product The quarterly Armenia GDP at constant 2005 prices in national currency (Dram). The time series is published by National Statistical Service of the Republic of Armenia on a quarterly basis. The series have been seasonally adjusted by OGResearch.

Central bank policy rate The refinancing rate has been used as a policy rate. The time series is published by the Central Bank of Armenia. It is available from the Central Bank of Armenia on a monthly basis.

Government expenditures Real government expenditures derived from real GDP measured at 2005 constant prices. The series are available from the Statistical Office of the Republic of Armenia on quarterly basis. As the GDP series the expenditure series were adjusted for the seasonal component.

Imports Real imports. The quarterly series is a component of the real GDP series measured at 2005 prices. Imports were seasonally adjusted by OGResearch.

Exports Real exports. The quarterly series is a component of the real GDP series measured at 2005 prices. Similar to imports exports were seasonally adjusted.

Net flow of remittances Remittances play significant role in the dynamics of macroeconomic aggregates. The series measured in billions of USD were obtained from the Central Bank of Armenia.

Consumer price index inflation CPI inflation obtained from the National Statistical Service of the Republic of Armenia.

Food consumer price index inflation Inflation in the food prices included in the CPI, obtained from the National Statistical Service of the Republic of Armenia.

Ex-food Consumer price index inflation Inflation in the prices excluding food prices, obtained from the National Statistical Service of the Republic of Armenia.

Inflation target The Central Bank of Armenia engaged in explicit inflation targeting since early 2006 fixing the central inflation target at 4 per cent within 1.5 per cent target bands.

Nominal exchange rate USD/EUR The mid-rate of USD/EUR exchange rate. The exchange rate is obtained from the Bloomberg database.

Nominal exchange rate RUB/EUR The mid-rate of RUB/EUR exchange rate. The exchange rate is obtained from the Bloomberg database.

U.S. Headline Consumer Price Index The seasonally adjusted data are obtained from the OECD database. The CPI is available on a monthly basis.

U.S. real gross domestic product U.S. GDP in chained 2005 Dollars. The seasonally adjusted data are available on a quarterly basis in the Bloomberg database.

Eurozone Harmonized Consumer Price Index The seasonally adjusted data are obtained from the OECD database. The CPI is available on a monthly basis.

Eurozone real gross domestic product Eurozone GDP in millions of Euros at 2000 prices. The seasonally adjusted data are available on a quarterly basis in the Bloomberg database.

Russia Consumer Price Index The seasonally adjusted data are obtained from the OECD database. The CPI is available on a monthly basis.

Russia real gross domestic product Russia GDP in millions of RUB at 2000 prices. The seasonally adjusted data are available on a quarterly basis in the Bloomberg database.

U.S. Fed Funds O/N interest rate The time series is obtained from the American Federal Reserve Board.

World oil price As an indicator of the world oil price, the price of Brent Crude oil in U.S. Dollars is used. The oil price is obtained from the Bloomberg database.

World food price The time series with the world food price index is obtained from the Food and Agricultural Organization of the United Nations. The index is provided with a monthly frequency, a base period is the average of 2002–2004.

Deposits interest rates Nominal 1 year deposits interest rates obtained from the Central Bank of Armenia.

Credits interest rates Nominal 1 year interest rates on credits available from the Central Bank of Armenia.

Terms of trade An index of export prices divided by an index of import prices. Important items in exports and imports series such as metals, diamonds, and cereals were isolated with relevant weight to construct the measure. The series are available from the Central Bank of Armenia.

3.2 Dynamic properties

Impulse response functions

This section describes dynamic properties of the core model through impulse response to structural shocks. We provide only a selection of shocks that were found important in explaining Armenian data. The rest can be obtained on request.

The model is calibrated so that it fits reasonably well the recent Armenian history including the beginning of the inflation targeting regime (see Chapter 5.1). However, that is not necessarily the most reasonable calibration to use in a forecast for the future. While the structural economic changes may have long lags, monetary policy behavior, its transmission and perception by the public can change fairly quickly. Indeed, a usual part of a forecast exercise is to experiment with various policy parameters and expectation formation mechanisms in order to identify “robust” (if not optimal) policy responses to current pressures and risks. In our model, such experiments should focus on the following features that are no doubt going to change in a near future:

- The target level of an exchange rate change. Going forward, the authorities are probably going to rely less on interventions, as more standard transmission mechanisms settle in.
- Forward looking policy. In the current calibration policy looks only one quarter ahead. This is very extreme and policy horizons are likely to be longer in practice.
- Interest rate reaction function. In the past, the policy rate has been relatively stable responding to shocks gradually. In the current calibration the policy rate reacts to the output gap and the exchange rate, underscoring the flexible IT strategy. However, the policy rate may become more active in the future as well as less responsive to the exchange rate, for instance.

To illustrate the importance of these features we analyze below the model response to a change in the inflation target under two calibrations: the current one and one with a more active and forward-looking interest rates.

Private expenditure shock(ε^d). Figure 3.1.

Higher demand increases inflation by raising the marginal costs of both Phillips curves. However, output is up only half of demand, as some additional demand dissipates through exports. Inflation increase is relatively small and so is the interest rate reaction. Higher interest rates strengthen the nominal (and real) exchange rates, bringing inflation down and reducing the overheating - directly, via imported inflation, and indirectly, by reducing the real marginal costs and output (through substitution effects).

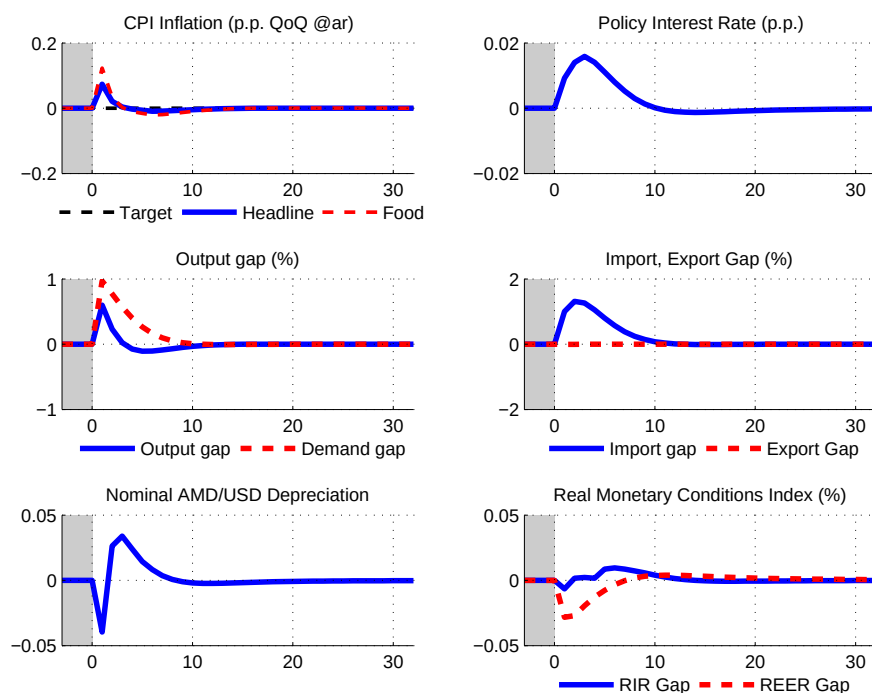


Figure 3.1: Private Expenditure Gap Shock

Ex-food inflation shock ($\varepsilon^{\pi^{xf}}$). Figure 3.2.

Higher inflation triggers a mild response from policy rates – not enough to raise real interest rates, but enough for a small nominal appreciation. Falling real interest rates boost demand, but strong real exchange rate diverts even more demand to imports, and domestic output falls as a result. Negative output gap and over-valued real exchange rate are the main forces bringing inflation down.

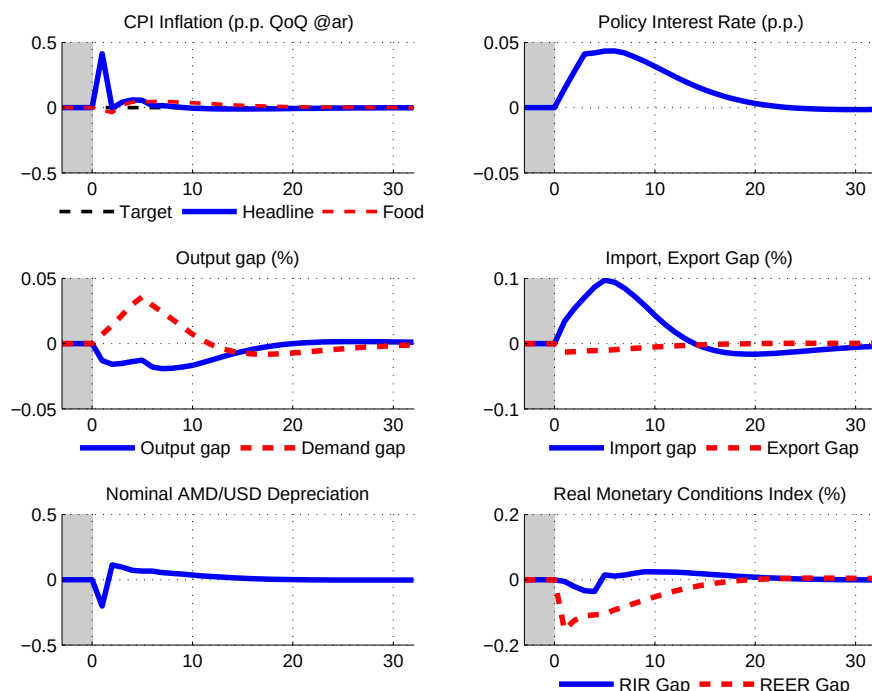


Figure 3.2: Ex-food Inflation Shock

Policy shock (ε^i). Figure 3.3.

An unexpected policy shock brings down demand and output - directly via real interest rates as well as indirectly through real exchange rate appreciation. Falling output as well as the strong real exchange rate put inflation down too. Imports temporarily increase, as the substitution effect initially drives some domestic demand to foreign goods. Later on the substitution effect gives way to the effect of declining demand.

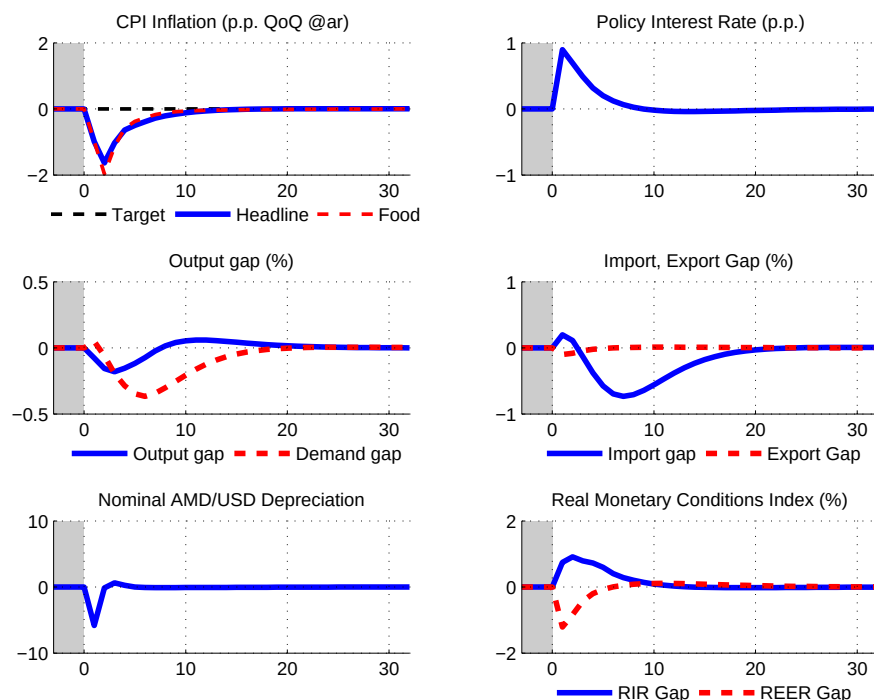


Figure 3.3: Policy Shock

Shock in Real world price of food trend($\varepsilon \Delta \bar{r}^p^{food}$). Figure 3.4.

This shock presents the reaction of the economy to a permanent level shift in the world price of foods (relative to US CPI). Domestic inflation increases and the price level ultimately too. Policy reaction is very small and the negative real rates give a small boost to domestic demand and output. The nominal exchange actually depreciates on impact - the higher price level ultimately demands a weaker exchange and the policy rate reaction is too small to overcome this effect.

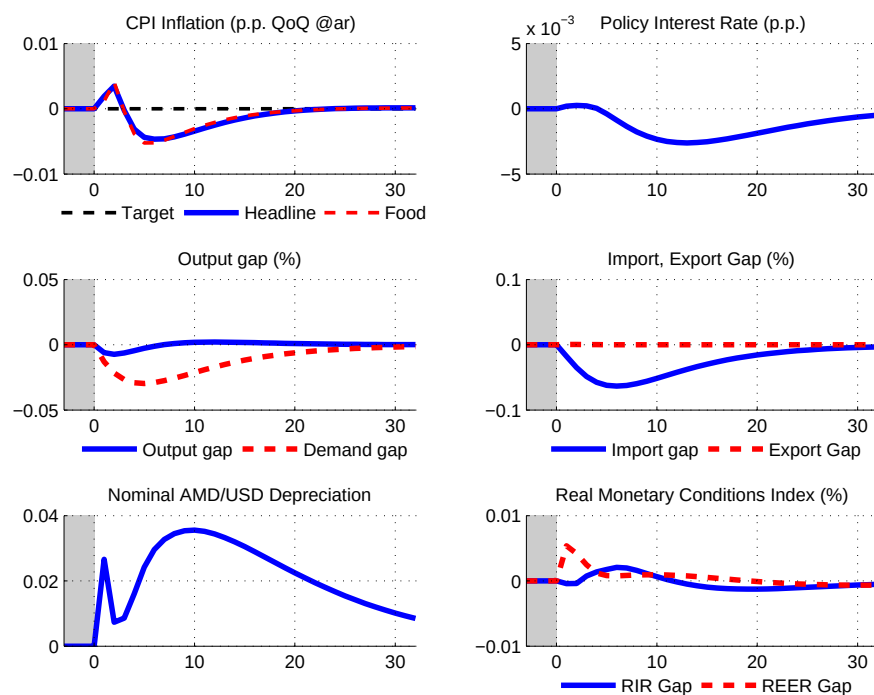


Figure 3.4: Shock in Real World Price of Food Trend

Foreign interest rate shock ($\varepsilon^{i*FED-US}$). Figure 3.5.

Higher foreign rates reduce the interest rate differential triggering exchange rate depreciation on impact. Depreciation feeds to inflation both directly as well as through the weaker real exchange rate and higher output. Demand is even higher than output thanks to negative real rates. Policy reaction gradually removes the stimulus from weak real interest and real exchange rate.

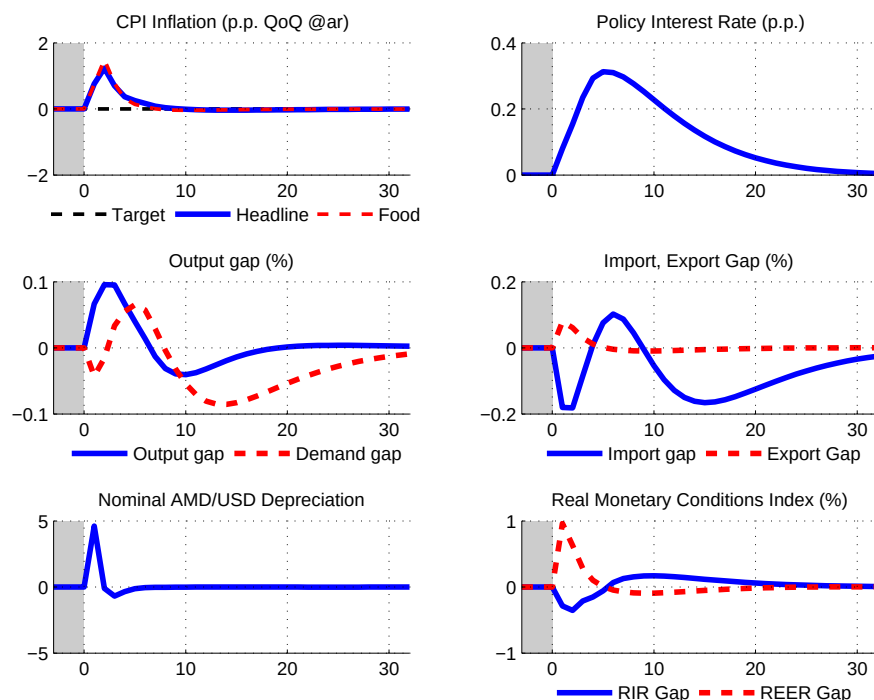


Figure 3.5: Foreign Interest Rate Shock

Inflation target shock ($\varepsilon^{\pi^{tar}}$). Figure 3.6.

The shock decreases the target level of inflation by 1 percentage point on impact. In order to get disinflation going the monetary authority is mainly relying on its exchange rate intervention mechanism and lets the target exchange rate appreciate sharply. As a consequence, the nominal exchange rate moves too, tightening monetary conditions, bringing down both output and inflation. The decline in inflation is almost immediate, and inflation widely undershoots the new target level in reflection of the high exchange rate pass-through. Given the strength of this exchange rate intervention mechanism, the policy rate can move down fairly fast, because inflation expectations decline rapidly ensuring the real interest rate increase and reinforce the output decline. The output loss needed to bring 1 percent permanent disinflation (the sacrifice ratio) is about 0.25 percent – relatively low, underscoring the flexible nature of the inflation process in Armenia that is very responsive to the exchange rate.

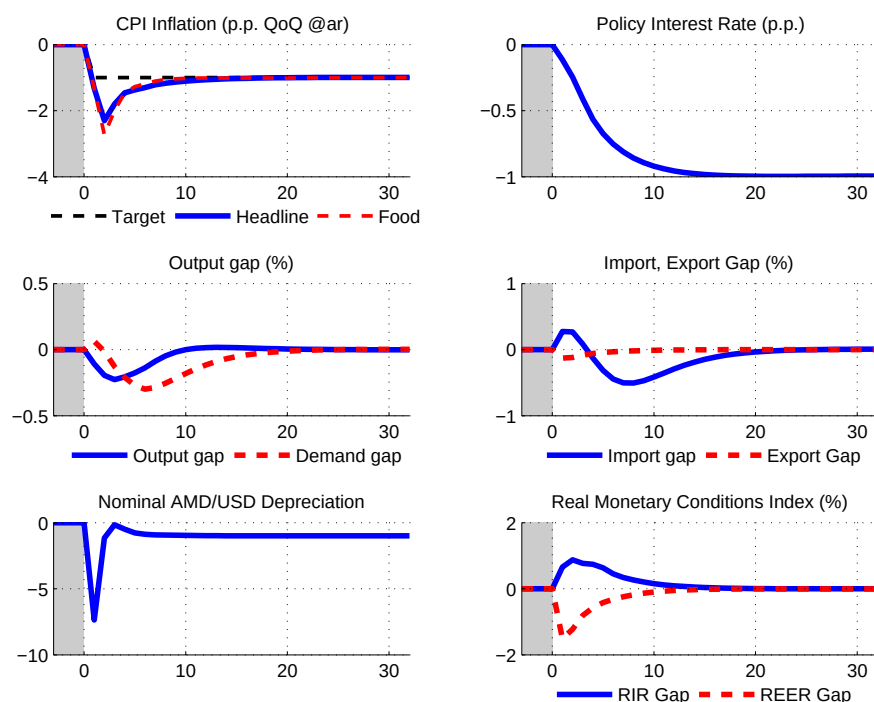


Figure 3.6: Inflation Target Shock

By contrast, Figure 3.7 shows the model response to the same shock under the following modifications: (i) the policy rate auto-regression coefficient is reduced from 0.9 to 0.7, (ii) the lead in the expected deviation from target is increased from one quarter to four, and

(iii) the target exchange rate change is dropped. In short, the policy rate is more active and forward-looking, and the central bank does not use direct FX market interventions.

Evidently, the exchange rate channel is much weaker in this simulation and inflation does not undershoot the new target nearly as much. Policy rates now play a more important role than before: although they do not rise at the beginning, their fall is more gradual, as they need to ensure that the real interest rates are positive enough to bring down output and inflation. The sacrifice ratio remains similar, because the calibration of structural characteristics and transmission channels stays the same. Only the mix of instruments needed to achieve the given loss in output is different.

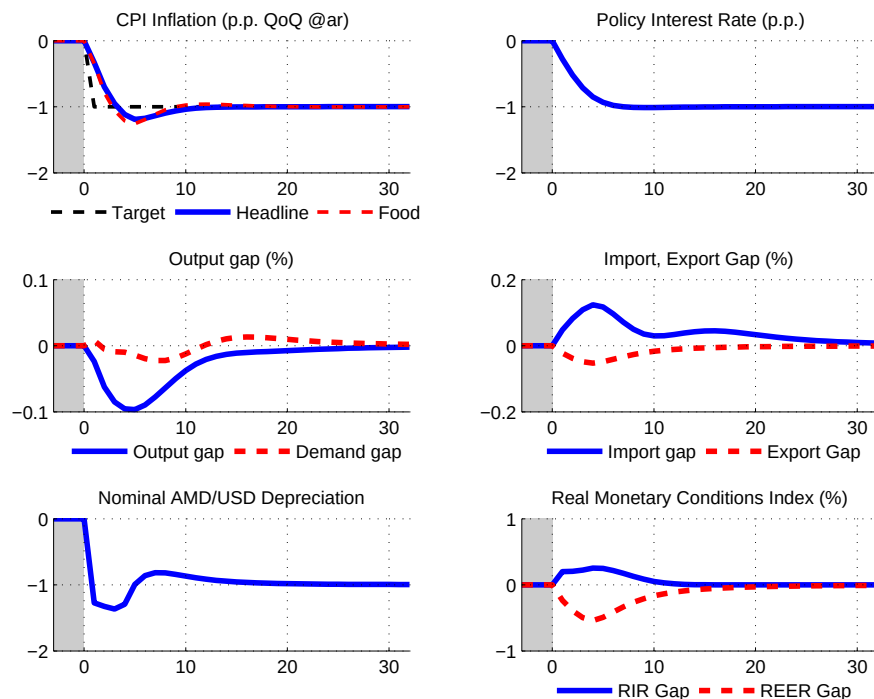


Figure 3.7: Inflation Target Shock

Exchange rate shock ($\varepsilon^{AMD/USD}$). Figure 3.8.

The exchange rate depreciates on impact, driving up both inflation and domestic output. Trade balance improves due to the substitution effect of the real exchange rate. Policy rates increase causing the exchange rate to partially come back. High inflation eventually strengthens the real exchange rate, bringing output and inflation back to their original levels.

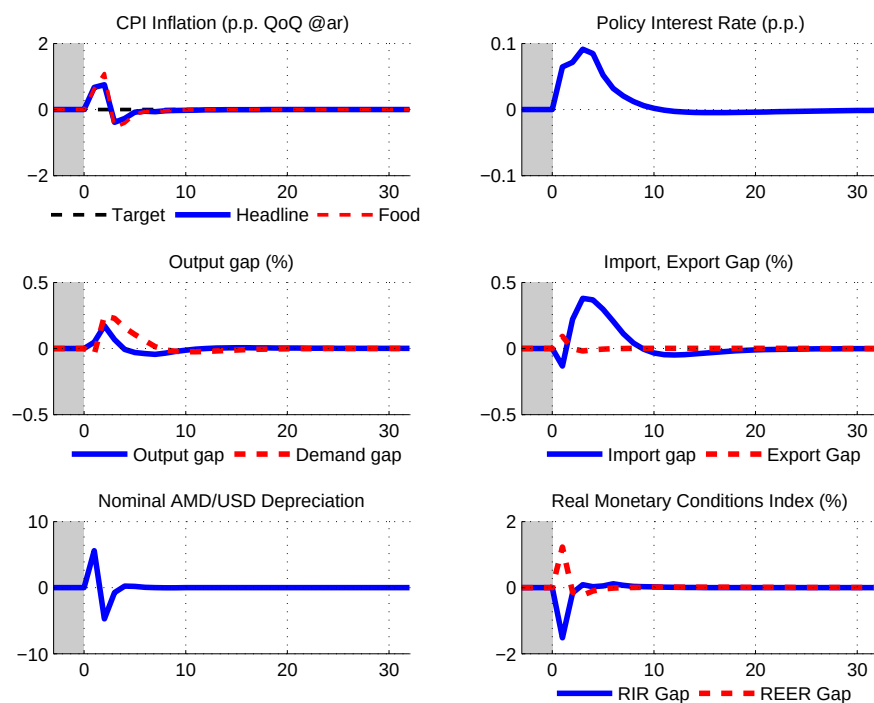


Figure 3.8: UIP (Exchange Rate) Shock

Variance decomposition

This section presents the variance decompositions of the main Armenian variables: the nominal policy interest rate, the nominal AMD per USD exchange rate, YoY CPI inflation, and the output gap. The decompositions are presented in tables and standardized charts.

The decompositions characterize the stochastic properties of the structural model and are crucial for the computation of the forecast confidence bands.¹

They are also a useful tool for model calibration and policy analysis. The model implied unconditional volatility of macroeconomic variables should correspond to the historically observed volatilities or economic intuition. Inspecting that a variable's volatility is due to some unexpected shocks signals problems in the calibration of the transmission mechanism or the shocks' standard errors. This affects the Kalman filter and its estimation of the initial forecast conditions. The decompositions can also help dissect the forecast uncertainty into more or less relevant macroeconomic shocks and focus the policy discussion accordingly.

Unconditional variances and their decompositions are often used in policy analysis as well. For instance, the model implied volatilities for the most important macroeconomic variables, such as inflation, output growth, interest rates or an exchange rate change, can be compared across different policy rules in an effort to find a normatively more "convenient" (if not optimal) constrained rule.

The notations of the shocks are summarized in the following table.

¹The confidence bands of a forecast are given by the conditional variances presented in this subsection and the uncertainty about the initial condition of the forecast. The latter is estimated by the Kalman filter.

Shock Abbreviation	Description
i	Monetary Policy Shock
dl_amd_usd	UIP Premium Shock
dl_cpi_f	Food Inflation Cost Push Shock
dl_cpi_xf	Ex-food Inflation Cost Push Shock
l_d_gap	Private Expenditures Shock
l_m_gap	Import Shock
l_x_gap	Export Shock
l_qremit_gap	Real Remittances Shock
l_qfood_gap	Real World Price of Food Shock
dl_usd_eur	USD per EUR Depreciation Shock
dl_usd_rub	USD per RUB Depreciation Shock
dl_cpi_ru	Russia Inflation Shock
dl_px2cpi_us	Relative Export Price to US CPI Shock

Note that the inflation target shock has been switched off for the computation of the variance decompositions. The reason is that this shock is by construction the main source of volatility of nominal variables and in the long-term it overrides the effects of other shocks.

Central bank's policy rate

Figure 3.9 shows the conditional variance decomposition of the short nominal interest rates.

For the first year, variance of the interest rate is highly dominated by the policy rate shock (*i*). Over time, it will make up for more than half of the total interest rate volatility, see the unconditional variance decomposition in Fig. 3.13. The volatility is also influenced by the shock in the AMD per USD exchange rate (*dl_amd_usd*) reflecting the presence of an exchange rate element in the central bank's reaction function. Inflation - cost-push shocks, such as the food inflation shock (*dl_cpi_f*), ex-food inflation shock (*dl_cpi_xf*), and world food price shock (*l_qfood_gap*) also play a role.

Nominal exchange rate

Figure 3.10 depicts the conditional variance decomposition of the $100 \cdot \log$ of the AMD/USD nominal exchange rate's level. The $100 \cdot \log$ transformation allows us to interpret standard errors in percents.

In the short-run, the most important shock driving the exchange rate volatility is the white noise UIP premium shock (*dl_amd_usd*). This shock loses its importance in favor of the

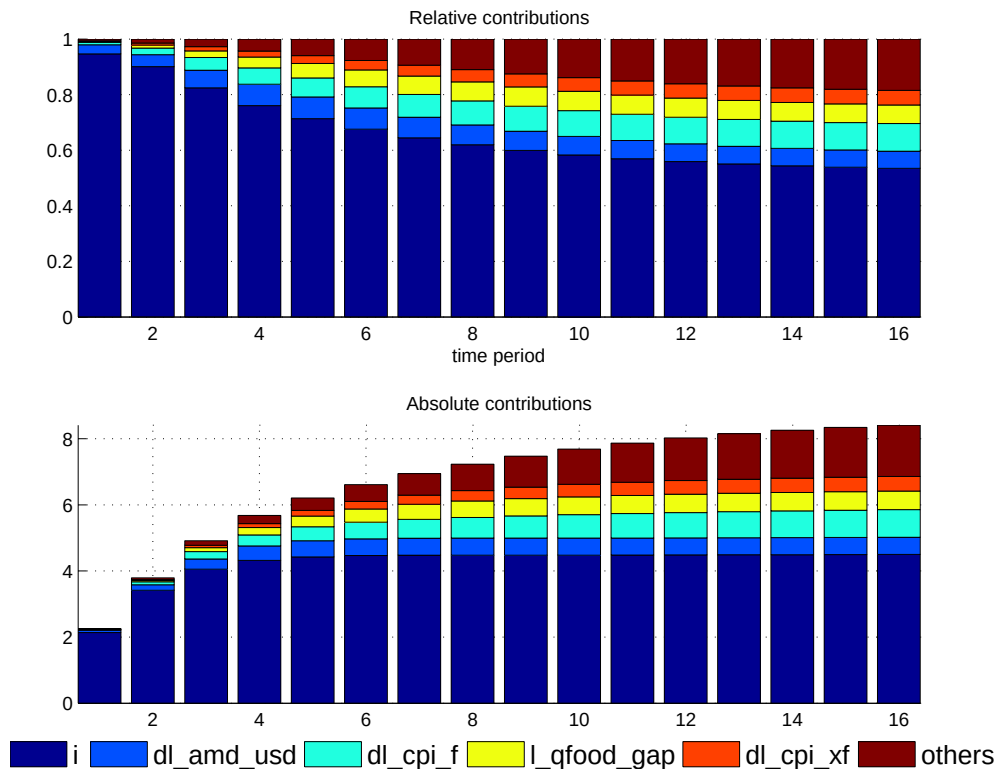


Figure 3.9: Conditional variance decomposition of the central bank's policy rate (percent p.a.)

monetary policy shock and the shock to relative export price that are contained in the group called *others*. Other important uncertainties are presented by shocks to the Euro and Russian ruble exchange rates, and Russian inflation.

YoY CPI inflation

Figure 3.11 depicts the conditional variance decomposition of the YoY CPI inflation rate. The most important shocks are both food and ex-food cost push shocks (*dl_cpi_food* and *dl_cpi_xf*) in the short-run. The inflation uncertainty also bears from the nominal exchange rate shock via imported inflation. They together explain about 80 % of volatility in the very first quarter, but their importance decreases to about 40 % in the long run.

From the second quarter onwards, the relative and absolute importance of the nominal interest rate shock (*i*) increases rapidly.

Output gap

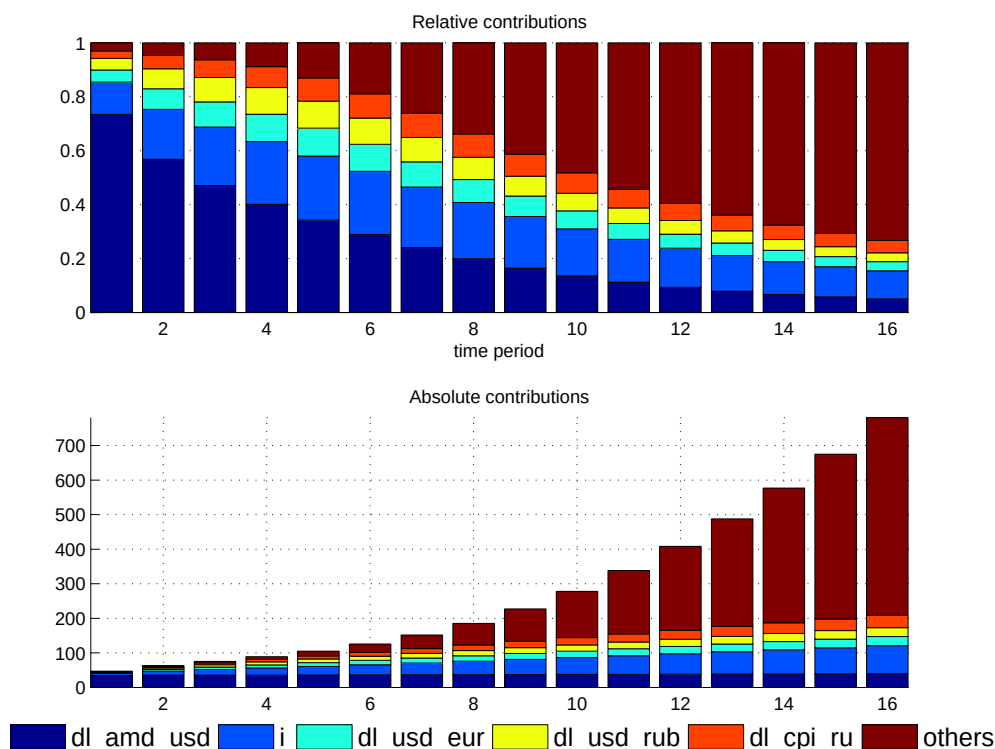


Figure 3.10: Conditional variance decomposition of the AMD/USD exchange rate (100*log)

Figure 3.12 depicts the conditional variance decomposition of the output gap. In the short and long run, the volatility is dominated only by demand shocks (shock to imports l_m_gap , private domestic expenditures l_d_gap , exports l_x_gap , and remittances l_qremit_gap). The share of these shocks in the unconditional variance of the output gap is almost 85 %.

The fourth most significant shock is the shock to relative export price to US CPI reflecting the terms of trade influence on domestic real activity, see Fig. 3.13,

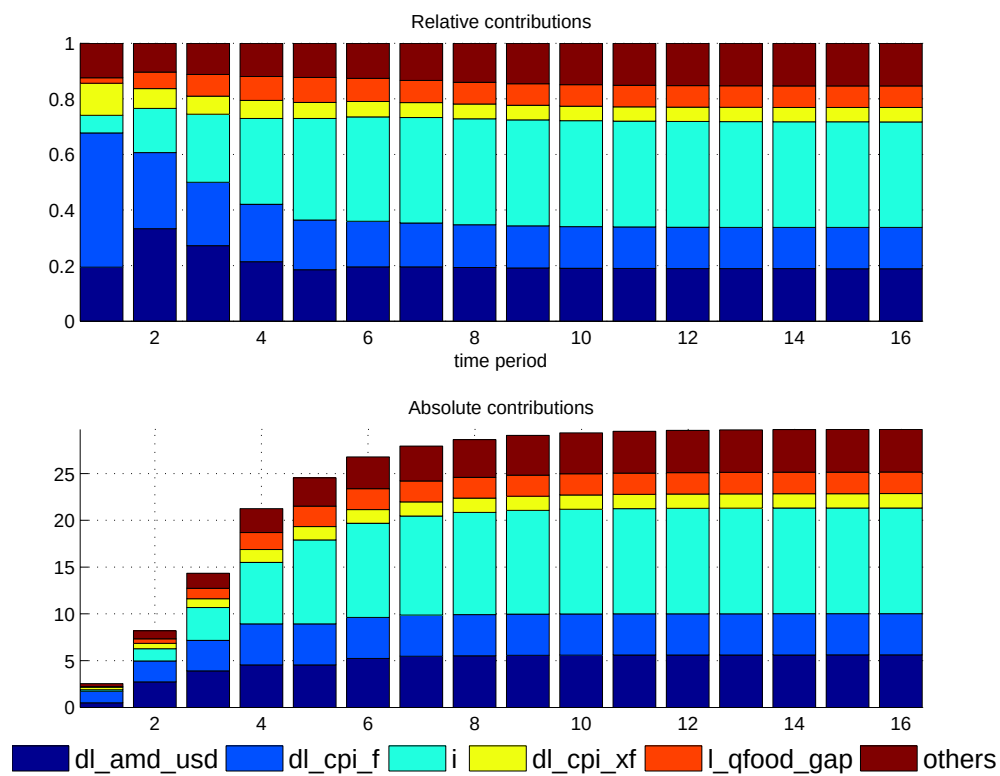


Figure 3.11: Conditional variance decomposition of CPI inflation (percent YoY)

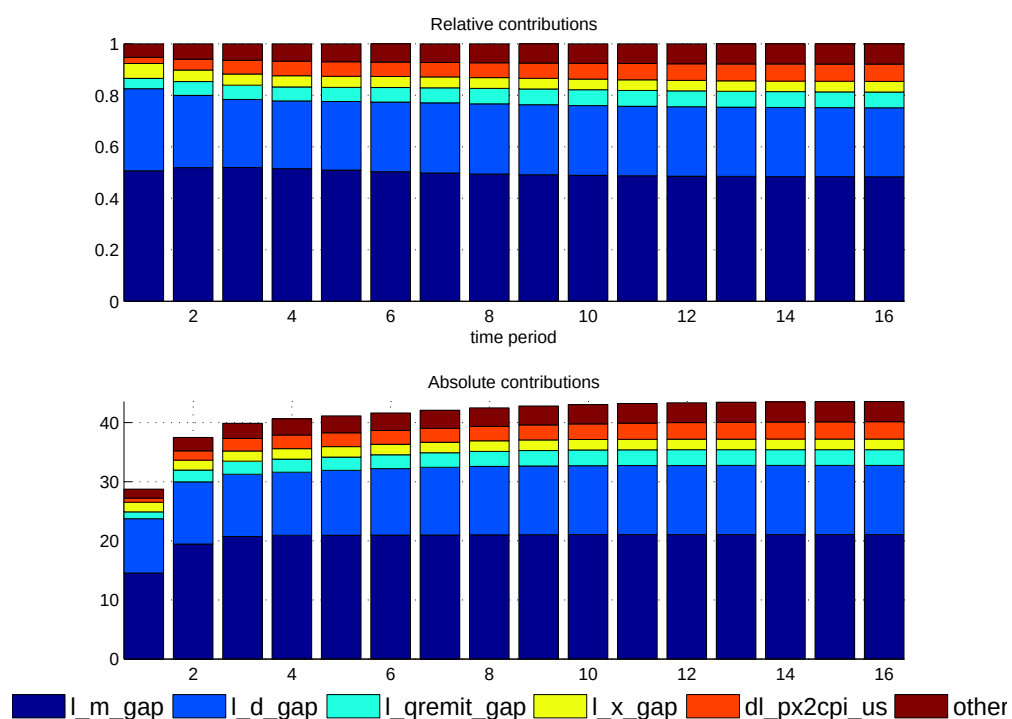


Figure 3.12: Conditional variance decomposition of the real GDP gap (percent)

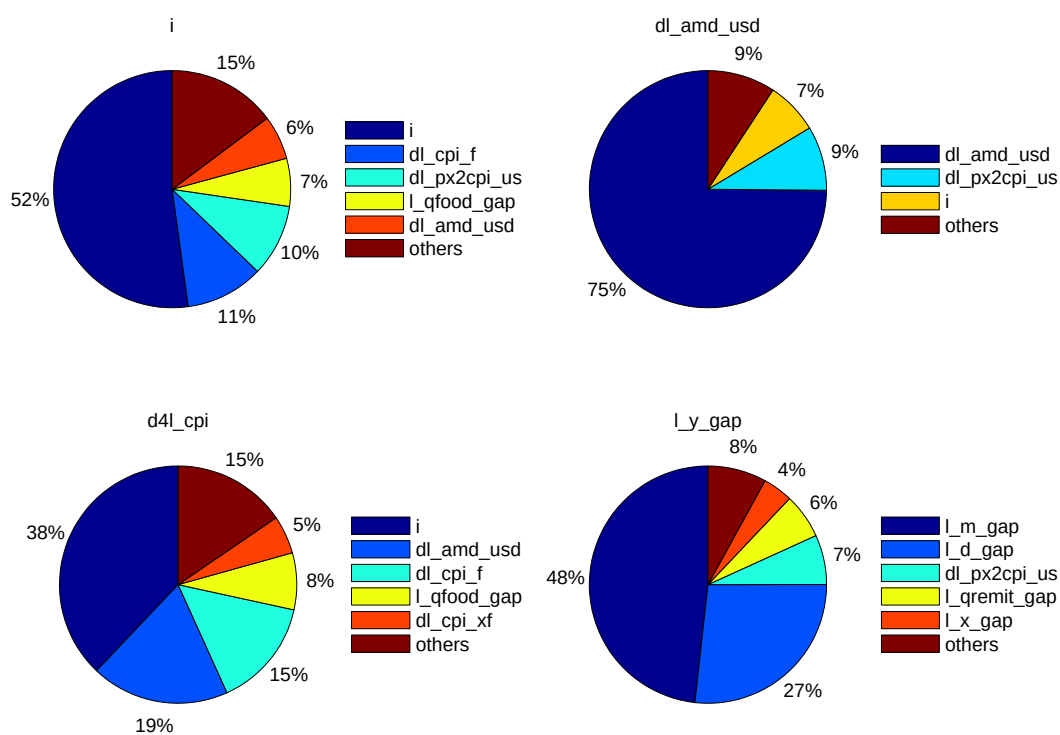


Figure 3.13: Unconditional variance decomposition

3.3 Interpretation of history

In this section we interpret the Armenian economic history by applying the model to historical data. Confrontation of the model interpretation with data and economic intuition is an important debugging tool that also helps in identification of some of the model parameters.

The model's structure separates long-term development of trends (also loosely referred to as 'equilibrium') from cyclical fluctuations of the respective gaps ('disequilibrium'). Therefore the following interpretation of history is based on the model-consistent estimations of these unobserved variables - trends and gaps - by a procedure known as Kalman filter. The procedure also estimates historical shocks hitting the economy and allows us to compare the reaction of the economy to that of the model.

In the following text we focus on the model's interpretation of selected historical episodes and show only the relevant estimation results. The full Kalman filter results can be obtained upon request.

We focus on the following historical periods:

- Rapid nominal and real appreciation in 2005–2007
- Adverse impact of global economic crisis in 2008–2009
- Gradual economic recovery in 2010

Rapid nominal and real appreciation in 2005–2007

The Armenian economy enjoyed double digit economic growth from early 2002 (see figure 3.14). The authorities had maintained a fixed exchange rate before 2005. Despite the high growth and fixed exchange rate inflation stayed low.

The situation changed rapidly at the end of 2004. The terms-of-trade suddenly improved – owing mainly to low prices of imported hydrocarbons and food products, and high prices of exported molybdenum, copper and other minerals. This accelerated economic growth further, giving a boost to mining industry and exports (all figure 3.14).

Besides, the economic boom in Russia elevated the amounts of remittances from the Armenian diaspora reducing poverty and boosting residential construction.

The strong capital inflows (including FDI), remittances and booming exports put strong appreciation pressures on the dram that culminated in relaxation of the exchange rate regime followed by a sharp appreciation in 2005. The dram continued to appreciate at a predictable pace for the rest of the period, making it one of the factors keeping a low inflation environment throughout the period despite the strong growth.

The nominal appreciation was keeping inflation close to the estimated target of 5-6%. Although the authorities continued to reduce the speed of nominal appreciation in the following years by FX interventions, Figures 3.20 and 3.21 shows how the imported element helped to tame non-food inflation despite the high growth and slowly rising domestic imbalances (pushing up the real marginal component of inflation).

The persistent real exchange rate appreciation is reflected in the estimation of the country risk premium – it is estimated to have declined from its peak at about 15 percent in 2005 to only 2 percent in early 2008. This is consistent with the soft evidence from that period – improving economic conditions, growing appetite of foreigners to invest into Armenian economy and declining dollarization (see figure 3.15). In line with the declining risk premium, the estimation ascribes much of the observed real appreciation to a trend rather than a disequilibrium (gap) movement. According to the estimation, the real exchange rate trend started appreciating rapidly in early 2005 (see figure 3.15), having been flat before. The abrupt change in the real exchange rate trend at the beginning of 2005 explains why Armenia could have enjoyed high growth, fixed nominal exchange rate and low inflation in the period before 2005.

Because much of the real appreciation after 2005 was equilibrium according to the estimation, it did not undermine growth (Fig 3.16). The actual real appreciation then pretty much followed the estimated equilibrium movement in 2005-2007. At the same time, both demand and output were growing faster than the potential, eventually substantially overheating the economy. With the neutral real exchange rate, the exports benefited from high foreign demand (see figure 3.18); while private demand was boosted by low real interest rates at first, and growing remittances later on (see figure 3.17).

The model's estimation shows a positive and rising domestic demand gap for the entire period (see figure 3), culminating by a 20% overhang just before the crisis. The output gap is estimated to have risen from close to zero in 2005 to almost 15% before the crisis. This is despite the fact that the estimated potential output growth has also risen in this period - corresponding to soft evidence of remittances and terms-of-trade boosting domestic investment and construction (see figure 3.19) The real appreciation speeded faster than equilibrium in 2007/8, but the overvalued real exchange rate did little in curbing the internal imbalances - instead, it fed into rising imports and external disequilibrium.

Adverse impact of global economic crisis 2008–2009

The Armenian economy was hit hard by the global economic crisis. Although the Armenian financial sector is relatively isolated from global markets and the banking system remained sound, the drop in foreign demand and receding FDIs were the main channels of economic downturn in Armenia, which saw output declining by more than 14 percent in 2009.

The estimation shows the real exchange rate as substantially overvalued in late 2008 and demand overheating. With such a predicament the economy quickly collapsed, once the external environment turned recessionary and the country risk premium started rising (see figure 3.15). The sharp decline of world demand in late 2008 hit Armenian exports (see figure 3.18). Falling world demand put commodity prices under pressure adversely affecting the Armenian terms-of-trade and worker's remittances (most of private transfers come from Russia whose business cycle closely depends on export of commodities). These factors contributed to a sharp drop in Armenian private demand in late 2008 (see figure 3.18), but the estimation shows that both the gap and potential were hit at the same time (see figure 3.19). Construction and mining sectors were hit most and aggregate domestic demand is estimated to have dropped from almost 15 percent of excess demand in early 2008 to minus 10 percent below (equilibrium) trend in early 2009 (see figure 3.16).

Rising uncertainties of foreign investors and recoiling FDIs put additional pressures on the Armenian exchange rate, while the central bank's commitment to a soft peg caused a large loss of international reserves. To partially equilibrate the large imbalances the authorities let the nominal exchange rate depreciate by close to 35 percent (w.r.t. USD) between early 2009 and early 2010.

Inflation in this period was driven mostly by the strong exchange rate pass-through and international commodity prices. Both food and non-food inflation increased in 2007-2008 following the soaring world oil and food prices, while exchange rate appreciation was partially offsetting – especially during 2007 (see figures 3.20 and 3.21). However, rising domestic demand - far in excess of its trend - fuelled inflation further in 2008.

The sharp fall of commodity prices in 2009, declining domestic demand and initially stable nominal exchange rate created a disinflationary environment and inflation dropped from more than 10 percent in late 2008 to only 2 percent in early 2009, well below the official inflation target (see figure 3.14).

Gradual economic recovery in 2010 and 2011

Despite unfavorable weather conditions, low external demand and difficult access to foreign financing, the Armenian economy returned to growth - albeit lackluster - in early 2010. The main factors of the rebound in private demand were low real interest rates and growing remittances (see figure 3.17). Government demand also contributed to the recovery, as the authorities - helped by international institutions - succeeded in running counter-cyclical policies.

However, the recovery is still very fragile - GDP growth has hovered close to zero since early 2010. Indeed, the estimation shows that potential output growth still remains below its long-term average and the risk premium above the pre-crisis level. This accords with soft-facts such as the low investment activity and near frozen construction sector (see figure 3.19). Dollarization also keeps high. In the meantime inflation has soared to double-digit levels. The exchange rate depreciation from 2009 coupled with soaring commodity prices has been taking its toll against the backdrop of recovering demand.

The high inflation in 2010/11 has tightened the real exchange rate. The estimation shows it has been overvalued since late 2010 and thus stymieing the fragile recovery through substitution effects. The coincidence of an overvalued real exchange rate and rapid recovery could potentially lead to accumulation of both internal and external imbalances - a situation similar to the pre-crisis period of 2007 and 2008. Output and demand were already close to their potential levels in early 2011 and their further growth is likely given the very low real interest rates, high public expenditure and remittances. At the same time, the overvalued real exchange rate signals that this new demand will fuel imports, while domestic investments (and potential) may remain at low levels for some time in the near future.

Against this backdrop, inflation risks might not be benign. Fuelled by commodity prices, inflation is already above the pre-crisis levels, although output and demand gap are nowhere near the pre-crisis levels. This points to the importance of nominal and real appreciation that were both keeping inflation on check in the pre-crisis period, but are absent now. In other words, according to the estimation the economy has much weaker fundamentals (lower potential output growth, little trend appreciation and high risk premium) than it had in 2007.

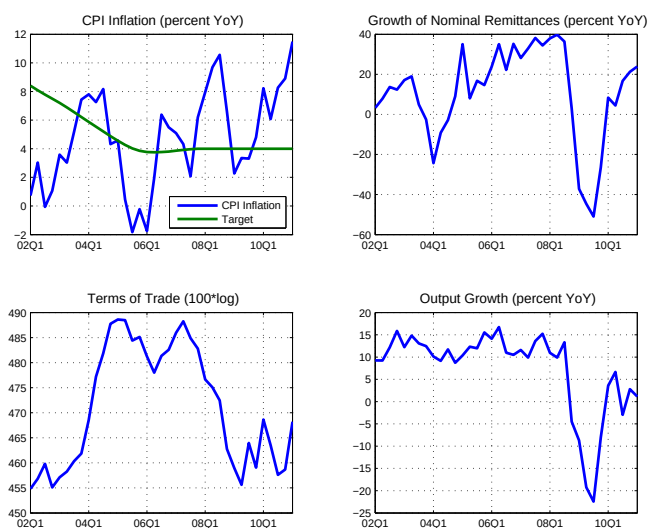


Figure 3.14: Selected Macroeconomic Indicators for Armenia

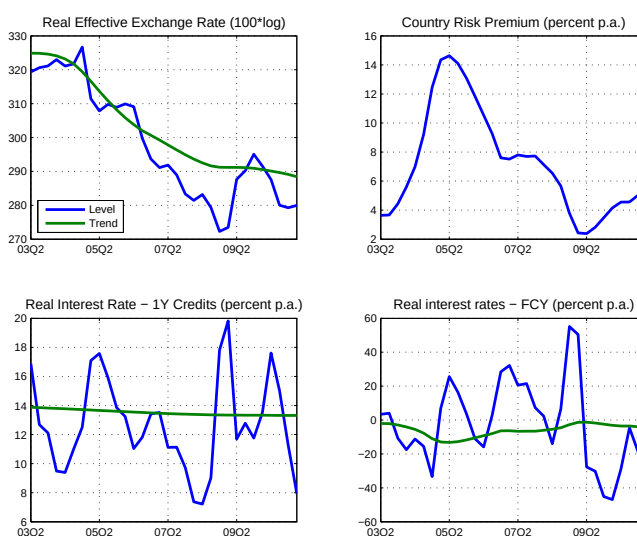


Figure 3.15: Real Trends in Armenia

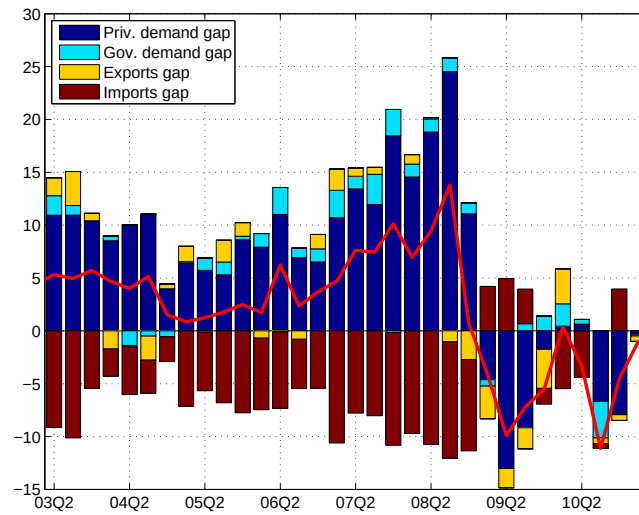


Figure 3.16: Output Gap Decomposition

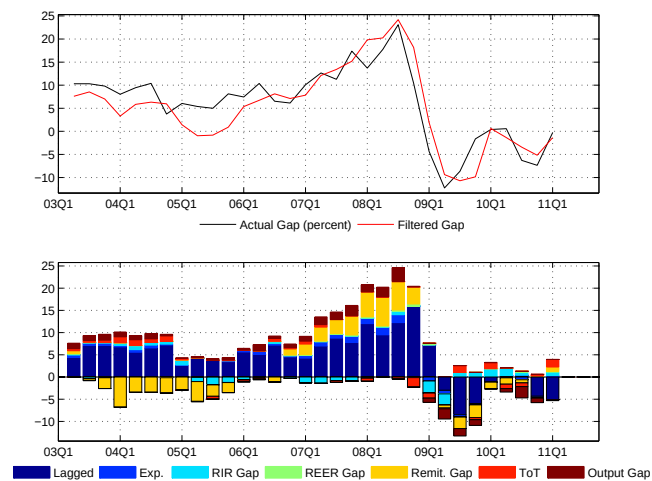


Figure 3.17: Private Demand Decomposition

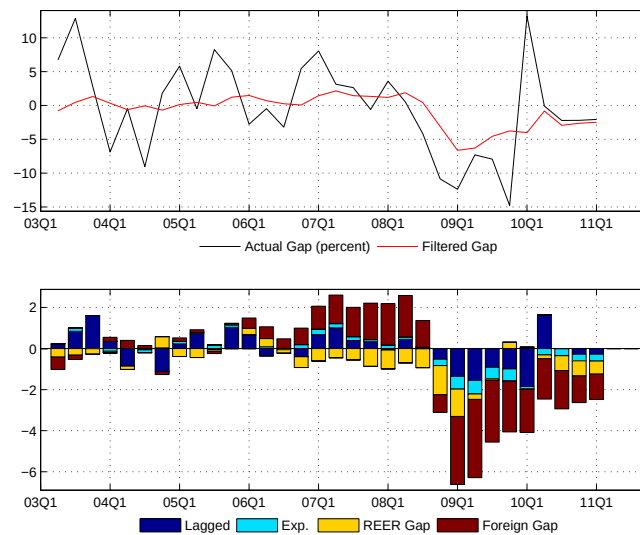


Figure 3.18: Exports Decomposition

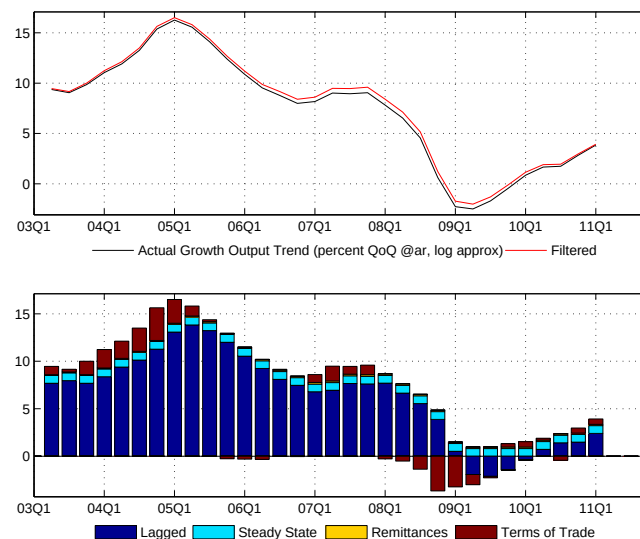


Figure 3.19: Potential Output Growth Decomposition

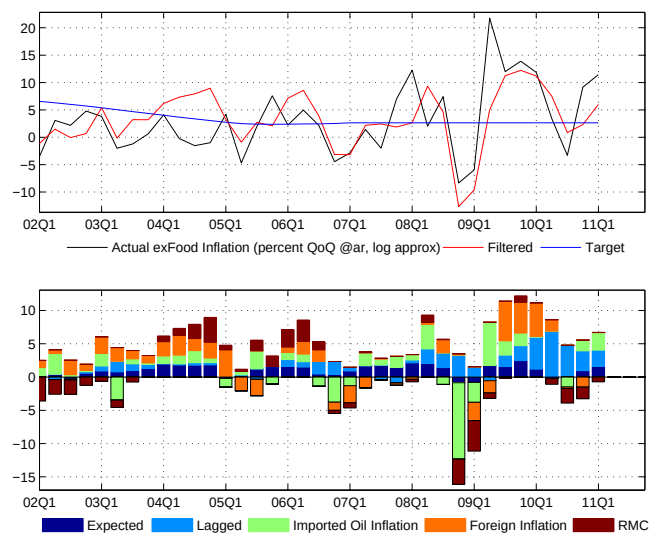


Figure 3.20: Decomposition of CPI Inflation Excluding Food Prices

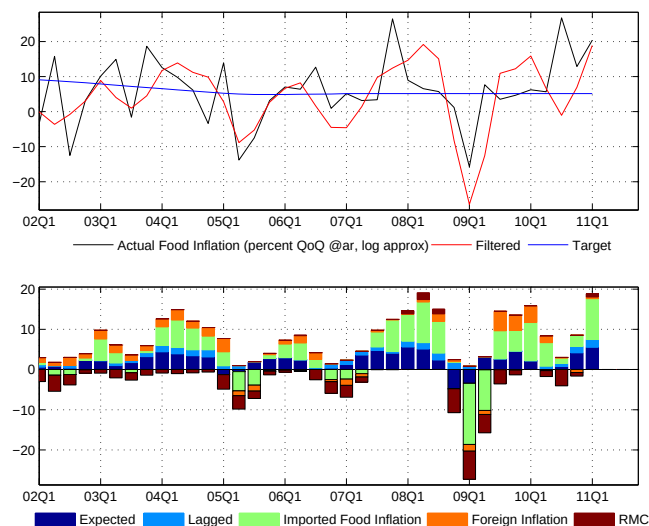


Figure 3.21: Decomposition of Food CPI Inflation

Chapter 4

Other Models in the FPAS

The core forecasting and simulation model in the FPAS framework can be complemented by other models to increase the forecasting accuracy, to extend the scope of the analysis or bring a richer theoretical structure to bear in analyzing shocks to the baseline projection.

In terms of forecasting accuracy, judgment and expert understanding of recent data are typically superior to any model. The technical environment of the FPAS has to ensure that the projection can successfully incorporate such judgment in the core model simulation. However, while superior to models, good data expertise takes time to build. In the absence of such expertise, forecasting accuracy can also benefit from data-driven models - especially in the near-term. There are a multitude of such data-driven approaches that central banks use (often concurrently) from simple univariate extrapolation methods to common factor models or neural networks. We present here one such approach that fits the Armenian data reasonably well - BVAR.

In terms of extending the scope of the core model, it is desirable to be able to generate consistent forecasts for the rest of the world to use as an input into the Armenian projection. A model-based framework for generating such external projections is provided by the Global Projection Model (GPM), developed at the Research Department of the IMF. GPM is capable of producing both baseline projections and scenario analyses for six major regions of the world, including the major advanced countries and the larger emerging-market economies.

Regarding theoretical structure, the longer-run or “equilibrium” values of some key variables in the core model are tied to stochastic trends, as opposed to being derived endogenously. Deeper structural models, while less useful from a baseline forecasting

perspective, can be used to perform scenario analysis around the baseline projection to provide insights into the economic forces at work in the medium and long runs, including how initial economic imbalances are resolved in terms of full stock-flow dynamics and associated relative price adjustments.

The next section of this chapter describes near-term forecasting model using a Bayesian VAR framework. This is followed by the discussion of the structure of the GPM model and how it can be used to provide forecasts of the external environment for a country like Armenia. Finally a small structural model is designed that has been calibrated to broadly replicate some of the main features of the Armenian economy. The model incorporates full stock-flow dynamics, as well as imperfect international asset substitutability, allowing it to be used for the analysis of a wide range of scenarios involving both policy innovations and other shocks to the baseline scenario.

4.1 Near-term forecast models

The near-term forecast (NTF) for Armenia is based on a reduced-form time-series model, namely the Bayesian Vector Autoregression (BVAR).

The main specific features of the Armenian NTF model are the following:

Quarterly frequency. Except interest rates, all BVAR variables are annualized quarterly growth rates (log-approximated) of foreign and domestic macroeconomic time series. Variables that are only available in quarterly frequency appear in BVAR only once - as averages for each quarter. Variables that are available in a monthly frequency appear in the BVAR three times: as the averages for the quarter, for the first two and the first month of the quarter respectively.

Range. The range is set from 2001Q1 till 2010Q3, because of the last real domestic economic activity indicator availability is in 2010Q3.

BVAR order. The lag of autoregression was set to two using the Akaike Information Criterion.

Priors. The estimation used the Litterman white noise priors with the weight equal to the number of observations less the BVAR order (the weight is the same for both orders).

Foreign block. The foreign block contains exogenous variables to the Armenian economy: world oil price inflation (Brent), world food inflation (UN FAO food price index), US CPI inflation, and Russian real GDP growth (only quarterly frequency data).

Domestic block. The domestic part includes all main macroeconomic variables and is composed of AMD/USD nominal exchange rate depreciation, two nominal interest rates (interbank repo rate and loan 6M interest rates in local currency), CPI inflation and its food price inflation part, and real GDP growth (only quarterly frequency data).

The list of time series included in BVAR is in table 4.1. The order of the series matches the one of the model.

Name of variable	Description
dl_oil	Brent Oil Price Inflation (QoQ, percent @ar)
dl_food	UN FAO Food Price Index Inflation (QoQ, percent @ar)
dl_y_ru	RU GDP Growth (QoQ, percent @ar)
dl_cpi_us	US CPI Inflation (QoQ, percent @ar)
dl_amd_usd	AMD/USD Depreciation (QoQ, percent @ar)
i_repo	Interbank Repo Rate (percent p.a.)
i_loan	Interest rate on AMD loans for 91–180 days (percent p.a.)
dl_cpi	CPI Inflation (QoQ, percent @ar)
dl_cpi_f	Food Price Inflation (QoQ, percent @ar)
dl_oil_1m	Brent Oil Price Inflation, 1M (QoQ, percent @ar)
dl_oil_2m	Brent Oil Price Inflation, 2M (QoQ, percent @ar)
dl_food_1m	UN FAO Food Price Index Inflation, 1M (QoQ, percent @ar)
dl_food_2m	UN FAO Food Price Index Inflation, 2M (QoQ, percent @ar)
dl_cpi_us_1m	US CPI Inflation, 1M (QoQ, percent @ar)
dl_cpi_us_2m	US CPI Inflation, 2M (QoQ, percent @ar)
dl_amd_usd_1m	AMD/USD Depreciation, 1M (QoQ, percent @ar)
dl_amd_usd_2m	AMD/USD Depreciation, 2M (QoQ, percent @ar)
i_repo_1m	Interbank Repo Rate, 1M (percent p.a.)
i_repo_2m	Interbank Repo Rate, 2M (percent p.a.)
i_loan_1m	Interest rate on AMD loans for 91–180 days, 1M (percent p.a.)
i_loan_2m	Interest rate on AMD loans for 91–180 days, 2M (percent p.a.)
dl_cpi_1m	CPI Inflation, 1M (QoQ, percent @ar)
dl_cpi_2m	CPI Inflation, 2M (QoQ, percent @ar)
dl_cpi_f_1m	Food Price Inflation, 1M (QoQ, percent @ar)
dl_cpi_f_2m	Food Price Inflation, 2M (QoQ, percent @ar)

Table 4.1: List of time series used in final version of NTF model

Alternative specification. The variables included in the BVAR have been chosen by comparing the forecasting performance of several alternative specifications including:

- *Foreign block*: US GDP growth, RUB/USD exchange rate, and RU CPI inflation
- *Interest rates*: refinancing rate, short and long T-bill, and client rates
- *CPI combinations*: various combinations of CPI and its food and ex-food components - food inflation tracks well foreign foods, while ex-food or total CPI is influenced by oil
- *Real activity*: GDP components (private consumption, government expenditures, imports, ...) and production by sectors (industry, construction, and services) in place of GDP growth. However, high volatility of these alternative series cannot be well captured and they track the Russian GDP less well than if one uses aggregate GDP growth.
- *Monetary aggregates*: money base, M1, M2
- *Remittances*: the series is too short for a meaningful estimation and out of sample tests.

All constraints imposed on the BVAR estimation are illustrated in Fig. 4.1. Each point in the graph corresponds to an element of the AR matrix. The order of variables in the vector is the same as in Tab. 4.1. The elements of the AR matrices fixed to zero imply that Armenian variables do not affect any external sector variables. Further, the first-month-of-quarter and second-month-of-quarter variables do not affect any variables directly. Instead, they are linked to them through the covariance matrix of shocks.

The estimates of the AR matrices, covariance matrix, first and second moments of data and VAR can be obtained on request.

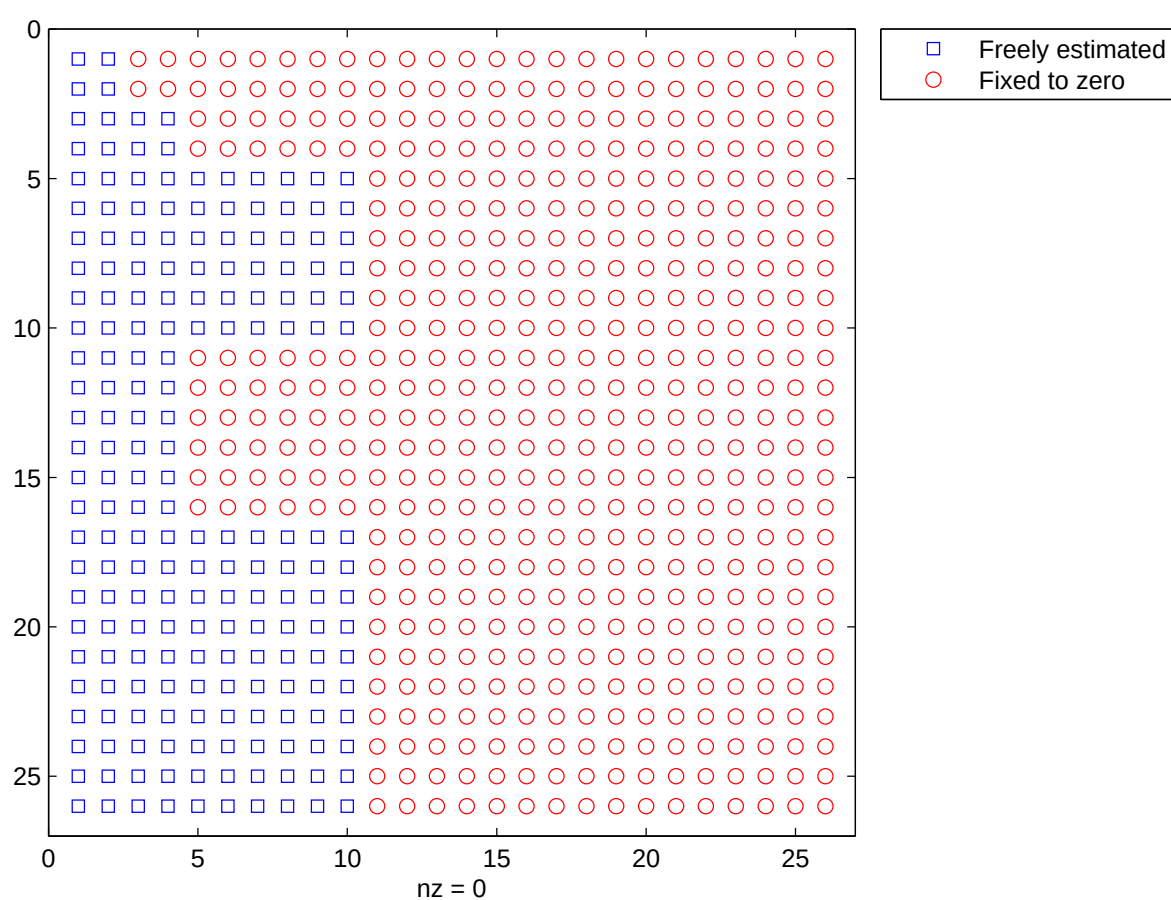


Figure 4.1: Visualization of constraints imposed on AR matrices in estimation

4.2 The Global Projection Model (GPM)

Modeling the external environment is a key element of any forecasting and policy analysis system, particularly for a country like Armenia that has large trade and financial flows with the rest of the world. While external developments are largely exogenous from Armenia's perspective, having a consistent framework for projections of the external environment is central to developing a plausible and defensible domestic forecast. It also plays a vital role in constructing alternative scenarios and assessing risks to the forecast.

GPM is a global projection and simulation model developed at the Research Department of IMF that provides such a framework. The current version includes six regions, specifically the U.S., Japan, the Euro area, emerging Asia, emerging Latin American countries (five inflation targeting countries), and "other".¹ The last group includes medium-sized developed economies and large emerging market countries (e.g. Russia). The model is designed to be compact and based on readily observable data to facilitate its use by a wide range of institutions in need of external forecasts.

The structure of GPM is based on theoretical underpinnings emanated from the new-Keynesian perspective of the monetary transmission mechanism, but interpreted with flexibility to obtain a realistic characterization of observed data. Parameterization is also eclectic, involving a mix of calibration and Bayesian estimation techniques. This approach is adopted, as opposed to (for instance) a pure VAR methodology, to provide the economic structure needed to analyze regional spillovers and create alternative scenarios. In other words, GPM is intended to be neither a "black box" nor a highly stylized general equilibrium model, but rather a flexible model capable of providing realistic forecasts with plausible economic foundations.

Each regional block has four observable endogenous variables: real GDP, CPI inflation, the policy interest rate, and the exchange rate. In addition, the blocks for the G-3 countries (the U.S, Japan, and the euro area) contain the unemployment rate and a measure of financial conditions. The latter variable is included to capture interactions between the real and the financial sectors of the economy.

Paths for GDP, the exchange rate, the real interest rate, and the unemployment rate are modeled as deviations from equilibrium values. The latter, in turn, are specified as stochastic trends, with shocks affecting both their levels and growth rates (for potential

¹Carabenciov, Ioan, Charles Freedman, Roberto Garcia-Saltos, Douglas Laxton, Ondrej Kamenik, and Petar Manchev, forthcoming, "GPM6-The Global Projection Model with Six Regions," IMF Working Paper (Washington: International Monetary Fund).

output, the unemployment rate, and the real exchange rate). Behavioral equations determine deviations from equilibrium values. The main elements of the behavioral equations are as follows:

- The output gap depends on the real interest rate, the real exchange rate, and output gaps in the other regions. In addition, it is affected by financial conditions measured by a bank lending tightening variable.
- Inflation is driven by the output gap, the change in the real exchange rate, and its own lead and lagged values to capture both inertia and forward-looking expectations.
- The policy interest rate is modeled as a Taylor-type equation, containing the deviation in expected inflation from its assumed target level and the output gap.
- The real exchange rate is driven by a modified uncovered interest parity condition, driven by real interest rate differentials and an exogenous risk premium. Exchange rate expectations can reflect inertia via the presence of the lagged exchange rate as well as its future value.
- The unemployment rate is an Okun-type relationship depending on the output gap.

Regional spillovers in GPM are felt directly via the presence of activity gaps in other regions in the domestic output equation, and indirectly through the role of exchange rate movements. Impulse response functions show that a shock to U.S. aggregate demand raises U.S. output by about 0.6 percent; inflation, the policy interest rate, and the real exchange rate all rise as a result. Output in the other regions increases as well, with the effect in the other G-3 regions being about 1/8 that in the U.S., and in the emerging market regions about 1/4. Confidence intervals can be constructed around both the baseline forecast and alternative scenarios produced using GPM, which is particularly valuable in illustrating the range of possible outcomes to policymakers in the form, for instance, of fan charts.

So far, GPM has begun to provide input to country desks in the formulation of their projections, thereby helping to improve the multilateral consistency of the IMF's World Economic Outlook (WEO).

4.3 A Small DGE Model for Armenia²

Forecasting and scenario analysis in the FPAS framework is usually done with quarterly models that fit the historical data well. Such models embody familiar economic linkages, but their structure and estimation is not tightly tied to the underlying theory. Medium-term properties are generally determined by closing gaps between the endogenous variables and exogenous trend or “equilibrium” values.

For analyzing structural shocks, especially in the medium and longer runs, these forecasting models can be complemented by simulation models based more closely on economic theory, commonly referred to as dynamic general equilibrium (DGE) models. Such a model has been constructed for Armenia to provide a consistent framework for considering the short- and long-run effects of shocks relative to a baseline path. As well as enhancing our intuition about the economic linkages at work, this model can be used to cross-check and adjust, where appropriate, forecasts and scenarios produced by more eclectic models.

The DGE model for Armenia consists of two regions: the home country (Armenia) and the rest of the world (ROW). Some distinctive features allow it to capture important aspects of the Armenian economy: imperfect substitutability between foreign and domestic assets; different growth rates between the home and ROW regions; transfers from abroad that affect household income and spending; and a large stock of foreign-currency denominated external debt. More generally, it incorporates full stock-flow dynamics in terms of asset accumulation, both physical and financial, as well as forward-looking expectations that are consistent with the model structure.

One use to which the model can be put is the analysis of the interaction between fiscal and monetary policy. An increase in government spending, for instance, raises output and inflation on impact through the usual channels. It also leads to the widening of an endogenous “risk premium” on Armenian assets given the fiscal deficit and rising government debt - this risk premium moderates the appreciation of the exchange rate that would occur if international assets were perfectly substitutable.

Monetary policymakers have to decide how to respond to the rise in inflation from the fiscal shock given the need to achieve their inflation objective. In the model, the reaction of monetary policy is characterized by a variant of the Taylor Rule, with the policy interest rate responding to both the deviation in inflation from target and in output from potential.

²For a more complete description of the small DGE model and its properties see Appendix C: “A Small DGE Model for Armenia.”

The specification is flexible, however, allowing policy to also respond to exchange rate movements and expectations of future inflation. Varying the parameters on these terms, as well as on the inflation and output gaps, allows an analysis of how the monetary response affects the size and timing of the effects of the fiscal shock. Policymakers can also use sterilized intervention to affect the exchange rate and interest rates given the assumption of imperfect international asset substitutability.

Other shocks of potential interest to policymakers are: an exogenous inflow of foreign capital, for instance lending to the government; an increase in the terms of trade arising from external factors; a domestic productivity growth shock; and an increase in remittances from abroad. In all of these cases, simulation results from the DGE model can be used in conjunction with the forecasting model, both to check for consistency of the scenarios and condition equilibrium paths.

Chapter 5

Forecast performance

5.1 Model historical forecasts

This section presents the performance of the structural FPAS model in terms of historical forecast errors. The historical forecasts of the core structural model were conditional on the historical developments of the external variables and CPI inflation target. The external variables on which we condition the forecasts are: U.S., Eurozone, and Russian CPI inflation, U.S., Eurozone, and Russian GDP gap, Fed funds rate, trend in the real Fed funds rate, USD/EUR and USD/RUB nominal exchange rate, world price of oil and its trend, world price of food and its trend, government expenditures, relative price of exports to US CPI, and country risk premium.

Figures 5.1, 5.2, 5.3, and 5.4 show the conditional historical forecasts for the nominal exchange rate, the policy rate, the YoY CPI inflation rate, and the YoY real GDP growth respectively. The calculation of the forecast error for the output gap, which is an unobserved variable, is based on its smoothed estimate, i.e. estimate conditional on the whole available sample. The sample period starts in 2002Q1.

In the historical forecasts exercise, we focus on the following interesting periods:

Period prior to 2005. Armenia was successful in keeping two digits annual real GDP growth accompanied with average annual CPI inflation between 3.5 and 8.0 % for years 2003–2005. In particular the authorities were successful in fighting high inflation in 2003 and subsequently the policy interest rate was reduced. The exchange rate was kept fixed during the period.

The historical forecasts track well the observed developments during this period. Since late 2003 the model predicts currency appreciation that only occurred after the authorities relaxed its grip on the exchange rate in 2005. The predicted appreciation is consistent with strong inflows of capital and that occurred in that period.

Period 2005–2007. When the currency was made more flexible in 2006, the exchange rate immediately strengthened and continued to appreciate steadily during the entire period. The policy interest rate was mildly rising. In 2006 Armenia announced a switch to an inflation targeting regime.

The overall forecast performance is fine, especially for AMD/USD exchange rate and nominal interest rate. In particular, the FPAS predicts well the total magnitude of nominal appreciation recorded in this period, see Fig. 5.1.

Crisis and post crisis. Remittances – an important driver of business cycle in Armenia – were growing rapidly until 2008, declined in 2009 and recovered again in 2010. The changing pattern of remittances was reflected in sharply declining GDP growth and nominal exchange rate depreciation of 2008 and 2009. The exchange rate weakened by almost 10 percentage points and GDP fell by 20

The model captures all these movements very well. The predicted extent of exchange rate depreciation almost exactly matches the observed movements. GDP growth forecasts are slightly more optimistic showing that the global crisis hit the Armenian economy more than the model would have expected based on the observed external developments. Expecting that the crisis will have more lasting effects on inflation, the model forecasts fail to predict the fast rebound in inflation that occurred in 2009.

Table 5.1 shows the comparison of the forecast root mean square errors for different horizons with the root mean square errors of the random walk (RW). The sample used in this analysis starts in 2002Q1.

Based on the table, we can make the following conclusions:

- The model's capacity to predict short-term is limited. This underscores the need for using judgment in forecasts.
- The model is very good in tracking the policy rates. At the two year horizon the model readily outperforms the random walk.
- The prediction ability for the nominal exchange rate is the same as the random walk benchmark in the short-run, but is improving with longer horizons. The model is also able to properly capture the long-run depreciation tendencies.
- The predictive capacity for inflation and GDP growth are also very good and much better than the random walk benchmark.

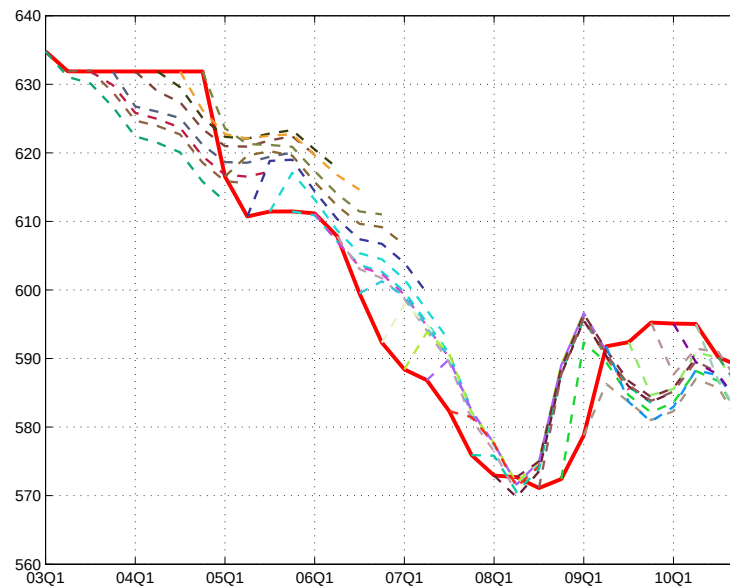


Figure 5.1: conditional historical simulations for the AMD/USD exchange rate ($100 \cdot \log$)

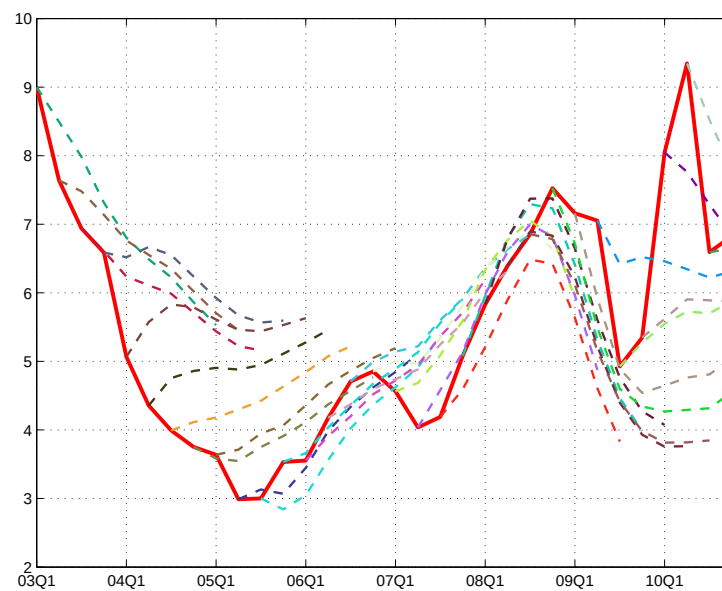


Figure 5.2: conditional historical simulations for the Central Bank's policy rate (% p.a., log approx.)

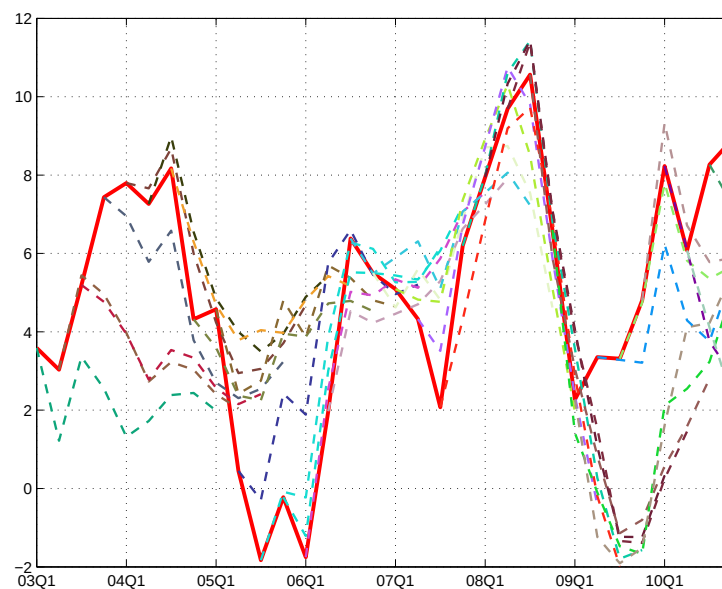


Figure 5.3: conditional historical simulations for the CPI inflation (YoY, %)

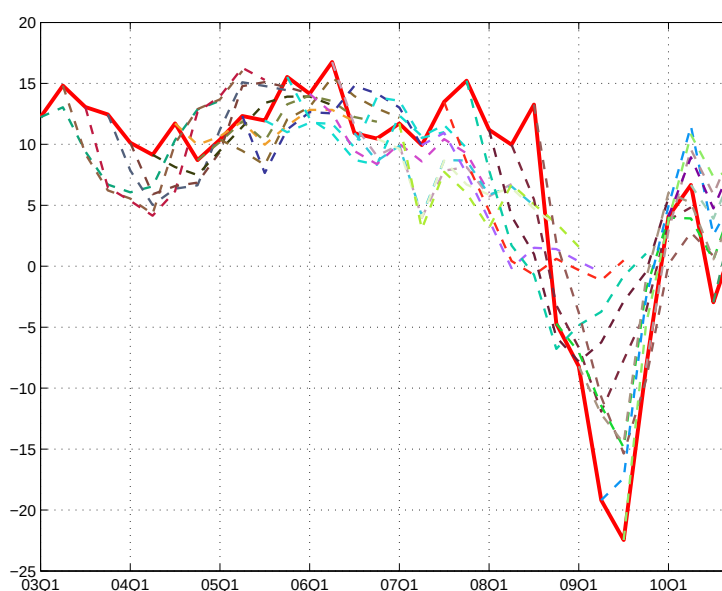


Figure 5.4: conditional historical simulations for the real GDP growth (YoY, %)

	1q	2q	3q	4q	5q	6q	7q	8q
Real GDP gap (percent)	1.015	0.861	0.852	0.810	0.745	0.689	0.621	0.570
Gov. demand gap (percent)	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Priv. demand gap (percent)	0.864	0.774	0.719	0.666	0.612	0.566	0.521	0.496
Real imports gap (percent)	1.127	1.113	1.072	0.992	0.911	0.862	0.804	0.746
Real exports gap (percent)	0.766	0.702	0.679	0.561	0.600	0.635	0.629	0.708
Real Ef Ex Rage gap (percent)	1.397	0.963	0.782	0.710	0.706	0.761	0.780	0.771
ROW output gap (percent)	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Real policy rate gap (percent)	1.040	0.753	0.699	0.706	0.748	0.875	0.921	0.791
Gap in remittances (percent)	1.001	0.883	0.769	0.684	0.627	0.535	0.489	0.449
Real GDP (percent QoQ @ar)	0.701	0.520	0.536	0.622	0.637	0.673	0.631	0.544
Gov. demand (percent QoQ @ar)	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Priv. demand (percent QoQ @ar)	0.740	0.507	0.497	0.528	0.660	0.652	0.563	0.529
Real imports (percent QoQ @ar)	0.845	0.611	0.545	0.542	0.638	0.793	0.782	0.586
Real exports (percent QoQ @ar)	0.450	0.668	0.752	0.631	0.772	0.807	0.701	0.915
Interbank Rate (percent p.a.)	0.802	0.710	0.645	0.671	0.682	0.692	0.701	0.692
1Y client deposit rates (percent p.a.)	1.015	1.230	1.220	1.157	1.095	1.067	1.053	1.028
1Y client lending rates (percent p.a.)	0.954	0.812	0.704	0.665	0.736	0.772	0.819	0.861
Nominal Ex. Rate AMD/USD (100*log)	1.386	0.940	0.739	0.627	0.556	0.512	0.486	0.461
Headline CPI (percent QoQ @ar)	0.762	0.855	0.736	0.681	0.717	0.700	0.714	0.705
Ex-food CPI (percent QoQ @ar)	0.724	0.680	0.730	0.659	0.750	0.971	0.849	0.819
Food CPI (percent QoQ @ar)	0.730	0.846	0.697	0.618	0.595	0.557	0.605	0.599
Imported food inflation (percent QoQ @ar)	0.736	0.410	0.389	0.412	0.452	0.509	0.542	0.488
Imported foreign inflation (percent QoQ @ar)	0.029	0.846	0.652	0.708	0.818	0.944	1.000	0.940
Net inflow or remittances (100*log, USD)	0.867	0.716	0.612	0.540	0.485	0.394	0.354	0.320
Prem	0.620	0.572	0.529	0.491	0.459	0.432	0.412	0.397
Headline CPI (percent YoY)	0.660	0.755	0.783	0.769	0.722	0.730	0.751	0.780
Real GDP (percent YoY)	0.719	0.604	0.528	0.470	0.409	0.458	0.536	0.581

Table 5.1: Comparison of conditional model historical forecasts (CMHF) with random walk forecasts (RW) in terms of RMSE ratios $\left(\frac{RMSE^{CMHF}}{RMSE^{RW}}\right)$

5.2 Near term forecast performance

The NTF forecasting performance is assessed by out-of-sample forecasts and their comparison with the forecasts of a benchmark model (random walk).

The out-of-sample historical forecasts show how the near term forecasts would have looked like with only the actual outturns of the external variables, but without any knowledge about future development of domestic variables. The reason for conditioning on external sector is that the NTF is not intended to produce forecast of variables included in the foreign block.

The historical forecasts for the main Armenian macroeconomic variables (AMD/USD depreciation, repo and loan interest rates, CPI and domestic food price inflation, and GDP growth) are shown in Fig. 5.5. Here the black line depicts the data and the colored lines represent the individual forecasts. Unstable BVAR models were discarded, therefore not all forecasts are shown. The reason for the instability for some early periods rests in the insufficient length of time series, which imposes constraints on the estimation of BVARs.

The results of BVAR's historical out-of-sample forecasts indicate good forecasting performance for all main macroeconomic variables. This gives us comfort in using the NTF model as a tool for near-term forecasting in the FPAS.

The numerical comparison of BVAR out-of-sample forecasts with the random walk benchmark is reported in Tab. 5.2. A value lower than unity means that BVAR forecasts performed better than those of the benchmark random walk model. The BVAR performs better for all variables on the horizon of up to two quarters. Except for the interbank repo and loan interest rates, the RMSE ratios for quarterly data are in favor of BVAR on longer horizons too.

From Tab. 5.2 we can see a good performance of the BVAR in case of the AMD/USD depreciation. The RMSE ratios are 0.75 and 0.72 for the first two quarters and then decrease to values about 0.39.

The out-of-sample forecasts of the interbank repo and loan interest rates are the most difficult variables for the BVAR to predict (reflected in a high RMSE ratios). However, the forecasts still match the overall dynamics of the variables – especially for the loan interest rate, see Fig. 5.5.

The BVAR forecasts of the CPI and domestic food inflations are very good. Based on the actual information about the prices of food and oil and about US CPI inflation the BVAR captures well both the dynamics as well as the magnitudes of Armenian CPI inflation.

	1q	2q	3q	4q	5q
dl_oil	0.000	0.000	0.000	0.000	0.000
dl_food	0.000	0.000	0.000	0.000	0.000
dl_y_ru	0.000	0.000	0.000	0.000	0.000
dl_cpi_us	0.000	0.000	0.000	0.000	0.000
dl_amd_usd	0.696	0.661	0.624	0.501	0.371
i_repo	0.847	0.815	0.998	1.439	1.095
i_loan	0.993	1.043	1.148	0.957	0.822
dl_cpi_f	0.727	0.682	0.771	0.721	0.705
dl_cpi	0.669	0.696	0.810	0.535	0.734
dl_y	0.835	0.776	0.620	0.554	0.575

Table 5.2: Comparison of out-of-sample forecasts with random walk (RW) benchmark in terms of RMSE ratios $\left(\frac{RMSE_{BVAR}}{RMSE_{RW}}\right)$

For instance, the BVAR anticipates the fall of the food inflation at the end of 2008 and also its subsequent rebound.

The forecasting performance in case of real GDP growth is significantly better than the performance of the random walk benchmark. The RMSE ratio is below 0.8 and it is decreasing at longer forecast horizons.

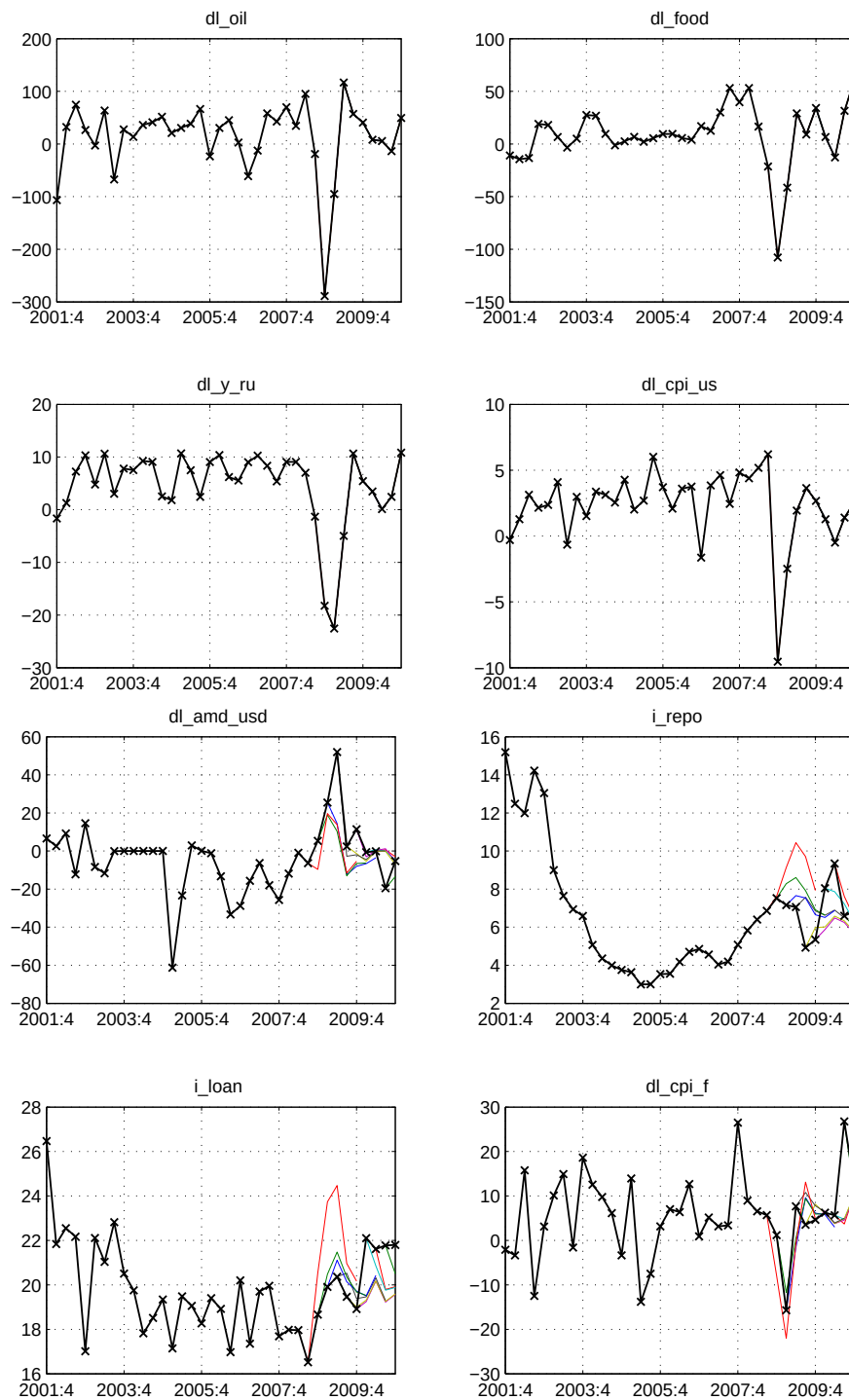


Figure 5.5: Out-of-sample near-term historical forecasts of Armenian macroeconomic variables

Bibliography

- [1] Olivier Blanchard, Francesco Giavazzi, and Filipa Sa. International investors, the U.S. current account, and the dollar. *Brookings Papers on Economic Activity*, 36(1):1–66, 2005.
- [2] Olivier J Blanchard. Debt, deficits, and finite horizons. *Journal of Political Economy*, 93(2):223–47, April 1985.
- [3] Philippe D Karam, Douglas Laxton, and Andrew Berg. A practical model-based approach to monetary policy analysis—overview. IMF Working Papers 06/80, International Monetary Fund, April 2006.
- [4] Philippe D Karam, Douglas Laxton, and Andrew Berg. Practical model-based monetary policy analysis—a how-to guide. IMF Working Papers 06/81, International Monetary Fund, April 2006.
- [5] Guy Meredith. Debt dynamics and global imbalances: Some conventional views reconsidered. IMF Working Papers 07/4, International Monetary Fund, January 2007.
- [6] Michael Mussa. *Exchange Rate Adjustments Needed to Reduce Global Payments Imbalances*. Institute for International Economics, Washington D.C., 2004.
- [7] Ara Stepanyan, Ashot Mkrtchyan, and Era Dabla-Norris. A New Keynesian model of the Armenian economy. IMF Working Papers 09/66, International Monetary Fund, March 2009.
- [8] Cedric Tille. Financial integration and the wealth effect of exchange rate fluctuations. Technical report, 2005.
- [9] Menahem E Yaari. Uncertain lifetime, life insurance, and the ‘theory of the consumer’. *Review of Economic Studies*, 32(2):137–50, April 1965.

Appendices

Appendix A

Structural model

List of structural model variables and parameters

Table A.1: Transition Variables (linear)

Variable	Model name	Description
y	l_y	Real GDP (100*log, bn AMD, 2005 prices)
g	l_g	Government Consumption (100*log, bn AMD, 2005 prices)
d	l_d	Private Expenditures (100*log, bn AMD, 2005 prices)
m	l_m	Real Imports (100*log, bn AMD, 2005 prices)
x	l_x	Real Exports (100*log, bn AMD, 2005 prices)
$\bar{y}$	l_{y_tnd}	Real GDP Trend (bn AMD, 2005 prices)
$\bar{g}$	l_{g_tnd}	Government Expenditures Trend
$\bar{d}$	l_{d_tnd}	Domestic Demand Trend
$\bar{m}$	l_{m_tnd}	Imports Trend
$\bar{x}$	l_{x_tnd}	Exports Trend
$\hat{y}$	l_{y_gap}	Output Gap (in percent)
$\hat{g}$	l_{g_gap}	Government Expenditures Gap (in percent)
$\hat{d}$	l_{d_gap}	Private Expenditures Gap (in percent)
$\hat{m}$	l_{m_gap}	Import Gap (in percent)
$\hat{x}$	l_{x_gap}	Export Gap (in percent)
i	i	Nominal Policy Interest Rate (percent p.a.)

continued on next page

Table A.1: Transition Variables (linear) continued

Variable	Model name	Description
r	r	Real Interest Rates (percent p.a.)
$\hat{r}$	r_gap	Real Interest Rates Gap (p.p. p.a.)
$\bar{r}$	r_tnd	Real Interest Rate Trend (percent p.a.)
i_d	i_d	Nominal 1Y deposits interest rate (percent, @ar)
i_l	i_l	Nominal 1Y credits interest rate (percent, @ar)
r_d	r_d	1Y deposits interest rate (percent @ar)
r_l	r_l	1Y credits interest rate (percent @ar)
r_f	r_f	Real FCY interest rate adjusted by FX growth (percent, @ar)
$\hat{r}_d$	r_d_gap	1Y deposits interest rate gap (percent @ar)
$\hat{r}_l$	r_l_gap	1Y credits interest rate gap (percent @ar)
$\hat{r}_f$	r_f_gap	Real FCY interest rate gap adjusted by FX growth (percent @ar)
$\bar{r}_d$	r_d_tnd	1Y deposits interest rate trend (percent @ar)
$\bar{r}_l$	r_l_tnd	1Y credits interest rate trend (percent @ar)
$\bar{r}_f$	r_f_tnd	Real FCY interest rate trend adjusted by FX growth (percent @ar)
$s^{AMD/USD}$	l_amd_usd	Nominal Exchange Rate AMD per USD (100*log)
$e_s^{AMD/USD}$	$e_l_amd_usd$	Expected nominal exchange rate 1Q ahead (AMD per USD, 100*log)
z	l_z	Real Effective Exchange Rate (100*log, depreciation up)
$\hat{z}$	l_z_gap	Real Effective Exchange Rate Gap (percent, depreciation up)
$\bar{z}$	l_z_tnd	Real Effective Exchange Rate Trend (100*log, depreciation up)
p	l_cpi	Consumer Price Index (100*log, prices in AMD)
p^{xf}	l_cpi_xf	CPI Excluding Food (100*log, prices in AMD)
p^f	l_cpi_f	Food Consumer Price Index (100*log, prices in AMD)
rmc^f	rmc_f	Real Marginal Costs: Food Prices (p.p.)

continued on next page

Table A.1: Transition Variables (linear) continued

Variable	Model name	Description
rmc^{xf}	rmc_xf	Real Marginal Costs: Ex-food Prices (p.p.)
p^{*food}	$l_food_im_f$	Imported Food Prices Adjusted by Balassa-Samuelson Effect (100*log, prices in USD)
p^{*oil}	$l_oil_im_xf$	Imported Oil Prices Adjusted by Balassa-Samuelson Effect (100*log, prices in USD)
e_{π^f}	$e_dl_cpi_f$	Expected Food CPI Inflation (percent, QoQ @ar)
$e_{\pi^{xf}}$	$e_dl_cpi_xf$	Expected Nonfood Inflation (percent, QoQ @ar)
e_{π}	e_dl_cpi	Expected Headline CPI Inflation (percent, QoQ @ar)
p^{*im}	$l_cpi_foreign_im$	Imported Prices Adjusted by B-S Effect (100*log)
$\pi^{*im,f}$	$dl_cpi_foreign_im_f$	Imported Prices Adjusted by B-S Effect in Food Sub-index (percent, QoQ @ar)
$\pi^{*im,xf}$	$dl_cpi_foreign_im_xf$	Imported Prices Adjusted by B-S Effect in Ex-food Sub-index (percent, QoQ @ar)
rp	l_rp	Relative Price of Food vs Ex-Food (100*log)
$\hat{rp}$	l_rp_gap	Relative Price Gap of Food vs Ex-Food (100*log)
$\bar{rp}$	l_rp_tnd	Relative Price Trend of Food vs Ex-Food (100*log)
$\Delta\bar{rp}$	dl_rp_tnd	Relative Price Trend Growth Rate (percent, QoQ)
$i^{*Fed-US}$	i_fed_us	World Nominal Interest Rate (Fed Funds rate, percent, p.a.)
$\bar{r}^{*Fed-US}$	$r_fed_us_tnd$	World Real Interest Rate Trend (percent p.a.)
p^*	$l_cpi_foreign$	Effective Foreign CPI (100*log, prices in USD)
$\pi^{*,ss}$	$ss_dl_cpi_foreign$	Steady-state Effective Foreign CPI (percent QoQ @ar)
p^{*US}	l_cpi_us	USA Headline CPI (100*log, USD)
p^{*EZ}	l_cpi_ez	Eurozone Headline CPI (100*log, EUR)
p^{*RU}	l_cpi_ru	Russia Headline CPI (100*log, RUB)
p^{oil}	l_oil	World Oil Prices (100*log, USD)

continued on next page

Table A.1: Transition Variables (linear) continued

Variable	Model name	Description
p^{food}	l_{food}	World Food Prices (2002-2004=100, FAO (UN),100*log)
rem	l_{remit}	Net inflow of remittances (100*log)
$\hat{y}^*$	$l_{y_foreign_gap}$	Foreign Output Gap (in percent)
$\hat{y}^{*US}$	$l_{y_us_gap}$	US Output Gap (percent)
$\hat{y}^{*EZ}$	$l_{y_ez_gap}$	Eurozone OutputGap (percent)
$\hat{y}^{*RU}$	$l_{y_ru_gap}$	Russia Output Gap (percent)
$s^{USD/EUR}$	l_{usd_eur}	Nominal USD per EUR (100*log)
$s^{USD/RUB}$	l_{usd_rub}	Nominal USD per RUB (100*log)
at	l_{tot}	Terms of trade (100*log)
qx	l_{px2cpi_us}	Relative price of exports to US CPI (100*log)
at_{ss}	ss_dl_tot	Steady state growth rate of terms of trade (percent, QoQ @ar)
$\bar{p}^x$	l_{px}	Export price level (100*log)
$\bar{i}$	i_tnd	Nominal Central Bank Interest Rate Trend (percent p.a.)
$\hat{\pi}^f$	$dev_dl_cpi_f$	Deviation of Food Inflation from Target (percent)
$\hat{\pi}^{xf}$	$dev_dl_cpi_xf$	Deviation of Ex-food Inflation from Target (percent)
$\hat{s}$	dev_dl_s	Deviation of Exchange Rate Depreciation from Target (per cent)
$prem$	$prem$	Country Risk Premium (percent, p.a.)
π^{tar}	dl_cpi_tar	CPI Inflation Target (percent, QoQ)
$\pi^{4,tar}$	$d4l_cpi_tar$	YoY CPI Inflation Target (percent, YoY)
$\pi^{tar,xf}$	$dl_cpi_xf_tar$	Ex-food Inflation Target (percent, QoQ)
$\pi^{tar,f}$	$dl_cpi_f_tar$	Food Inflation Target (percent, QoQ)
$\Delta s^{AMD/USD,tar}$	dl_s_tar	Exchange Rate Depreciation Target (QoQ percent @ar)
int	l_{int_proxy}	Proxy for FX Interventions in a Risk Premium Equivalent (percent, p.a.)
rp^{oil}	l_{qoil}	Real Price of Oil - Oil Price Divided by Foreign CPI in USD (100*log)
$\hat{r}p^{oil}$	l_{qoil_gap}	Real Price of Oil Gap (percent)
$\bar{r}p^{oil}$	l_{qoil_tnd}	Real Price of Oil Trend (100*log)

continued on next page

Table A.1: Transition Variables (linear) continued

Variable	Model name	Description
$\Delta \bar{r}^{oil}$	dl_qoil_tnd	Growth Rate of Real Price of Oil Trend (percent, QoQ @ar)
$r\bar{p}^{food}$	l_qfood	Real Price of Food - Food Price Index Divided by Foreign CPI in USD (100*log)
$\hat{r}p^{food}$	l_qfood_gap	Real Price of Food Gap (percent)
$\bar{r}p^{food}$	l_qfood_tnd	Real Price of Food Trend (100*log)
$\Delta \bar{r}p^{food}$	dl_qfood_tnd	Growth Rate of Real Price of Food Trend (percent, QoQ @ar)
$r\bar{w}p^{food}$	l_rfood	Real Price of World - Domestic Food Price in Domestic Currency (100*log)
$\hat{r}w\bar{p}^{food}$	l_rfood_gap	Real Price of World - Domestic Food Price in Domestic Currency Gap (percent)
$\bar{r}w\bar{p}^{food}$	l_rfood_tnd	Real Price of World - Domestic Food Price in Domestic Currency Trend (100*log)
$qrem$	l_qremit	Net inflow of remittances divided by foreign CPI in USD (100*log)
$\hat{q}r\bar{e}m$	l_qremit_gap	Real remittances gap (percent)
$\bar{q}r\bar{e}m$	l_qremit_tnd	Real remittances trend (100*log)
$\Delta \bar{q}r\bar{e}m$	dl_qremit_tnd	Growth of real remittances trend (percent QoQ)
$\Delta s^{AMD/USD}$	dl_amd_usd	Nominal AMD per USD Depreciation (QoQ percent @ar)
Δy	dl_y	Real GDP Growth (percent, QoQ @ar)
Δg	dl_g	Government Consumption Growth Rate (percent, QoQ)
Δx	dl_x	Real Exports Growth Rate (percent, QoQ)
Δm	dl_m	Real Imports Growth Rate (QoQ)
Δd	dl_d	Real Domestic Demand Growth Rate (QoQ)
π	dl_cpi	Headline CPI Inflation (percent, QoQ @ar)
π^f	dl_cpi_f	Food CPI Inflation (percent, QoQ @ar)
π^*	$dl_cpi_foreign$	Effective Foreign Inflation (percent, QoQ @ar)
π^{*US}	dl_cpi_us	USA Headline Inflation (percent, QoQ @ar)
π^{*EZ}	dl_cpi_ez	Eurozone Headline Inflation (percent, QoQ @ar)

continued on next page

Table A.1: Transition Variables (linear) continued

Variable	Model name	Description
π^{*RU}	dl_cpi_ru	Russia Headline Inflation (percent, QoQ @ar)
π^{*oil}	dl_oil	Inflation of Brent Crude Oil Prices (percent, QoQ @ar)
$\pi^{*food,im}$	$dl_food_im_f$	Inflation of Food Imports adjusted by Balassa-Samuelson Effect (percent, QoQ @ar)
$\pi^{*oil,im}$	$dl_oil_im_xf$	Inflation of Oil Imports (percent, QoQ @ar)
π^{*food}	dl_food	Inflation of World Food Prices (percent, QoQ @ar)
π^{xf}	dl_cpi_xf	Nonfood Inflation (percent, QoQ @ar)
Δz	dl_z	Real Effective Exchange Rate Growth (per cent, QoQ)
$\Delta i^{*Fed-US}$	di_fed_us	World Nominal Interest Rate Growth (percent, QoQ)
$\Delta s^{USD/EUR}$	dl_usd_eur	Depreciation of USD per EUR Exchange Rate (percent, QoQ @ar)
$\Delta s^{USD/RUB}$	dl_usd_rub	Depreciation of USD per RUB Exchange Rate (percent, QoQ @ar)
$\Delta \bar{d}$	dl_d_tnd	Domestic Demand Trend Growth (per cent, QoQ)
$\Delta \bar{g}$	dl_g_tnd	Government Consumption Trend Growth (per cent, QoQ)
$\Delta \bar{x}$	dl_x_tnd	Exports Trend Growth (per cent, QoQ)
$\Delta \bar{m}$	dl_m_tnd	Imports Trend Growth (per cent, QoQ)
$\Delta \bar{z}$	dl_z_tnd	Growth Rate of Real Effective Exchange Rate Trend (per cent, QoQ @ar)
$\Delta \bar{y}$	dl_y_tnd	Potential Growth of Real GDP (per cent, QoQ @ar)
$\Delta qrem$	dl_qrem	Growth of real remittances (percent QoQ)
Δat	dl_tot	Terms of trade growth (percent, QoQ)
Δqx	dl_px2cpi_us	Growth rate of Relative price of exports to US CPI (percent, QoQ, @ar)
π^{*im}	$dl_cpi_foreign_im$	Imported Prices Adjusted by B-S Effect (percent, QoQ @ar)
π^4	$d4l_cpi$	Consumer Price Index Inflation (per cent, YoY)

continued on next page

Table A.1: Transition Variables (linear) continued

Variable	Model name	Description
$\pi^{4,f}$	$d4l_cpi_f$	Food Consumer Price Index Inflation (per cent, YoY)
π^{*4}	$d4l_cpi_foreign$	Effective Foreign Inflation (percent, YoY @ar)
$\Delta^4 y$	$d4l_y$	Real GDP Growth (percent, YoY @ar)
$\Delta^4 d$	$d4l_d$	Real Domestic Demand Growth Rate (per cent, YoY)
$\Delta^4 g$	$d4l_g$	Government Consumption Growth Rate (per cent, YoY)
$\Delta^4 x$	$d4l_x$	Real Exports Growth Rate (percent, YoY)
$\Delta^4 m$	$d4l_m$	Real Imports Growth Rate (per cent, YoY)
$\Delta^4 s^{AMD/USD}$	$d4l_amd_usd$	Nominal AMD per USD Depreciation (YoY percent @ar)
$\pi^{4,xf}$	$d4l_cpi_xf$	Nonfood Inflation (percent, YoY)
$\Delta^4 z$	$d4l_z$	Real Effective Exchange Rate Growth (per cent, YoY)
$\pi^{*4,oil}$	$d4l_oil$	Inflation of Brent Crude Oil Prices (percent, YoY @ar)
$\pi^{*4,food}$	$d4l_food$	Inflation of World Food Prices (percent, YoY @ar)
π^{*4EZ}	$d4l_cpi_ez$	Eurozone Headline Inflation (per cent, YoY @ar)
π^{*4RU}	$d4l_cpi_ru$	Russia Headline Inflation (per cent, YoY @ar)
$\Delta^4 \bar{d}$	$d4l_d_tnd$	Domestic Demand Trend Growth (per cent, YoY)
$\Delta^4 \bar{g}$	$d4l_g_tnd$	Government Consumption Trend Growth (per cent, YoY)
$\Delta^4 \bar{x}$	$d4l_x_tnd$	Exports Trend Growth (per cent, YoY)
$\Delta^4 \bar{m}$	$d4l_m_tnd$	Imports Trend Growth (per cent, YoY)
$\Delta^4 \bar{z}$	$d4l_z_tnd$	Growth Rate of Real Effective Exchange Rate Trend (Per cent, YoY @ar)
$\Delta^4 \bar{y}$	$d4l_y_tnd$	Potential Growth of Real GDP (percent, YoY @ar)
Δ_{4at}	$d4l_tot$	Terms of trade growth (percent, YoY)
$\hat{d}^e$	El_d_gap	Expected Domestic Demand gap (in percent)

continued on next page

Table A.1: Transition Variables (linear) continued

Variable	Model name	Description
${}^e\hat{g}$	<i>El_g_gap</i>	Expected government Expenditures Gap (in percent)
${}^e\hat{x}$	<i>El_x_gap</i>	Expected Export Gap (in percent)
${}^e\hat{m}$	<i>El_m_gap</i>	Expected Import Gap (in percent)
${}^e\pi$	<i>Edl_cpi</i>	Expected Headline CPI Inflation (percent, QoQ @ar)
${}^e\pi^*$	<i>Edl_cpi_foreign</i>	Expected Effective Foreign Inflation (percent, QoQ @ar)
${}^e\pi^{tar}$	<i>Edl_cpi_tar</i>	Expected Implied Inflation Target (percent, QoQ)
${}^e\Delta\bar{z}$	<i>Edl_z_tnd</i>	Expectation of Growth Rate of REER Trend 1q ahead (per cent, QoQ @ar)
$term^d$	<i>c_term_d</i>	Term premium 1Y deposits vs. policy rate (percent p.a.)
$term^l$	<i>c_term_l</i>	Margin of banking system: 1Y deposits vs. 1Y credits (percent p.a.)

Table A.2: Transition Variables (log-linear)

Variable	Model name	Description
$S^{AMD/USD}$	<i>amd_usd</i>	Nominal Exchange Rate (AMD per USD) (AMD per USD)

Table A.3: Non-model variables

Variable	Name	Description
Δy^{ind}	<i>dl_y_ind</i>	Real GDP Growth for Industrial Production (percent, QoQ @ar)
Δy^{agr}	<i>dl_y_agr</i>	Real GDP Growth for Agriculture Sector (percent, QoQ @ar)
Δy^{ind}	<i>dl_y_constr</i>	Real GDP Growth for Construction Sector (percent, QoQ @ar)

continued on next page

Table A.3: Non-model variables continued

Variable	Name	Description
dep^{fx}	dep_fx	Percentage of Deposits in FX (percent)
$loans^{fx}$	$loans_fx$	Percentage of Loans in FX (percent)
$remit$	$remit$	Net Inflow of Remittances (USD)
$remit_out$	$remit_out$	Outflow of Remittances (USD)
$remit_in$	$remit_in$	Inflow of Remittances (million USD)
π^{serv}	dl_cpi_serv	CPI Inflation for Services (percent, QoQ @ar)
z	z	Real Effective Exchange Rate (index, 1997=1)
ca	ca	Current Account (million USD)

Table A.4: Transition Shocks

Variable	Model name	Description
$\varepsilon^{\hat{d}}$	$shock_l_d_gap$	Domestic Demand Shock
$\varepsilon^{\hat{g}}$	$shock_l_g_gap$	Government Demand Shock
$\varepsilon^{\hat{x}}$	$shock_l_x_gap$	Export Demand Shock
$\varepsilon^{\hat{m}}$	$shock_l_m_gap$	Import Demand Shock
$\varepsilon^{\hat{y}}$	$shock_l_y_gap$	Output Gap Shock
$\varepsilon^{\hat{\pi}^f}$	$shock_dl_cpi_f$	Food Inflation Markup Shock
$\varepsilon^{\hat{\pi}^{xf}}$	$shock_dl_cpi_xf$	NonFood Inflation Markup Shock
ε^{π}	$shock_dl_cpi$	Domestic Inflation Shock
$\varepsilon^{\pi^{tar}}$	$shock_dl_cpi_tar$	Implied Inflation Target Shock, Inflation Target Change
ε^i	$shock_i$	Monetary Shock
ε^{i_d}	$shock_i_d$	1Y deposits interest rate shock
ε^{i_l}	$shock_i_l$	1Y credits interest rate shock
$\varepsilon^{s^{AMD/USD}}$	$shock_dl_amd_usd$	Exchange Rate Shock
$\varepsilon^{s_{tar}}$	$shock_dl_s_tar$	Shock in target of nominal exchange rate growth
$\varepsilon^{\Delta \bar{y}}$	$shock_dl_y_tno$	Steady State Output Growth Shock
$\varepsilon^{\Delta \bar{g}}$	$shock_dl_g_tno$	Government Demand Trend Shock
$\varepsilon^{\Delta \bar{x}}$	$shock_dl_x_tno$	Export Trend Shock
$\varepsilon^{\Delta \bar{m}}$	$shock_dl_m_tno$	Import Trend Shock
$\varepsilon^{\Delta \bar{z}}$	$shock_dl_z_tno$	Real Exchange Rate Shock (depreciation up)

continued on next page

Table A.4: Transition Shocks continued

Variable	Model name	Description
$\varepsilon^{\bar{r}}$	<i>shock_r_tnd</i>	Real interbank inter
$\varepsilon^{\Delta \bar{r}p}$	<i>shock_dl_rp_tnd</i>	Import Prices to Overall CPI Shock (eq.)
$\varepsilon^{\Delta \bar{r}p^{oil}}$	<i>shock_dl_qoil_tnd</i>	Real Price of Oil Shock (trend)
$\varepsilon^{\hat{r}p^{oil}}$	<i>shock_l_qoil_gap</i>	Real Price of Oil Shock (gap)
$\varepsilon^{\Delta \bar{r}p^{food}}$	<i>shock_dl_qfood_tnd</i>	Real Price of Food Shock (trend)
$\varepsilon^{\hat{r}p^{food}}$	<i>shock_l_qfood_gap</i>	Real Price of Food Shock (gap)
$\varepsilon^{\Delta \bar{r}em}$	<i>shock_dl_qremit_tnd</i>	Shock to real remittances growth trend
$\varepsilon^{\bar{r}em}$	<i>shock_l_qremit_gap</i>	Shock to real remittances gap
$\varepsilon^{\pi^{US}}$	<i>shock_dl_cpi_us</i>	US Inflation Shock
$\varepsilon^{\pi^{EZ}}$	<i>shock_dl_cpi_ez</i>	EZ Inflation Shock
$\varepsilon^{\pi^{RU}}$	<i>shock_dl_cpi_ru</i>	RU Inflation Shock
$\varepsilon^{i^{*Fed-US}}$	<i>shock_i_fed_us</i>	US Fed Funds Rate Shock
$\varepsilon^{\hat{y}^{US}}$	<i>shock_l_y_us_gap</i>	US Demand Shock
$\varepsilon^{\hat{y}^{EZ}}$	<i>shock_l_y_ez_gap</i>	EZ Demand Shock
$\varepsilon^{\hat{y}^{RU}}$	<i>shock_l_y_ru_gap</i>	RU Demand Shock
$\varepsilon^{\Delta s^{USD/EUR}}$	<i>shock_dl_usd_eur</i>	USD per EUR Exchange Rate Shock
$\varepsilon^{\Delta s^{USD/RUB}}$	<i>shock_dl_usd_rub</i>	USD per RUB Exchange Rate Shock
$\varepsilon^{\bar{r}^{*US}}$	<i>shock_r_fed_us_tnd</i>	US Real Interest Rate Trend Shock
ε^{qx}	<i>shock_dl_px2cpi_us</i>	Relative export price to US cpi shock

Table A.5: Steady-State Parameters

Variable	Value	Description
π_{ss}	4	Steady state of inflation rate
Δz_{ss}	-3	Steady state real exch. rate growth
r_{ss}	2	Steady state of Real Interest Rate
π_{ss}^{*US}	2.5	Steady state of US inflation
π_{ss}^{*EZ}	1.9	Steady state of Eurozone inflation
π_{ss}^{*RU}	7	Steady state of Russia inflation
$\Delta s_{ss}^{USD/EUR}$	0.6	Steady state growth rate of USD/EUR exchange rate
$\Delta s_{ss}^{USD/RUB}$	0.5	Steady state growth rate of USD/RUB exchange rate
$r_{ss}^{*Fed-US}$	1	Steady state US FED real interest rate
Δg_{ss}	5	Steady state growth rate of government expenditures

continued on next page

Table A.5: Steady-State Parameters continued

Variable	Value	Description
Δx_{ss}	3	Steady state growth rate of exports
Δm_{ss}	5	Steady state growth rate of imports
$\Delta \bar{y}_{ss}$	5.44	Steady state real output growth
π^{*qoil}_{ss}	6.33333	Steady state inflation of real oil prices
π^{*qfood}_{ss}	-1.66667	Steady state inflation of real food prices
$\Delta \bar{y}_{ss}$	2.88667	Steady state real remittances growth
$\Delta \bar{q}x_{ss}$	1	Steady state relative price of exports in terms of US CPI
$\Delta \bar{r}p_{ss}$	2.5	Steady State of Relative Price Trend Growth Rate

Table A.6: Transition Parameters

Variable	Value	Description
μ_{US}	0.333333	Weight of US output gap in foreign output gap
μ_{EZ}	0.333333	Weight of Eurozone output gap in foreign output gap
a_{10}	0.9	Parameter at lagged private expenditure gap in IS curve
a_{11}	0.2	Parameter at real interest rate in IS curve
a_{12}	0.3	Parameter in IS curve
a_{13}	-0.1	Parameter at real exchange rate gap in IS equation
a_{14}	0.78	Parameter at remittances in IS equation
a_{15}	0.05	Parameter at terms of trade in IS equation
a_{20}	0.5	AR parameter at lagged govt. demand gap in govt. demand eq.
a_{20}	0.5	AR parameter at lagged govt. demand gap in govt. demand eq.
a_{41}	0.7	Parameter on lagged import gap in imports demand equation
a_{42}	1	Parameter on lagged import gap in imports demand equation
a_{43}	0.2	Parameter on lagged import gap in imports demand equation
b_1	0.9	Parameter on lagged nominal interest rate in monetary policy rule
b_2	1.2	Parameter on food inflation deviation in monetary policy rule

continued on next page

Table A.6: Transition Parameters continued

Variable	Value	Description
b_3	1.2	Parameter on ex-food inflation deviation in monetary policy rule
b_4	0.1	Parameter on exchange rate depreciation deviation in monetary policy rule
b_5	0.1	Parameter on output gap in monetary policy rule
a_{31}	0.5	Parameter on lagged export gap in export demand equation
a_{32}	3	Parameter on exchange rate gap in export demand equation
a_{33}	0.1	Parameter in export demand equation
j_5	0.3	Parameter on expected ex-food inflation in ex-food inflation equation
j_6	0.04	Parameter on oil price inflation in ex-food inflation equation
j_7	0.2	Parameter on imports price inflation in ex-food inflation equation
j_8	0.3	Parameter on real marginal cost in ex-food inflation equation
j_1	0.4	Parameter on expected food inflation in food inflation equation
j_2	0.3	Parameter on imported food price inflation in food inflation equation
j_3	0.15	Parameter on imports price inflation in food inflation equation
j_4	0.4	Parameter on real marginal cost in food inflation equation
ω	0.5391	Weight of food inflation in aggregate domestic inflation
a_1	0.1583	Weight of government expenditures gap in output gap
a_2	0.25	Weight of exports gap in output gap
a_3	0.47	Weight of imports gap in output gap
k_1	0.1	Parameter from real marginal cost for food equation
k_2	0.4	Parameter from real marginal cost for food equation
k_3	0.8	Parameter from real marginal cost for ex-food equation
k_4	0.1	Parameter from real marginal cost for ex-food equation
η	0.7	Parameter on expectation term in UIP equation
h_{31}	0.5	Parameter in deposits rate equation
h_{32}	0.5	Parameter in credits rate equation

continued on next page

Table A.6: Transition Parameters continued

Variable	Value	Description
w_{US}	0.38	AR parameter in US inflation
w_{EZ}	0.7	AR parameter in Eurozone inflation
w_{RU}	0.8	AR parameter in Russia inflation
ρ_{US}	0.86	AR parameter in US output gap
ρ_{EZ}	0.8	AR parameter in Eurozone output gap
ρ_{RU}	0.8	AR parameter in Russia output gap
f_1	0.85	AR parameter for US Fed Funds rate
f_2	0.924	AR parameter for US real rate
f_4	0.8	AR parameter on relative price growth trend
f_5	0.9	AR parameter on relative food price growth trend
f_6	0.9	AR parameter on relative oil price growth trend
f_7	0.75	AR parameter on relative food price gap equation
f_8	0.75	AR parameter on relative oil price gap equation
c_{21}	0.95	AR parameter in real remittances trend growth equation
c_{31}	0.5	Parameter on lagged remittance gap in real remittances gap equation
c_{32}	2	Parameter on output gap in real remittances gap equation
γ	1	Weight of expected nominal exchange rate in expectation equation
α	0.9999	AR parameter for inflation target
h_{11}	0.95	AR parameter for real interest rate trend growth
h_{12}	0.02	Parameter in real interest rate trend growth equation
h_{21}	0.85	AR parameter for real exchange rate trend growth
h_{22}	0.05	Parameter in real exchange rate trend growth equation
h_{61}	0.75	AR parameter for exports trend growth
h_{71}	0.85	AR parameter for imports trend growth
h_{51}	0.85	AR parameter for government expenditure trend growth
h_{41}	0.85	AR parameter for output trend growth
h_{62}	0.05	Parameter in exports trend growth equation
h_{72}	0.05	Parameter in imports trend growth equation
h_{52}	0.05	Parameter in government expenditures trend growth equation
h_{42}	0.05	Parameter in output trend growth equation
h_{43}	0.1	Parameter in output trend growth equation

continued on next page

Table A.6: Transition Parameters continued

Variable	Value	Description
h_{63}	0.1	Paramater in output trend growth equation

List of structural model equations

Demand Side :

$$\hat{d}_t = a_{10} \cdot \hat{d}_{t-1} + (1 - a_{10}) \cdot E_t[\hat{d}_{t+1}] - a_{11} \cdot \hat{r}_{t-1} + a_{12} \cdot (\hat{y}_t + a_{14} \cdot q\hat{r}\hat{e}m_t + a_{13} \cdot \hat{z}_t - \hat{d}_t) + a_{15} \cdot (\Delta at_t - at_{sst}) + \varepsilon_t^{\hat{d}} \quad (\text{A.1a})$$

$$\hat{g}_t = a_{20} \cdot \hat{g}_{t-1} + (1 - a_{20}) \cdot E_t[\hat{g}_{t+1}] + a_{20} \cdot (\hat{y}_t - \hat{g}_t) + \varepsilon_t^{\hat{g}} \quad (\text{A.1b})$$

$$\hat{x}_t = a_{31} \cdot \hat{x}_{t-1} + (1 - a_{31}) \cdot E_t[\hat{x}_{t+1}] + a_{32} \cdot (\hat{y}_t^* - \hat{x}_t + a_{33} \cdot \hat{z}_t) + \varepsilon_t^{\hat{x}} \quad (\text{A.1c})$$

$$\hat{m}_t = a_{41} \cdot \hat{m}_{t-1} + (1 - a_{41}) \cdot E_t[\hat{m}_{t+1}] + a_{42} \cdot (\hat{y}_t - 0 \cdot \hat{m}_t - a_{43} \cdot \hat{z}_t) + \varepsilon_t^{\hat{m}} \quad (\text{A.1d})$$

$$\hat{y}_t = (1 - a_1 - a_2 + a_3) \cdot \hat{d}_t + a_1 \cdot \hat{g}_t + a_2 \cdot \hat{x}_t - a_3 \cdot \hat{m}_t + \varepsilon_t^{\hat{y}} \quad (\text{A.1e})$$

$$\Delta y_t = (1 - a_1 - a_2 + a_3) \cdot \Delta d_t + a_1 \cdot \Delta g_t + a_2 \cdot \Delta x_t - a_3 \cdot \Delta m_t \quad (\text{A.1f})$$

Supply Side :

$$\pi_t = \omega \cdot \pi_t^f + (1 - \omega) \cdot \pi_t^{xf} + \varepsilon_t^\pi \quad (\text{A.2a})$$

$$\pi_t^f = j_1 \cdot \pi_t^f + (1 - j_1 - j_2 - j_3) \cdot \pi_{t-1}^{4,f} + j_2 \cdot \pi_{t-1}^{*food,im} + j_3 \cdot \pi_t^{*im,f} + j_4 \cdot rmc_t^f + \varepsilon_t^{\pi^f} \quad (\text{A.2b})$$

$$rmc_t^f = (1 - k_1 - k_2) \cdot \hat{y}_t + (k_1 + k_2) \cdot (\hat{z}_t - (1 - \omega) \cdot \hat{r}p_t) + k_2 \cdot \hat{r}p_t^{food} \quad (\text{A.2c})$$

$$\pi_t^{xf} = j_5 \cdot \pi_t^{xf} + (1 - j_5 - j_6 - j_7) \cdot \pi_{t-1}^{4,xf} + j_6 \cdot \pi_t^{*oil,im} + j_7 \cdot \pi_t^{*im,xf} + j_8 \cdot rmc_t^{xf} + \varepsilon_t^{\pi^{xf}} \quad (\text{A.2d})$$

$$rmc_t^{xf} = (1 - k_3 - k_4) \cdot \hat{y}_t + (k_3 + k_4) \cdot (\hat{z}_t + \omega \cdot \hat{r}p_t) + k_4 \cdot \hat{r}p_t^{oil} \quad (\text{A.2e})$$

Inflation target :

$$\pi_t^{tar,f} = \pi_t^{tar} + (1 - \omega) \cdot \Delta \bar{r} p_t \quad (\text{A.3a})$$

$$\pi_t^{tar,xf} = \pi_t^{tar} - \omega \cdot \Delta \bar{r} p_t \quad (\text{A.3b})$$

$$\pi_t^{tar} = \alpha \cdot \pi_{t-1}^{tar} + (1 - \alpha) \cdot \pi_{ss} + \varepsilon_t^{\pi^{tar}} \quad (\text{A.3c})$$

$$\pi_t^{4,tar} = (\pi_t^{tar} + \pi_{t-1}^{tar} + \pi_{t-2}^{tar} + \pi_{t-3}^{tar})/4 \quad (\text{A.3d})$$

Monetary Policy Rule :

$$\bar{i}_t = \bar{r}_t + E_t[\pi_{t+1}^{tar}] \quad (\text{A.4a})$$

$$\Delta s_t^{AMD/USD,tar} = 0.3 \cdot \Delta s_{t-1}^{AMD/USD,tar} + (1 - 0.3) \cdot (\Delta z_t - \pi_t^* + \pi_t^{tar}) + \varepsilon_t^{s_t^{tar}} \quad (\text{A.4b})$$

$$\hat{\pi}_t^f = E_t[\pi_{t+1}^{4,f}] - \pi_t^{tar,f} \quad (\text{A.4c})$$

$$\hat{\pi}_t^{xf} = E_t[\pi_{t+1}^{4,xf}] - \pi_t^{tar,xf} \quad (\text{A.4d})$$

$$\hat{s}_t = \Delta s_t^{AMD/USD} - \Delta s_t^{AMD/USD,tar} \quad (\text{A.4e})$$

$$i_t = b_1 \cdot i_{t-1} + (1 - b_1) \cdot (\bar{i}_t + b_2 \cdot \omega \cdot \hat{\pi}_t^f + b_3 \cdot (1 - \omega) \cdot \hat{\pi}_t^{xf} + b_4 \cdot \hat{s}_t + b_5 \cdot \hat{y}_t) \quad (\text{A.4f})$$

Nominal UIP :

$$s_t^{AMD/USD} = \eta \cdot (s_t^{eAMD/USD} - (i_t - i_t^{*Fed-US} - prem_t)/4) + (1 - \eta) \cdot (s_{t-1}^{AMD/USD} + \Delta s_t^{AMD/USD,tar}/4) + \varepsilon_t^{s^{AMD/USD}} \quad (\text{A.5a})$$

$$s_t^{eAMD/USD} = \gamma \cdot E_t[s_{t+1}^{AMD/USD}] + (1 - \gamma) \cdot (s_{t-1}^{AMD/USD} + 2/4 \cdot (\Delta z_t - \pi_t^* + \pi_t^{tar})) \quad (\text{A.5b})$$

$$int_t = s_t^{AMD/USD} - (s_t^{eAMD/USD} - (i_t - i_t^{*Fed-US} - prem_t)/4) \quad (\text{A.5c})$$

Real UIP :

$$\bar{r}_t - \bar{r}_t^{*Fed-US} = \Delta \bar{z}_t + prem_t - \mu_{EZ} \cdot (\pi_{ss}^{*EZ} - \pi_{ss}^{*US} + \Delta s_{ss}^{USD/EUR}) - (1 - \mu_{US} - \mu_{EZ}) \cdot (\pi_{ss}^{*RU} - \pi_{ss}^{*US} + \Delta s_{ss}^{USD/RUB}) \quad (A.6a)$$

$$\bar{r}_t = h_{11} \cdot \bar{r}_{t-1} + (1 - h_{11}) \cdot r_{ss} - h_{12} \cdot (\Delta qrem_t - \Delta \bar{y}_{ss}) + \varepsilon_t^{\bar{r}} \quad (A.6b)$$

$$\Delta \bar{z}_t = h_{21} \cdot \Delta \bar{z}_{t-1} + (1 - h_{21}) \cdot \Delta z_{ss} - h_{22} \cdot (\Delta_4 at_t - at_{sst}) + \varepsilon_t^{\Delta \bar{z}} \quad (A.6c)$$

$$i_{dt} = term_t^d + h_{31} \cdot i_{dt-1} + (1 - h_{31}) \cdot (i_t + E_t[i_{t+1}] + E_t[i_{t+2}] + E_t[i_{t+3}]) / 4 \quad (A.6d)$$

$$i_{lt} = term_t^l + h_{32} \cdot i_{lt-1} + (1 - h_{32}) \cdot i_{dt-1} + \varepsilon_t^{i_l} \quad (A.6e)$$

$$\bar{r}_{dt} = \bar{r}_t + term_t^d / (1 - h_{31}) \quad (A.6f)$$

$$\bar{r}_{lt} = \bar{r}_t + term_t^l / (1 - h_{32}) + term_t^d / (1 - h_{31}) \quad (A.6g)$$

$$\bar{r}_{ft} = \bar{r}_t - prem_t \quad (A.6h)$$

Domestic Definitions and Identities :

$$z_t = s_t^{AMD/USD} + p_t^* - p_t \quad (A.7a)$$

$$r_t = i_t - {}^e\pi_t \quad (A.7b)$$

$$r_{dt} = i_{dt} - {}^e\pi_t \quad (A.7c)$$

$$r_{lt} = i_{lt} - {}^e\pi_t \quad (A.7d)$$

$$r_{ft} = i_t^{*Fed-US} + ({}^e s_t^{AMD/USD} - s_t^{AMD/USD}) \cdot 4 - {}^e\pi_t \quad (A.7e)$$

$$\hat{r}_t = 0.25 \cdot \hat{r}_{dt} + 0.25 \cdot \hat{r}_{lt} + (1 - 0.5 - 0.25) \cdot \hat{r}_{ft} \quad (A.7f)$$

Relative Price Food vs. Ex-food:

$$rp_t = p_t^f - p_t^{xf} \quad (\text{A.8a})$$

$$rp_t = \hat{r}p_t + \bar{r}p_t \quad (\text{A.8b})$$

$$\Delta \bar{r}p_t = (\bar{r}p_t - \bar{r}p_{t-1}) \cdot 4 \quad (\text{A.8c})$$

$$\Delta \bar{r}p_t = f_4 \cdot \Delta \bar{r}p_{t-1} + (1 - f_4) \cdot \Delta \bar{r}p_{ss} + \varepsilon_t^{\Delta \bar{r}p} \quad (\text{A.8d})$$

Foreign Demand:

$$\hat{y}_t^* = \mu_{US} \cdot \hat{y}_t^{*US} + \mu_{EZ} \cdot \hat{y}_t^{*EZ} + (1 - \mu_{US} - \mu_{EZ}) \cdot \hat{y}_t^{*RU} \quad (\text{A.9a})$$

$$\hat{y}_t^{*US} = \rho_{US} \cdot \hat{y}_{t-1}^{*US} + \varepsilon_t^{*US} \quad (\text{A.9b})$$

$$\hat{y}_t^{*EZ} = \rho_{EZ} \cdot \hat{y}_{t-1}^{*EZ} + \varepsilon_t^{*EZ} \quad (\text{A.9c})$$

$$\hat{y}_t^{*RU} = \rho_{RU} \cdot \hat{y}_{t-1}^{*RU} + \varepsilon_t^{*RU} \quad (\text{A.9d})$$

Foreign Prices :

$$\pi_t^{*im} = 0.5 \cdot \pi_{t-1}^{*im} + (1 - 0.5) \cdot (\pi_{t-1}^* + \Delta s_{t-1}^{AMD/USD} - \Delta \bar{z}_t) \quad (\text{A.10a})$$

$$\pi_t^{*im,f} = \pi_t^{*im} + (1 - \omega) \cdot \Delta \bar{r}p_t \quad (\text{A.10b})$$

$$\pi_t^{*im,xf} = \pi_t^{*im} - \omega \cdot \Delta \bar{r}p_t \quad (\text{A.10c})$$

$$p_t^* = \mu_{US} \cdot p_t^{*US} + \mu_{EZ} \cdot (p_t^{*EZ} + s_t^{USD/EUR}) + (1 - \mu_{US} - \mu_{EZ}) \cdot (p_t^{*RU} + s_t^{USD/RUB}) \quad (\text{A.10d})$$

$$\begin{aligned} \pi_t^{*,ss} &= \mu_{US} \cdot \pi_{ss}^{*US} + \mu_{EZ} \cdot (\pi_{ss}^{*EZ} + \Delta s_{ss}^{USD/EUR}) \\ &\quad + (1 - \mu_{US} - \mu_{EZ}) \cdot (\pi_{ss}^{*RU} + \Delta s_{ss}^{USD/RUB}) \end{aligned} \quad (\text{A.10e})$$

$$\pi_t^{*US} = w_{US} \cdot \pi_{t-1}^{*US} + (1 - w_{US}) \cdot \pi_{ss}^{*US} + \varepsilon_t^{*US} \quad (\text{A.10f})$$

$$\pi_t^{*EZ} = w_{EZ} \cdot \pi_{t-1}^{*EZ} + (1 - w_{EZ}) \cdot \pi_{ss}^{*EZ} + \varepsilon_t^{*EZ} \quad (\text{A.10g})$$

$$\pi_t^{*RU} = w_{RU} \cdot \pi_{t-1}^{*RU} + (1 - w_{RU}) \cdot \pi_{ss}^{*RU} + \varepsilon_t^{*RU} \quad (\text{A.10h})$$

Foreign Interest Rates:

$$i_t^{*Fed-US} = f_1 \cdot i_{t-1}^{*Fed-US} + (1 - f_1) \cdot (\bar{r}_t^{*Fed-US} + \pi_{ss}^{*US}) + \varepsilon_t^{i^{*Fed-US}} \quad (A.11a)$$

$$\bar{r}_t^{*Fed-US} = f_2 \cdot \bar{r}_{t-1}^{*Fed-US} + (1 - f_2) \cdot r_{ss}^{*Fed-US} + \varepsilon_t^{\bar{r}^{*US}} \quad (A.11b)$$

Relative Home-Foreign Food Prices:

$$rwp_t^{food} = p_t^{food} + s_t^{AMD/USD} - p_t^f \quad (A.12a)$$

$$r\hat{w}p_t^{food} = \hat{z}_t + \hat{r}p_t^{food} - (1 - \omega) \cdot \hat{r}p_t \quad (A.12b)$$

$$rwp_t^{food} = r\bar{w}p_t^{food} + r\hat{w}p_t^{food} \quad (A.12c)$$

World Food Prices :

$$\pi_t^{*food,im} = 0.5 \cdot \pi_{t-1}^{*food,im} \quad (A.13a)$$

$$+ (1 - 0.5) \cdot (\pi_t^{*food} + \Delta s_t^{AMD/USD} - \Delta \bar{z}_t - \Delta \bar{r}p_t^{food} + (1 - \omega) \cdot \Delta \bar{r}p_t)$$

$$rp_t^{food} = p_t^{food} - p_t^* \quad (A.13b)$$

$$rp_t^{food} = \bar{r}p_t^{food} + \hat{r}p_t^{food} \quad (A.13c)$$

$$\Delta \bar{r}p_t^{food} = f_5 \cdot \Delta \bar{r}p_{t-1}^{food} + (1 - f_5) \cdot \pi^{*qfood}_{ss} + \varepsilon_t^{\Delta \bar{r}p^{food}} \quad (A.13d)$$

$$\hat{r}p_t^{food} = f_7 \cdot \hat{r}p_{t-1}^{food} + \varepsilon_t^{\hat{r}p^{food}} \quad (A.13e)$$

$$\Delta \bar{r}p_t^{food} = 4 \cdot (\bar{r}p_t^{food} - \bar{r}p_{t-1}^{food}) \quad (A.13f)$$

World Oil Prices :

$$\pi_t^{*oil,im} = \pi_t^{*oil} + \Delta s_t^{AMD/USD} - \Delta \bar{z}_t - \Delta \bar{r}p_t^{oil} - \omega \cdot \Delta \bar{r}p_t \quad (A.14a)$$

$$rp_t^{oil} = p_t^{oil} - p_t^* \quad (A.14b)$$

$$rp_t^{oil} = \bar{r}p_t^{oil} + \hat{r}p_t^{oil} \quad (A.14c)$$

$$\Delta \bar{r}p_t^{oil} = f_6 \cdot \Delta \bar{r}p_{t-1}^{oil} + (1 - f_6) \cdot \pi^{*qoil}_{ss} + \varepsilon_t^{\Delta \bar{r}p^{oil}} \quad (A.14d)$$

$$\hat{r}p_t^{oil} = f_8 \cdot \hat{r}p_{t-1}^{oil} + \varepsilon_t^{\hat{r}p^{oil}} \quad (A.14e)$$

$$\Delta \bar{r}p_t^{oil} = 4 \cdot (\bar{r}p_t^{oil} - \bar{r}p_{t-1}^{oil}) \quad (A.14f)$$

Remittances:

$$qrem_t = rem_t - p_t^* \quad (A.15a)$$

$$qrem_t = qr\bar{e}m_t + qr\hat{e}m_t \quad (A.15b)$$

$$\Delta qr\bar{e}m_t = c_{21} \cdot \Delta qr\bar{e}m_{t-1} + (1 - c_{21}) \cdot \Delta \bar{y}_{ss} + \varepsilon_t^{\Delta qr\bar{e}m} \quad (A.15c)$$

$$qr\hat{e}m_t = c_{31} \cdot qr\hat{e}m_{t-1} + c_{32} \cdot \hat{y}_t^* - 0.2 \cdot \hat{z}_t + \varepsilon_t^{qr\hat{e}m} \quad (A.15d)$$

$$\Delta qr\bar{e}m_t = 4 \cdot (qr\bar{e}m_t - qr\bar{e}m_{t-1}) \quad (A.15e)$$

Foreign Exchange Rates:

$$\Delta s_t^{USD/EUR} = \Delta s_{ss}^{USD/EUR} + \varepsilon_t^{\Delta s^{USD/EUR}} \quad (A.16a)$$

$$\Delta s_t^{USD/RUB} = \Delta s_{ss}^{USD/RUB} + \varepsilon_t^{\Delta s^{USD/RUB}} \quad (A.16b)$$

Trend-gap identities:

$$y_t = \bar{y}_t + \hat{y}_t \quad (\text{A.17a})$$

$$d_t = \bar{d}_t + \hat{d}_t \quad (\text{A.17b})$$

$$g_t = \bar{g}_t + \hat{g}_t \quad (\text{A.17c})$$

$$x_t = \bar{x}_t + \hat{x}_t \quad (\text{A.17d})$$

$$m_t = \bar{m}_t + \hat{m}_t \quad (\text{A.17e})$$

$$r_{dt} = \bar{r}_{dt} + \hat{r}_{dt} \quad (\text{A.17f})$$

$$r_{lt} = \bar{r}_{lt} + \hat{r}_{lt} \quad (\text{A.17g})$$

$$r_{ft} = \bar{r}_{ft} + \hat{r}_{ft} \quad (\text{A.17h})$$

$$z_t = \bar{z}_t + \hat{z}_t \quad (\text{A.17i})$$

Growth rates of GDP trends:

$$\Delta \bar{y}_t = h_{41} \cdot \Delta \bar{y}_{t-1} + (1 - h_{41}) \cdot \Delta \bar{y}_{ss} + h_{42} \cdot (\Delta q r \bar{e} m_t - \Delta \bar{y}_{ss}) + h_{43} \cdot (\Delta_4 at_t - at_{ss}) + \varepsilon_t^{\Delta \bar{y}} \quad (\text{A.18a})$$

$$\Delta \bar{g}_t = h_{51} \cdot \Delta \bar{g}_{t-1} + (1 - h_{51}) \cdot \Delta g_{ss} + h_{52} \cdot (\Delta q r \bar{e} m_t - \Delta \bar{y}_{ss}) + \varepsilon_t^{\Delta \bar{g}} \quad (\text{A.18b})$$

$$\Delta \bar{x}_t = h_{61} \cdot \Delta \bar{x}_{t-1} + (1 - h_{61}) \cdot \Delta x_{ss} + h_{62} \cdot (\Delta q r \bar{e} m_t - \Delta \bar{y}_{ss}) + h_{63} \cdot (\Delta_4 at_t - at_{ss}) + \varepsilon_t^{\Delta \bar{x}} \quad (\text{A.18c})$$

$$\Delta \bar{m}_t = h_{71} \cdot \Delta \bar{m}_{t-1} + (1 - h_{71}) \cdot \Delta m_{ss} + h_{72} \cdot (\Delta q r \bar{e} m_t - \Delta \bar{y}_{ss}) + \varepsilon_t^{\Delta \bar{m}} \quad (\text{A.18d})$$

$$at_t = p_t^x - (0.25 \cdot p_t^{oil} + 0.5 \cdot p_t^* + (1 - 0.5 - 0.25) \cdot p_t^{food}) \quad (\text{A.18e})$$

$$at_{ss} = \Delta \bar{q} \bar{x}_{ss} + \pi_{ss}^{*US} - (0.25 \cdot (\pi_{ss}^{*qoil} + \pi_t^*) + 0.5 \cdot \pi_t^{*,ss} + (1 - 0.5 - 0.25) \cdot (\pi_{ss}^{*qfood} + \pi_t^*)) \quad (\text{A.18f})$$

$$\Delta q x_t = 0.8 \cdot \Delta q x_{t-1} + (1 - 0.8) \cdot \Delta \bar{q} \bar{x}_{ss} + \varepsilon_t^{qx} \quad (\text{A.18g})$$

$$q x_t = p_t^x - p_t^{*US} \quad (\text{A.18h})$$

Growth Rates:

$$\Delta s_t^{AMD/USD} = 4 \cdot (s_t^{AMD/USD} - s_{t-1}^{AMD/USD}) \quad (\text{A.19a})$$

$$\Delta y_t = 4 \cdot (y_t - y_{t-1}) \quad (\text{A.19b})$$

$$\Delta g_t = 4 \cdot (g_t - g_{t-1}) \quad (\text{A.19c})$$

$$\Delta x_t = 4 \cdot (x_t - x_{t-1}) \quad (\text{A.19d})$$

$$\Delta m_t = 4 \cdot (m_t - m_{t-1}) \quad (\text{A.19e})$$

$$\Delta d_t = 4 \cdot (d_t - d_{t-1}) \quad (\text{A.19f})$$

$$\pi_t = 4 \cdot (p_t - p_{t-1}) \quad (\text{A.19g})$$

$$\pi_t^f = 4 \cdot (p_t^f - p_{t-1}^f) \quad (\text{A.19h})$$

$$\pi_t^* = 4 \cdot (p_t^* - p_{t-1}^*) \quad (\text{A.19i})$$

$$\pi_t^{*US} = 4 \cdot (p_t^{*US} - p_{t-1}^{*US}) \quad (\text{A.19j})$$

$$\pi_t^{*EZ} = 4 \cdot (p_t^{*EZ} - p_{t-1}^{*EZ}) \quad (\text{A.19k})$$

$$\pi_t^{*RU} = 4 \cdot (p_t^{*RU} - p_{t-1}^{*RU}) \quad (\text{A.19l})$$

$$\pi_t^{*oil} = 4 \cdot (p_t^{oil} - p_{t-1}^{oil}) \quad (\text{A.19m})$$

$$\pi_t^{*food,im} = 4 \cdot (p_t^{*food} - p_{t-1}^{*food}) \quad (\text{A.19n})$$

$$\pi_t^{*oil,im} = 4 \cdot (p_t^{oil} - p_{t-1}^{oil}) \quad (\text{A.19o})$$

$$\pi_t^{*food} = 4 \cdot (p_t^{food} - p_{t-1}^{food}) \quad (\text{A.19p})$$

$$\pi_t^{xf} = 4 \cdot (p_t^{xf} - p_{t-1}^{xf}) \quad (\text{A.19q})$$

$$\Delta z_t = 4 \cdot (z_t - z_{t-1}) \quad (\text{A.19r})$$

$$\Delta i_t^{*Fed-US} = 4 \cdot (i_t^{*Fed-US} - i_{t-1}^{*Fed-US}) \quad (\text{A.19s})$$

$$\Delta s_t^{USD/EUR} = 4 \cdot (s_t^{USD/EUR} - s_{t-1}^{USD/EUR}) \quad (\text{A.19t})$$

$$\Delta s_t^{USD/RUB} = 4 \cdot (s_t^{USD/RUB} - s_{t-1}^{USD/RUB}) \quad (\text{A.19u})$$

$$\Delta \bar{d}_t = 4 \cdot (\bar{d}_t - \bar{d}_{t-1}) \quad (\text{A.19v})$$

$$\Delta \bar{g}_t = 4 \cdot (\bar{g}_t - \bar{g}_{t-1}) \quad (\text{A.19w})$$

$$\Delta \bar{x}_t = 4 \cdot (\bar{x}_t - \bar{x}_{t-1}) \quad (\text{A.19x})$$

$$\Delta \bar{m}_t = 4 \cdot (\bar{m}_t - \bar{m}_{t-1}) \quad (\text{A.19y})$$

$$\Delta \bar{z}_t = 4 \cdot (\bar{z}_t - \bar{z}_{t-1}) \quad (\text{A.19z})$$

$$\Delta \bar{y}_t = 4 \cdot (\bar{y}_t - \bar{y}_{t-1}) \quad (\text{A.20a})$$

$$\Delta qrem_t = 4 \cdot (qrem_t - qrem_{t-1}) \quad (\text{A.20b})$$

$$\Delta at_t = 4 \cdot (at_t - at_{t-1}) \quad (\text{A.20c})$$

$$\Delta qx_t = 4 \cdot (qx_t - qx_{t-1}) \quad (\text{A.20d})$$

$$\pi_t^{*im} = 4 \cdot (p_t^{*im} - p_{t-1}^{*im}) \quad (\text{A.20e})$$

$$\pi_t^4 = p_t - p_{t-4} \quad (\text{A.20f})$$

$$\pi_t^{4,f} = p_t^f - p_{t-4}^f \quad (\text{A.20g})$$

$$\pi_t^{*4} = p_t^* - p_{t-4}^* \quad (\text{A.20h})$$

$$\Delta^4 y_t = y_t - y_{t-4} \quad (\text{A.20i})$$

$$\Delta^4 d_t = d_t - d_{t-4} \quad (\text{A.20j})$$

$$\Delta^4 g_t = g_t - g_{t-4} \quad (\text{A.20k})$$

$$\Delta^4 x_t = x_t - x_{t-4} \quad (\text{A.20l})$$

$$\Delta^4 m_t = m_t - m_{t-4} \quad (\text{A.20m})$$

$$\Delta^4 s_t^{AMD/USD} = s_t^{AMD/USD} - s_{t-4}^{AMD/USD} \quad (\text{A.20n})$$

$$\pi_t^{4,xf} = p_t^{xf} - p_{t-4}^{xf} \quad (\text{A.20o})$$

$$\Delta^4 z_t = z_t - z_{t-4} \quad (\text{A.20p})$$

$$\pi_t^{*4,oil} = p_t^{oil} - p_{t-4}^{oil} \quad (\text{A.20q})$$

$$\pi_t^{*4,food} = p_t^{food} - p_{t-4}^{food} \quad (A.20r)$$

$$\pi_t^{*4EZ} = p_t^{*EZ} - p_{t-4}^{*EZ} \quad (A.20s)$$

$$\pi_t^{*4RU} = p_t^{*RU} - p_{t-4}^{*RU} \quad (A.20t)$$

$$\Delta^4 \bar{d}_t = \bar{d}_t - \bar{d}_{t-4} \quad (A.20u)$$

$$\Delta^4 \bar{g}_t = \bar{g}_t - \bar{g}_{t-4} \quad (A.20v)$$

$$\Delta^4 \bar{x}_t = \bar{x}_t - \bar{x}_{t-4} \quad (A.20w)$$

$$\Delta^4 \bar{m}_t = \bar{m}_t - \bar{m}_{t-4} \quad (A.20x)$$

$$\Delta^4 \bar{z}_t = \bar{z}_t - \bar{z}_{t-4} \quad (A.20y)$$

$$\Delta^4 \bar{y}_t = \bar{y}_t - \bar{y}_{t-4} \quad (A.20z)$$

$$\Delta_4 at_t = at_t - at_{t-4} \quad (A.21a)$$

Expectations:

$${}^e\pi_t^f = 0.5 \cdot (E_t[\pi_{t+1}] + (1 - \omega) \cdot E_t[\Delta r \bar{p}_{t+1}]) + (1 - 0.5) \cdot E_t[\pi_{t+1}^{xf}] \quad (A.22a)$$

$${}^e\pi_t^{xf} = 0.5 \cdot (E_t[\pi_{t+1}] - \omega \cdot E_t[\Delta r \bar{p}_{t+1}]) + (1 - 0.5) \cdot E_t[\pi_{t+1}^{xf}] \quad (A.22b)$$

$${}^e\pi_t = 0.5 \cdot E_t[\pi_{t+1}] + (1 - 0.5) \cdot \pi_t^4 \quad (A.22c)$$

$${}^e\hat{d}_t = E_t[\hat{d}_{t+1}] \quad (A.22d)$$

$${}^e\hat{g}_t = E_t[\hat{g}_{t+1}] \quad (A.22e)$$

$${}^e\hat{x}_t = E_t[\hat{x}_{t+1}] \quad (A.22f)$$

$${}^e\hat{m}_t = E_t[\hat{m}_{t+1}] \quad (A.22g)$$

$${}^e\pi_t = E_t[\pi_{t+1}] \quad (A.22h)$$

$${}^e\pi_t^* = E_t[\pi_{t+1}^*] \quad (\text{A.22i})$$

$${}^e\pi_t^{tar} = E_t[\pi_{t+1}^{tar}] \quad (\text{A.22j})$$

$${}^e\Delta\bar{z}_t = E_t[\Delta\bar{z}_{t+1}] \quad (\text{A.22k})$$

$$100 \cdot \log(S_t^{AMD/USD}) = s_t^{AMD/USD} \quad (\text{A.22l})$$

Appendix B

Free model historical forecasts

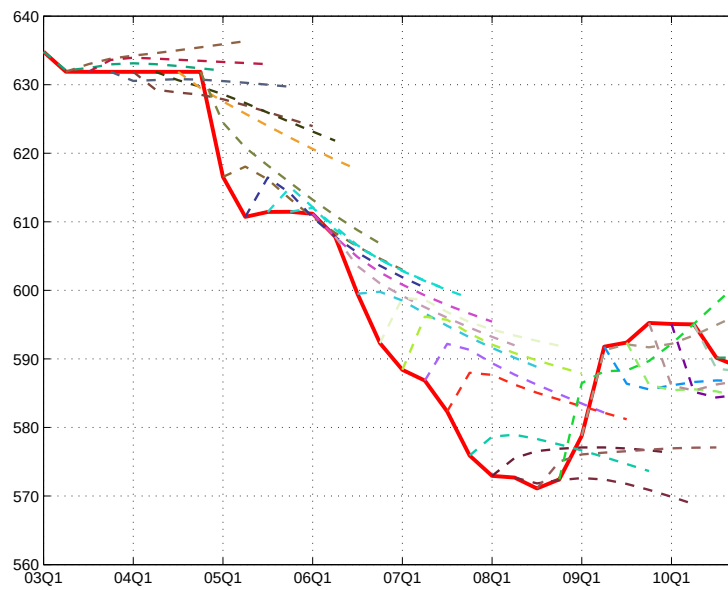


Figure B.1: Free historical simulations for the AMD/USD exchange rate ($100 \cdot \log$)

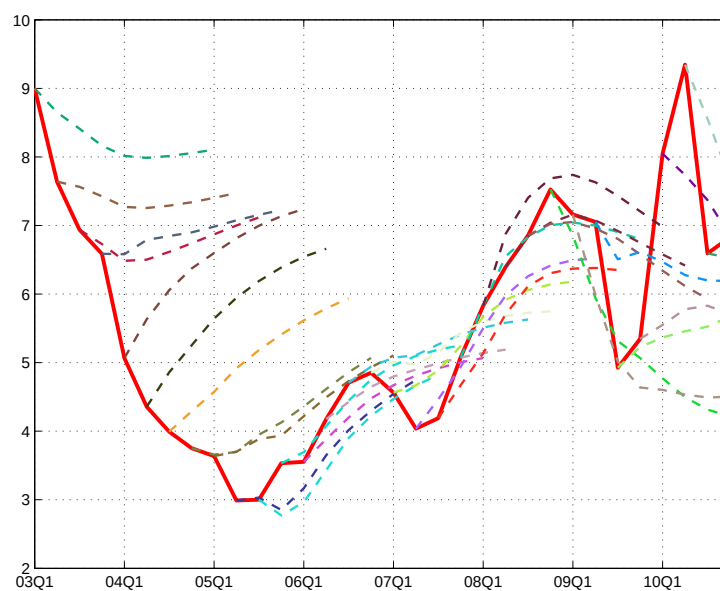


Figure B.2: Free historical simulations for the Central Bank's policy rate (% , p.a., log approx.)

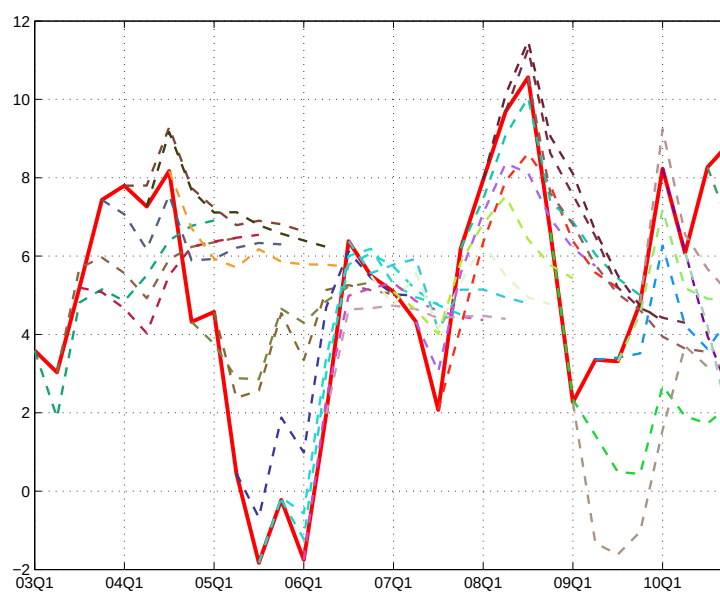


Figure B.3: Free historical simulations for the CPI inflation (YoY, %)

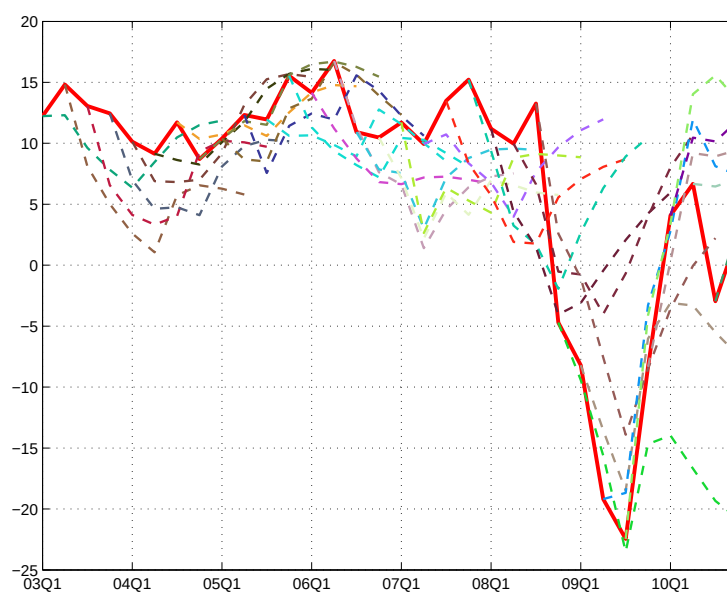


Figure B.4: Free historical simulations for the output gap (%)

Appendix C

A Small DGE model of Armenia

C.1 Introduction

This paper describes a small non-linear dynamic general equilibrium (DGE) model calibrated to capture some of the main features of the Armenian economy and its relations with the rest of the world. The purpose of the model is to provide a consistent framework for considering the short- and long-run effects of shocks relative to an exogenous baseline path. Thus, the model is not intended to be used as a stand-alone forecasting tool; rather its purpose is to provide a vehicle for testing and refining intuition about how shocks affect the economy under various assumptions regarding, e.g., fiscal and monetary policies, expectations formation, and the degree of international capital mobility.

The model is designed primarily to perform deterministic simulations given changes in the exogenous variables, either anticipated or unanticipated. In this sense, it “tells a story” about the role of relative price, activity, and stock adjustments in resolving tensions that arise from economic shocks. This distinguishes it from linearized stochastic DGE models, which are more appropriate for analyzing issues of volatility in the endogenous variables compared with observed data.¹

Modeling is generally a “work in progress,” and the structure of this model has been kept deliberately simple, while preserving the consistency of the DGE framework, to facilitate adapting it to changing needs. At the same time, the structure embodies a flexible calibration that allows a wide range of outcomes to be generated given different

¹See, for instance, [7] for an application of a stochastic DGE model to the Armenian economy.

views about the parameters. A notable feature of this model, relative to other open-economy DGE models, is the incorporation of imperfect substitutability between home and foreign assets. This allows an endogenous characterization of the difference in expected yields on home and foreign assets, and also provides a role for intervention in foreign exchange markets to affect the exchange rate and interest rates.

Another interesting aspect of relaxing the assumption of perfect international capital mobility is its implications for the effects of fiscal shocks. In particular, the government's decision to borrow either domestically or externally to finance fiscal deficits leads to "portfolio" effects on financial variables, in addition to the usual real effects generated by changes in spending. The portfolio impact also depends on whether external financing is from "official" sources at concessional interest rates or from private markets at rates that reflect the country risk premium. Thus the model embodies a flexible specification for government financing that allows such portfolio effects to be analyzed under different assumptions about financing sources.

The next section describes the structure of the model, followed by a discussion of how the model calibration was adapted to capture specific features of the Armenian economy. Some illustrative shocks are then described, focusing on the impact of different assumptions about policy and other parameters. The final section provides concluding remarks and directions for future work.

C.2 Model Structure

The model consists of two regions: the home country (Armenia) and the rest of the world (ROW). Its structure is generally consistent with that of recent DGE models, with some distinctive features that allow it to more accurately reflect the Armenian context. Perhaps the most notable of these is imperfect substitutability between foreign and domestic assets, allowing portfolio effects to play a role in driving movements in the exchange rate and interest rates. In particular, it allows the introduction of a substantial, time-varying risk premium or "excess return" on Armenian external liabilities.

The following model description closely follows the actual coding and mnemonics to increase its usefulness as a reference document for users. Data in the model are annual. Future-period values of variables (indicated by a $t + 1$ subscript) are model-consistent in the simulations using a forward-looking algorithm. Upper-case mnemonics generally refer to variables that can change over time (either exogenous or endogenous), while lower-case mnemonics refer to model parameters that are set in the initial calibration.

The mnemonics are those used for the home country in the model; there are equivalent relationships for the ROW, with _F appended to the mnemonic.

Production and Investment

Each region produces a single output, employing labor and capital in a CES production function. The output of each region is an imperfect substitute for that of the other region. With trade between the two regions, domestic absorption (i.e. consumption plus investment plus government spending) consists of a bundle of both the home and foreign goods, as discussed below.

The price of domestic output is the numeraire in each region, and variables are expressed in real terms relative to this numeraire. Variables that consist of a bundle of domestic and imported goods are valued at the relative price of absorption to the numeraire (PC_t). The relative price of the outputs of the two regions is given by the real exchange rate (RER_t), defined such that a rise in RER indicates appreciation of the home-country real exchange rate (i.e., a rise in the price of its output relative to ROW output).

Total labor supply in each region is proportional to the population (POP_t) plus an index of labor-augmenting technical progress ($THETA_t$), which grows at rate G_t . Output is assumed to be at its potential level (YP_t) when labor and capital (K_t) are employed at “normal” levels (i.e. levels consistent with an absence of upward or downward pressure on inflation). Potential output is then given by the production function:

$$YP_t = (a_0 K_t^{-ces1} + (1 - a_0)(THETA_t POP_t)^{-ces1})^{-1/ces1},$$

where a_0 is a parameter related to the baseline capital share in output and $ces1$ determines the elasticity of substitution between labor and capital.² In the limiting case where $ces1$ is zero, the CES production function becomes a Cobb-Douglas function with an elasticity of substitution between labor and capital of one; as $ces1$ rises, the elasticity of substitution falls; a value of one, for instance, yields an elasticity of substitution of one half.

When the elasticity of substitution between labor and capital is not equal to one, the capital share (AA_t) is time varying, depending on changes in the capital-to-output ratio and the elasticity of substitution in production:

²The elasticity of substitution is given by $1/(1 + ces1)$.

$$AA_t = a_0(Y P_t / K_t)^{ces1}$$

The capital stock is assumed to be privately held (i.e., government capital does not enter the production function). It depreciates at rate k_0 , and is augmented in each period by gross private investment (INV_t) net of costs of adjusting the capital stock (KC_t), as defined below. The capital stock at the beginning of period t is then:

$$K_t = (1 - k_0)K_{t-1} + d_0 INV_{t-1} - KC_{t-1}.$$

Changing the rate of investment from that consistent with maintaining a constant capital-to-output ratio is assumed to entail costs to firms. These costs are specified as a quadratic function of the investment/capital ratio in the current period relative to the ratio that would maintain a constant ratio of capital to potential output. In particular, the investment/capital ratio that would be consistent with a constant capital/output ratio equals the ratio of potential output in period t to period $t - 1$ ($DY P_t$) minus one plus the capital depreciation rate. The total cost of adjusting the capital stock can then be expressed as:

$$KC_t = 1/2k_1((INV_t/K_t) - (DY P_t - 1 + k_0))^2 K_t,$$

where k_1 is a parameter that determines the adjustment cost.

The associated marginal cost of installing an additional unit of investment (KMC_t) can be obtained by differentiating the above equation with respect to investment, giving:

$$KMC_t = k_1((INV_t/K_t) - (DY P_t - 1 + k_0)).$$

The optimal rate of capital investment in period t is derived from Tobin's Q framework. Specifically, the present value of the future stream of income accruing to a unit of installed capital at time t - i.e., the "market value" of a unit of capital (KV_t) - is given by the discounted value of the net income accruing to capital, per unit of capital, in all future periods:

$$K_{t+1} = \sum_{z=t}^{\infty} \left\{ \frac{AA_z Y Z / (PC_z K_z) - k_0}{\prod_{s=1}^{z-1} (1 + RPC_s + RISK_s)} \right\}$$

The numerator in the above expression is the capital share in output, valued in units of absorption, divided by the capital stock, less the rate of capital depreciation. The denominator is the discount factor, which is a function of the real interest rate defined in terms of changes in the absorption deflator (RPC_t) and a subjective risk premium ($RISK_t$) specific to investment in physical capital as opposed to financial assets.

Substituting the period-ahead value of capital, KV_{t+1} , into the above expression allows it to be written more simply as:

$$KV_t = KV_{t+1}/(1 + RPC_t + RISK_t) + AA_tYY_t/(PC_tK_t) - k0.$$

The zero excess profit condition for competitive firms implies that the market value of the capital stock in period t+1 equal the price of a unit of investment in period t, plus the marginal adjustment cost of installing a new unit of capital. As the price of investment is unity, defined in units of absorption, this gives:

$$KV_{t+1} = 1 + KMC_t.$$

It is useful to look at this equation in terms of the model dynamics that determine the period-t investment rate: as the market value of a unit of capital in period t+1 increases, the investment rate in period t will also rise, pushing up the marginal adjustment cost to the point at which it equals the difference between the price of a unit of investment in period t and the value of a unit of installed capital in period t+1.

Households

Households are assumed to be born at rate BTH_t and die at rate DTH_t , giving aggregate population growth ($DPOP_t$) of:

$$DPOP_t = \frac{1 + BTH_t}{1 + DTH_t}$$

Consumption is based on the framework of [2] and [9], with allowance also being made for “rule-of-thumb” households that consume out of current income. In the Blanchard-Yaari framework, current consumption is a time-varying fraction of total wealth, which in turn is the sum of human wealth (WH_t) and financial wealth (WF_t).

Human wealth is the discounted value of household income, excluding income from assets (YH_t). The latter equals: the labor share in output less taxes (GTX_t); net transfers from abroad (TR_t); and a portion of the subjective risk premium on the domestic capital stock. The risk premium is assumed to be distributed in a lump-sum form to households by a central “risk-pooling” agency, which could be interpreted as a banking system that charges a spread between lending and deposit rates for intermediating risk. Only a portion of the risk premium accrues to domestic households, as the banking system is assumed to be partially owned by foreigners. The expression for household income then is:

$$YH_t = ((1 - AA_t)YY_t - GTX_t + TR_t)/PC_t + (1 - r1)RISK_t K_t$$

Labor income, taxes, and transfers are divided by the absorption deflator to express them in units of consumption; the parameter $r1$ determines the share of the capital risk premium accruing to domestic households.

Human wealth is obtained as the discounted value of current and future household income, with the discount rate equaling the sum of the population growth rate, the real interest rate, and the death rate.³

$$WH_t = YH_t + WH_{t+1}/(DPOP_t + RPC_t + DHT_t)$$

Financial wealth is the sum of the market value of the capital stock ($KV_t K_t$), net foreign assets (NFA_t), and net government debt (GB_t), expressed in units of absorption:⁴

$$WF_t = KV_t K_t + (NFA_t + GB_t)/PC_t \quad (C.1)$$

Consumption by Blanchard-Yaari households (CBY_t) is a time-varying, forward-looking fraction of total wealth (GAM_t), depending on the path of interest rates, the rate of household time preference (ρ), and the probability of death. As discussed in [5], this framework yields an equilibrium real interest rate that is implausibly high for typical parameter

³The population growth rate is included so that human wealth reflects only of the future labor income of those currently alive.

⁴Only a portion of government debt represents net real wealth because the associated tax burden is subtracted from labor income.

values, including a positive rate of time preference. One way of obtaining a real interest rate that is more typical of observed values is to invoke the “relative income hypothesis,” whereby households derive utility from the level of their consumption relative to that of other households, as opposed to the absolute level of consumption. This adjustment causes an additional term in the productivity growth rate, G_t , to be added to the numerator in the expression for GAM_t , giving:

$$CBY_t = GAM_t(WH_t + WF_t) \quad (C.2)$$

$$GAM_t = \frac{2 - (1 + rho + DHT_t)^{-c3}(1 + RPC_t + DTH_t)^{(c3-1)}(1 + G_t)^{(1-c3)}}{(1 + 1/GAM_{t+1})}$$

In addition to forward-looking households, some households are assumed to exhibit “rule-of-thumb” behavior in the short run in that their consumption is determined only by changes in current household non-asset income, expressed as a share of potential output. Over time, this specification ensures that consumption of income-constrained households converges to that of forward-looking households, yielding long-run model properties that are consistent with the Blanchard-Yaari framework. Finally, the model allows for households to exhibit “habit persistence,” interpreted as dependence of current consumption on its lagged value, adjusted for growth in potential output. The relationship for total household consumption can then be expressed as

$$C_t = \left(\frac{YH_t/YP_t}{YH_{t-1}/YP_{t-1}}\right)^{c5} CBY_t^{(1-c5)/(1-c1)} \left(C_{t-1} \frac{YP_t}{YP_{t-1}}\right)^{(c5+c1-c5c1)} \quad (C.3)$$

The parameter $c5$ determines the share of rule-of-thumb households, while $c1$ determines the degree of habit persistence in consumption. Setting both of these to zero would yield a pure Blanchard-Yaari framework where total consumption equals CBY_t .

Inflation

Inflation (PI_t , defined in terms of growth in the output price index) is determined by an expectations-augmented Phillips curve. Current-period inflation is determined by a weighted average of its lagged and expected future values, the gap between actual and

potential output, and the difference between absorption inflation (PIC_t) and output price inflation:

$$PI_t = pi1 PI_{t-1} + (1 - pi1)PI_{t+1}^e + pi2 \log(Y_t/Y P_t) + pi3(PIC_t - PI_t) \quad (C.4)$$

Perhaps the only noteworthy aspect of this typical New Keynesian specification is the inclusion of the difference between absorption and output inflation. This term captures upward (and downward) pressures on factor costs due to changes in final demand prices that can arise, for instance, from wage setting behavior when wages are explicitly or implicitly linked to consumer prices.⁵

Government

Each region has a government that spends, taxes, and issues debt.⁶ Primary government spending (GP_t) is determined by an exogenous target ratio of government spending ($G0_t$) to potential output times potential output:

$$GP_t = G0_{t-1} Y P_t \quad (C.5)$$

Tax revenues (GTX_t) are determined by an endogenous tax rate (GTR_t) times actual output (the relationship determining the tax rate is described below). Taxes are assumed to be non-distortionary. Tax revenues then are:

$$GTX_t = GTR_t Y Y_t. \quad (C.6)$$

Spending can be financed either by taxes or issuing government debt. Two types of government debt exist in the model: domestic debt issued in the currency of the home

⁵The difference between these two inflation rates will be proportional to movements in the real exchange rate (see below), and the equation could instead be specified using the change in the exchange rate.

⁶Any utility derived from government spending is assumed to be separable from private consumption in the underlying household utility function.

country (GBD_t), and external debt issued in the currency of the ROW region (GBX_t). The yield on domestic debt is the home country real interest rate, R_t . The yield on external debt, in contrast, can be at either “concessional” or “market” interest rates. External debt issued at concessional interest rates yields the foreign real interest rate (R_{F_t}), whereas external debt issued at market rates yields the foreign real interest rate plus the home country risk premium or “excess return” (XR_t), which is discussed in more detail below. The share of external debt issued at concessional interest rates is determined by the parameter $gbx1$.

The government can also hold external assets (GAX_t), interpreted as official reserve holdings. As the government sector in the model is based on a general government definition, these foreign assets include reserves held at the central bank. The total net debt of the government (GB_t) can then be defined as:

$$GB_t = (GBX_t - GAX_t)/RER_t + GBD_t$$

In terms of dynamics, net government debt in period t is determined by the gap between primary spending and taxes in period $t - 1$, as well as the sum of the lagged values of the components of net debt adjusted for interest costs and exchange rate movements (i.e., valuation effects):

$$GB_t = GP_{t-1}PC_{t-1} - GTX_{t-1} + (1 + R_{t-1})GBD_{t-1} + \frac{(1 + R_{F_{t-1}} + (1 - gbx1)XR_{t-1})GBX_{t-1} - (1 + R_{F_{t-1}})GAX_{t-1}}{RER_t}$$

The valuation effects of exchange rate movements are reflected in the use of the current-period as opposed to lagged exchange rate in the denominator of the second line in this expression. Appreciation of the real exchange rate will tend to lower the net value of government debt to the extent that external liabilities exceed government holdings of foreign assets. Similar valuation effects arise in the dynamics of the overall net foreign assets of the country, as discussed below.

While the above expression determines the change between periods $t - 1$ and t in net government debt, the composition of the change between the three components remains to be determined. Both the external debt and holdings of external assets are assumed to increase at exogenous ratios to changes in domestic potential output (expressed in

units of foreign currency). Shock terms (denoted by the prefix SHK_) are added to these equations to facilitate imposing exogenous policy actions to change external debt or assets. Domestic debt is determined residually via the definition of net government debt given above. Thus:

$$\begin{aligned} GBX_t &= GBX_{t-1} + GBX0_t(YP_t - YP_{t-1})RER_{t-1} + SHK_GBX_t \\ GAX_t &= GAX_{t-1} + GAX0_t(YP_t - YP_{t-1})RER_{t-1} + SHK_GAX_t \end{aligned}$$

To force long-run government solvency and rule out “Ponzi games”, the tax rate is assumed to adjust to achieve a target ratio for public debt to potential GDP ($GB0_t$), with convergence of the actual to target debt ratio determined by the gaps between the current and lagged debt ratios and their target levels:

$$GTR_t = GTR_{t-1} + gb1((GB_t/YP_t) - GB0_t) + gb2((GB_{t-1}/YP_{t-1}) - GB0_{t-1})$$

Monetary Policy

Monetary policy is modeled using a modified Taylor Rule, where the nominal interest (I_t) responds, inter alia, to the difference between actual and target inflation (PIT_t):

$$I_t = I_{t-1} + i2((1 - i5)PIC_t + i5 PIC_{t+1} - PIT_t) + i3 \log\left(\frac{YY_t}{YP_t}\right) - i4 \log\left(\frac{RER_t}{RER_{t-1}}\right).$$

Absorption inflation is used to reflect the fact that inflation targets are typically set in terms of consumption as opposed to output prices. The parameter $i2$ determines the response to the inflation gap, while $i5$ is a “switch” that can be used to have monetary policy respond to forecast as opposed to current-period inflation. The reaction function also includes the gap between actual and potential output, consistent with the Taylor specification. Less typically, it includes the change in the real exchange rate, allowing a role for monetary policy to respond to exchange rate movements. Setting the parameter on the exchange-rate term to zero yields a standard Taylor Rule, whereas a high value causes monetary policy to be dominated by exchange-rate smoothing objectives.⁷

⁷The change in the real exchange rate could be replaced by the deviation in the real exchange rate from a target level to generate an exchange-rate targeting as opposed to smoothing regime.

Finally, adjustment in the nominal interest rate is assumed to take place relative to its lagged value, as opposed to an underlying “equilibrium” interest rate. Using an equilibrium interest rate in a simulation model causes problems insofar as its value changes, both over time and in response to shocks, in a way that is difficult to endogenize without introducing substantial complexity to the model structure. In the event, the use of the lagged interest rate yields good simulation properties, and may also be more typical of the actual approach taken to interest-rate adjustment by central banks.

In addition to the use of the short-term interest rate as monetary policy instrument, it is possible to generalize the relationship for government holdings of foreign assets (i.e. official reserves) such that they respond to the excess return on domestic assets and/or movements in the real exchange rate:

$$GAX_t = GAX_0 Y P_t RER_t - gax1(XR_t - xr_0)Y P_t + gax2(RER_t/RER_{t-1})Y P_t.$$

Setting the parameters $gax1$ and $gax2$ to positive values (as opposed to zero in the default calibration) then allows reserve movements to be used as a smoothing mechanism to reduce volatility in the excess return or the exchange rate.⁸

External trade, prices, and assets

Private consumption, investment and government spending consist of both domestic and ROW output. The share of imports in private consumption and investment varies according to the relative price of ROW to domestic output. The response of import shares is determined via CES functions that bundle the home and ROW components of spending into aggregate consumption and investment. Specifically:

$$C_t = (im_0^{1/c_2} C H_t^{(c_2-1)/c_2} + (1 - im_0)^{1/c_2} I M C_t^{(c_2)/c_2})^{c_2/(c_2-1)},$$

and

$$INV_t = (im_0^{1/c_2} I N V H_t^{(c_2-1)/c_2} + (1 - im_0)^{1/c_2} I M I_t^{(c_2)/c_2})^{c_2/(c_2-1)},$$

⁸Any intervention of this type is effectively “sterilized”, in that the stock of domestic government debt automatically adjusts in an offsetting way to keep net government debt unchanged.

where CH_t and $INVH_t$ are the home-produced components of consumption and investment respectively, while IMC_t and IMI_t are the imported components. The baseline share of imports is determined by the parameter $im0$, which is assumed to be the same for both consumption and investment (although this could be generalized). The elasticity of substitution between the domestic and imported components is determined by the parameter $c2$ (see the above discussion of the production function), which again is assumed to be the same for consumption and investment.

The first-order conditions for the optimal spending allocation across domestic and imported goods yield the following relationships:

$$IMC_t = \left(\frac{im0}{1 - im0} \right) CH_t RER_t^{c2},$$

and

$$IMI_t = \left(\frac{im0}{1 - im0} \right) INVH_t RER_t^{c2},$$

The import component of government spending is a fixed share of total spending equal to the baseline import share in private spending ($im0$). Total imports (IM_t), then, are:

$$IM_t = IMC_t + IMI_t + im0 GOVP_t.$$

Exports of each region (EX_t) equal the imports of the other region. The trade balance (TB_t) equals exports minus imports, where the latter are valued in units of domestic output:

$$TB_t = EX_t - IM_t / RER_r.$$

The domestic absorption price relative to the output price is determined by $im0$, $c2$, and the real exchange rate:

$$PC_t = (im0^{1/c2} RER_t^{(c2-1)/c2} + (1 - im0)^{1/c2})^{c2/(c2-1)}.$$

In terms of financial linkages between regions, introducing imperfect asset substitutability in the model requires specifying relationships for gross holdings of home assets by the ROW and vice versa. The specification for these relationships follows that used in [1] (henceforth BGS). Specifically, it is assumed that the share of ROW assets in domestic financial wealth depends on the yield gap between home and foreign assets: a higher yield (XR_t) on home assets reduces holdings of foreign assets, while a lower yield raises them.

The excess return equals the difference between domestic and foreign interest rates adjusted for the expected change in the real exchange rate, with expected exchange rate appreciation raising the return on home assets.

$$XR_t = \left(\frac{1 + R_t}{1 + R_{Ft}} \right) \left(\frac{RER_{t+1}^e}{RER_t} \right) - 1.$$

The relationship for the share of ROW liabilities in home financial wealth can then be expressed as:

$$\frac{XL_{Ft}/RER_t}{WF_t} = \text{beta0} - \text{bxr1} XR_t$$

The left-hand side gives the external liabilities of the ROW (XL_{Ft}), expressed in units of home output, as a share of home financial wealth. On the right-hand side, the parameter beta0 determines the baseline share, while bxr1 determines the response of the share to the excess return.⁹ The higher is bxr1 , the more sensitive is the share to the excess return; at the limit, an infinite value for bxr1 implies perfect substitutability between home and ROW assets, whereas zero implies that the asset share is fixed at its baseline level.

The above equation has been expanded in the model code to capture phenomena that are relevant to Armenia. In particular, official holdings of foreign reserves are subtracted from the external liabilities of the ROW in the private portfolio share. This provides a role for changes in official reserve holdings to affect the excess return, and thus the real exchange rate. A second change is to introduce a non-zero baseline excess return

⁹Dynamic stability of the model requires home bias in portfolio shares, i.e. a value for beta0 of less than one half.

on domestic assets into the model, consistent with the relatively high real interest rates observed in Armenia in recent years compared with the ROW, giving:

$$\frac{(XL_{Ft} - GAX_t)/RER_t}{WF_t} = \text{beta0} - \text{bxr1}(XR_t - \text{xr0})$$

where xr_0 is the baseline level of the excess return on home assets.

Similarly, the portion of external government debt that is financed at concessional interest rates is subtracted from total Armenian external liabilities in the portfolio share of the ROW region. This is because concessional external debt is assumed to be issued by official creditors or other governments, and thus is not a component of the external debt held in private portfolios. This gives:

$$\frac{(XL_t RER_t - \text{gbx1} GBX_t)}{WF_{Ft}} = \text{beta_f0} - \text{bxr_f1}(XR_t - \text{xr0})$$

Net foreign assets of the home country (NFA_t) are defined as:

$$NFA_t = XL_{Ft}/RER_t - XL_t,$$

In addition, there is an equation describing the evolution of net foreign assets from period $t - 1$ to t in terms of the trade balance, transfers from abroad, the capital risk premium accruing to foreigners, asset income, and valuation effects arising from exchange rate movements. As discussed in BGS (2005), this relationship can be written as:

$$\begin{aligned} NFA_t = & TB_{t-1} + TR_{t-1} - r1RISK_{t-1}K_{t-1}PC_{t-1} \\ & + \frac{1 + R_{Ft-1}XL_{Ft-1}}{RER_t} \\ & - (1 + R_{t-1})(XL_{t-1} - GBX_{t-1}/RER_{t-1}) \\ & - \frac{(1 + R_{Ft-1} + (1 - \text{gbx1})XR_{t-1})GBX_{t-1}}{RER_t} \end{aligned}$$

Valuation effects from exchange rate movements arise because external liabilities denominated in the ROW currency are valued at the current-period as opposed to lagged real exchange rate, as are the government external liabilities.¹⁰

Transfers from abroad are equal to a fixed ratio to ROW output expressed in terms of domestic currency:

$$TR_t = \frac{tr0 \ YY_F_t}{RER_t}$$

The expected future exchange rate in the model can be specified to be either purely forward-looking, or to contain elements of current and past exchange rate levels:

$$ER_{t+1}^e = rer1ER_t + rer2ER_{t-1} + (1 - rer1 - rer2)ER_{t+1}$$

The parameters *rer1* and *rer2* determine the weights on the backward-looking and current components of exchange rate expectations.

C.3 Calibrating the Model for Armenia

Calibrating an open-economy DGE model of this type to capture key features of the Armenian economy is challenging. Such models generally assume an initial steady state in which real growth rates are the same across regions, trade is balanced, and domestic and international interest rates are equalized. None of these conditions applies to Armenia in recent years, requiring significant changes to the typical model calibration and construction of the baseline data.

To keep the discussion concise, the description of the calibration focuses on the baseline values that are of particular relevance to these aspects of Armenia's situation, as opposed to a general discussion of the model parameterization (including for the ROW). A more complete treatment of these issues, and associated references to the literature, can be found in [5].

¹⁰See [8] for a discussion of valuation effects arising from exchange rate movements.

It should also be emphasized that the calibration is preliminary, subject to the (often limited) available information on variables such as Armenia's external assets and liabilities. At the same time, the flexibility of the model structure and method of creating the baseline data make it relatively straightforward to modify the calibration to incorporate richer data sources and alternative views as to the behavioral parameters.

Real growth and demographics

One notable phenomenon specific to Armenia is the high GDP growth observed in recent years (notwithstanding the contraction in 2009). Based on the October 2010 WEO data, average real growth in Armenia during 2003-11 is expected to stand at close to $7\frac{1}{2}$ percent, compared with growth in the ROW of slightly over $3\frac{1}{2}$ percent using PPP weights, and less than 3 percent using current exchange rates. Growth is projected to slow in Armenia to just over 4 percent through the WEO projection horizon through 2015, while remaining stable in the ROW.

In light of these stylized facts, the baseline calibration is set to yield potential real growth of $4\frac{1}{2}$ percent in Armenia and 3 percent in the ROW. With population growth assumed to be 1 percent per year in both regions, this implies a rate of labor-augmenting productivity growth of $3\frac{1}{2}$ percent per year in Armenia and 2 percent in the ROW. As this difference in productivity growth would not be expected to continue indefinitely, the Armenian rate is reduced gradually starting in 2020 such that it converges to the ROW rate over the following 50 years.

The size of the Armenian economy relative to that of the ROW is also based on WEO data. At PPP weights, the ROW region is roughly 5,000 times larger than Armenia - obviously this is a huge difference in scale, and feedback effects from Armenia to the ROW are effectively zero.

Calibration: Demographics and Growth		
Description	Model Variable	Value
Birth rate	BTH, BTH_F	0.03
Death rate	DTH, DTH_F	0.02
Population growth	$POPDOT - 1, POPDOT_F - 1$	0.01
Productivity growth: Armenia	$THETA(t) / THETA(t-1) - 1$	0.035
ROW	$THETA_F(t) / THETA_F(t-1) - 1$	0.020
Real output growth: Armenia	$YP(t)/YP(t-1) - 1$	0.045
ROW	$YP_F(t)/YP_F(t-1) - 1$	0.030
Relative size: ROW/Armenia output	YP_F/YP	5000

Production and Investment

Given population and productivity growth, calibration of the production, capital, and investment equations plays an important role in determining: (i) the baseline share of investment in GDP, (ii) the response of investment to changes in the real interest rate, and (iii) the baseline real interest rate. The calibration of household and external variables also influence these values, as discussed below.

In the production sector, the relevant parameters are: the capital share; the elasticity of capital/labor substitution; the capital depreciation rate; the risk premium on capital; and the investment adjustment cost. The production parameters were set to be equal in the two regions in the baseline calibration, using typical values for models of this nature. The investment adjustment cost was set to yield responses of investment to changes in real interest rates and output that appeared plausible in light of the evidence for advanced economies.

Calibration: Production and Investment		
Description	Model Parameter/Variable	Value
Capital share: Armenia	AA	0.30
ROW	AA_F	0.30
Elasticity of capital/labor substitution	$1 / (1+ces1)$	0.50
Capital depreciation rate	$d0, d_f0$	0.07
Risk premium on capital stock	$risk, risk_f$	0.02
Investment adjustment cost	$k0, k_f0$	10
Investment/output ratio: Armenia	INV / YP	0.24
ROW	INV_F / YP_F	0.20
Capital/output ratio: Armenia	K / YP	1.9
ROW	K_F / YP_F	2.5

Households

A distinctive feature of the Armenian economy is a high ratio of consumption to output, averaging some 80 percent during 2003-10, notwithstanding high real interest rates. In part this reflects significant transfers from abroad. In addition, however, the model requires a relatively high rate of time preference for Armenia to raise the consumption/income ratio - it was set of 0.02 as opposed to zero in the ROW, yielding a share of consumption in output of 72 percent. While this is still below the observed level, it proved difficult to raise consumption further in the face of the national income accounts constraint that domestic absorption equal domestic output plus the trade deficit. With the baseline trade deficit in the model being smaller than that observed in the actual data, as discussed below, some components of absorption are necessarily lower than observed levels.

Calibration: Households		
Description	Model Parameter/Variable	Value
Rate of time preference: Armenia	ρ	0.02
ROW	ρ_f	0.00
Share of “rule of thumb” households	c_5, c_{5_f}	0.50
Parameter on lagged consumption	c_1, c_{1_f}	0.50
Elasticity of substitution: home/imported goods	c_2, c_{2_f}	1.3
Elasticity of intertemporal substitution	c_3, c_{3_f}	0.3
Consumption/output ratio: Armenia	C / YP	0.72
ROW	C_F / YP_F	0.55

External

Distinctive features of Armenia’s external position in recent years are: high transfers from abroad; a high ratio of imports to output; and an associated large current account deficit. In light of this evidence, the share of transfers in ROW income was set to yield a baseline ratio of transfers to Armenian income of 0.16. This transfer income incorporates both conventional transfers and labor income from domestic residents working abroad. A high ratio of imports was obtained by setting the share of imports in absorption at 0.40. Net foreign assets were set at a ratio of -0.50 to output, based on the value obtained by cumulating the current account deficit from 2000 to 2010.

The baseline excess return, or gap between domestic and ROW real interest rates, was set at 0.04, giving a domestic real interest rate of about $7\frac{1}{2}$ percent compared with $3\frac{1}{2}$ percent in the ROW. The parameter controlling the sensitivity of portfolio shares to the excess return on domestic assets was set at 0.5, implying a relatively low degree of capital mobility (see BGS (2005) and [5]).

Calibration: External Sector		
Description	Model Parameter/Variable	Value
Import share in absorption: Armenia	<i>im0</i>	0.40
Ratio of transfer income from abroad/output	<i>TR/YP</i>	0.16
Excess return on home assets	<i>xr0</i>	0.04
Response of debt shares to excess return	<i>bxr1, bxr_f1</i>	0.5
Ratio of imports/output: Armenia	<i>IM/YP</i>	0.45
Ratio of exports/output: Armenia	<i>IM/YP</i>	0.33
Ratio of trade balance/output: Armenia	<i>TB/YP</i>	-0.12
Ratio of current account/output: Armenia	<i>CA/YP</i>	-0.04
Ratio of NFA /GDP: Armenia	<i>NFA/YP</i>	-0.50
Real interest rate: Armenia	<i>R</i>	0.080
ROW	<i>R_F</i>	0.040

Fiscal policy

In the fiscal sector, parameters are set to yield shares of government spending and debt in the model that are typical of Armenian and ROW observed values. The adjustment parameters for the response of the tax rate to the gap between actual and target government debt were set to yield smooth paths for the debt stock without excessive over-shooting - they imply convergence of the actual to the target debt stock in about 7 years.

Calibration: Fiscal Policy		
Description	Model Parameter	Value
Public spending/output: Armenia	<i>govp0</i>	0.17
ROW	<i>govp_f0</i>	0.20
Public debt/output	<i>gdbt0, gdbt_f0</i>	0.30
External public debt/output	<i>gbx0</i>	0.30
Share of official debt in external debt	<i>gbx1</i>	1.0
Government holdings of foreign assets/output	<i>gax0</i>	0.15
Response of tax rate to current debt gap	<i>gdbt1, gdbt_f1</i>	0.30
Response of tax rate to lagged debt gap	<i>gdbt2, gdbt_f2</i>	-0.25

Inflation and monetary policy

In the inflation equation, equal weights are imposed on lagged and expected future inflation in the baseline calibration, and a parameter on the output gap that is typical of the response of inflation to excess demand for developed economies. The parameter on absorption inflation of 0.10 would correspond to a parameter on the change in the real exchange rate of about 0.45, given the baseline import share in output of 45 percent.

In the Taylor Rule characterization of monetary policy, the target inflation rate for Armenia is set at 4 percent, and at $2\frac{1}{2}$ percent in the ROW. The response of the nominal interest rate to deviations in inflation from target is 1.5 in both regions, consistent with the standard Taylor value. The parameter on the output gap is also set to the typical Taylor value of 0.5. The baseline parameter on the exchange rate is set at zero, implying that the exchange rate is not a direct policy objective when setting interest rates.

Calibration: Inflation and Monetary Policy		
Description	Model Parameter	Value
Weight on lagged inflation in Phillips Curve	<i>pi1</i>	0.50
Weight on output gap in Phillips Curve	<i>pi2</i>	0.20
Weight on absorption inflation in Phillips Curve	<i>pi3</i>	0.10
Target inflation rate: Armenia	<i>pit1</i>	0.040
ROW	<i>pit_f1</i>	0.025
Parameter on inflation gap in Taylor Rule	<i>i2, i2_f</i>	1.5
Parameter on output gap in Taylor Rule	<i>i3, i_f3</i>	0.5
Parameter on exchange rate in Taylor Rule	<i>i4</i>	0
Parameter on future inflation in Taylor Rule	<i>i5, i_f5</i>	0

Partial responses to interest rate and exchange rate shocks

Here we describe model responses to exogenous shocks to the interest rate and exchange rate. They are partial responses, in that feedback effects from the economy back onto the interest rate and exchange rate are excluded. Such partial responses can be useful in comparing the model properties with prior views about the behavior of the economy.

The interest rate shock involves an increase of 1 percentage point in domestic interest rates in the model the household, production, and investment sectors in the model remain endogenous, but other sectors are exogenized to remove feedback effects. The

results are shown in the following table, where the changes in spending are expressed as a ratio to output. Both consumption and investment fall in response to higher interest rates, but the immediate impact is mostly felt on investment: in the first year of the shock, investment declines by 1.3 percent as a share of output, while consumption declines by only 0.2 percent. The shock reduces total spending by 1.5 percent of output in the first year. The effects build over time, until by the fourth year spending falls by 4.2 percent of output.

Change in ratio to output	Year 1	Year 2	Year 3	Year 4
Consumption	-0.006	-0.010	-0.013	-0.015
Investment	-0.007	-0.007	-0.007	-0.007
Total absorption	-0.013	-0.017	-0.020	-0.022

For the exchange rate shock, total output is held constant, thus the effects on trade are generated through the substitution elasticities in the model and valuation changes due to the exchange rate movement itself. The following table shows the responses to a 10 percent appreciation in the real exchange rate for imports, exports, and the trade balance, again expressed as ratios to output as the effects are relatively constant over time, only the first-year impact is shown. Holding total domestic consumption exogenous, imports increase by slightly over 3 percent of output, while exports decline by $3\frac{1}{2}$ percent. The trade balance deteriorates by about 2 percent of GDP, reflecting the favorable valuation impact of the improved terms of trade. When consumption is endogenous, it rises in response to higher real incomes, and imports increase by more than in the previous case, with the trade balance now showing a deterioration of 3 percent of GDP.¹¹

Change in ratio to output	Consumption exogenous	Consumption endogenous
Real imports	0.030	0.038
Real exports	-0.029	0.030
Real Trade balance	-0.059	-0.068
Nominal trade balance	-0.016	-0.024

Construction of baseline data

Given the above calibration, baseline data for the endogenous variables are constructed using a steady-state version of the dynamic model. In the steady-state model, leads and lags on the endogenous variables are replaced by their contemporaneous values, adjusted where appropriate by steady-state growth in that variable. For example, occurrences of the lagged level of output (YY_{t-1}) in the dynamic model are replaced in the

¹¹See [6] for a discussion of the effects of exchange rate movements on trade balances.

steady-state model by current output (Y_t) divided by steady-state growth in output. In cases where variables are stationary in the steady state, e.g. the real exchange rate, leads and lags in the dynamic model are replaced by current values.

Certain parameters, such as rates of time preference, need to be set to be consistent with the data calibration described above. For this purpose, the steady-state model is “flipped” such that these parameters are derived endogenously in the baseline calibration, conditional on values for data such as NFA. After their baseline values are derived in this way, these parameters are exogenized in the model simulations.

With the differences in the parameterization of the model for Armenia and the ROW, for instance in terms of real growth rates, the baseline data are not in a true steady state. To address this issue, a second step was necessary to create baseline data that are fully consistent with the dynamic model. In particular, the unshocked dynamic model was simulated using the steady-state data as starting and ending values for the endogenous variables. These simulations to generate the baseline data were started 20 years before the start of the shocked simulations described below to allow divergences between the steady-state and dynamic starting point to decay.

C.4 Illustrative Shocks

Here we present the results of some illustrative shocks to the model using the default parameterization described above. The shocks start in 2011 and are generally assumed to be permanent. The simulation algorithm solves for the forward-looking variables such that their values are consistent with the model structure and path of the exogenous variables, thus the results are consistent with rational expectations.

Fiscal expansion

The first shock is to fiscal policy in the form of a permanent increase in primary government spending of 5 percent of GDP, combined with a rise in the target for government debt of 30 percent of GDP. Raising the government debt target slows the response of taxes via the tax reaction function, generating larger fiscal deficits in the initial years of the simulation. In particular, the fiscal deficit increases by slightly over 6 percent of GDP in the first year, declining to zero over the following 10 years as the tax rate rises to stabilize the debt-to-GDP ratio.

The impact on some macro variables for the first 5 years of the simulation is shown in figure C.1. Output rises by $2\frac{1}{4}$ percent in the first year, implying a government spending multiplier of slightly less than one half. One factor that reduces the impact multiplier is the assumed ratio of imports in government spending of 40 percent, resulting in “leakage” of the spending shock through higher imports. Indeed, this channel contributes to most of the deterioration in the trade balance of about 2 percent of GDP.

The real interest rate rises by about 1 percentage point in the first year and over 2 percentage points in the second, as monetary policy responds both to higher output and inflation. The excess return on home assets also rises, given the increase in the supply of these assets in the face of imperfect asset substitutability. The exchange rate appreciates on impact, but only slightly.

This shock leads to a significant cycle in the economy over the following 4 years, with output falling below its baseline level by 2013, with associated reversals in interest rates and the trade balance. In the long run, output stabilizes at 0.7 percent below its original level, with both the real interest rate and excess return on domestic assets rising by $\frac{1}{2}$ percentage point.

One aspect of the results that may be surprising is the modest impact on the real exchange rate of the fiscal expansion. When the model is run under the assumption of perfect capital mobility, the exchange rate appreciates by significantly more - about $2\frac{1}{2}$ percent

by 2012. This channel does more of the work in offsetting the fiscal expansion, as the real interest rate peaks at only about 1 percentage point above baseline.

This simulation has been run under the assumption that the resulting government deficit is financed internally. It is also possible to assume that the deficit is financed by external borrowing from official sources at concessional interest rates. In this case, the output multiplier is somewhat larger, as the domestic real interest rate falls in response to a reduction in the excess return on domestic assets, and the exchange rate appreciates by significantly more. The contrast in results, depending on the source of government financing, illustrates the importance of portfolio effects in an environment of imperfect capital mobility.

Sterilized intervention

Here the shock is to official holdings of foreign reserves in the form of a sale of foreign reserves matched by an equivalent purchase of domestic assets - in other words, sterilized intervention. The intervention is a one-time event, with the sale of foreign reserves

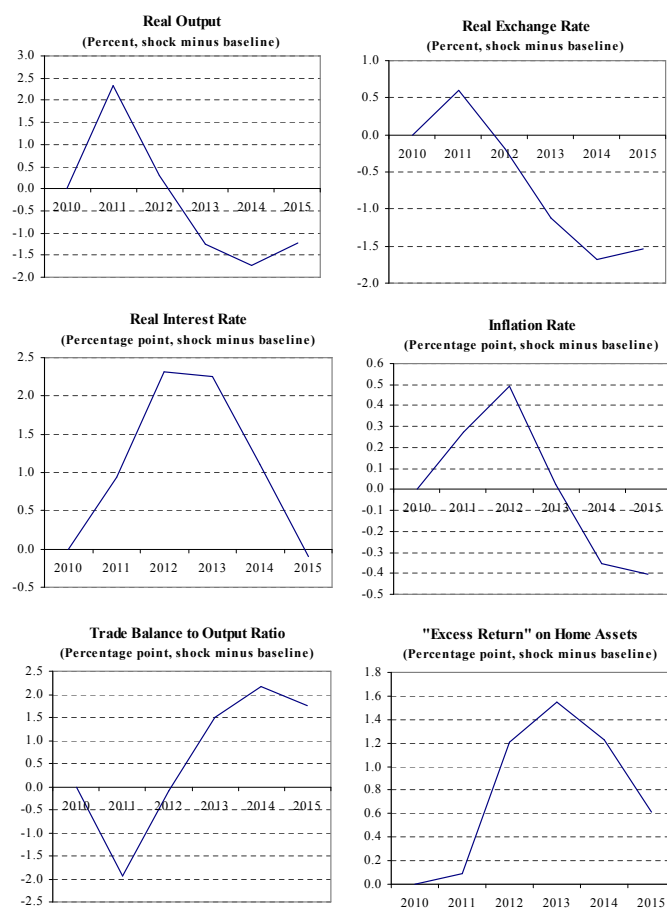


Figure C.1: Impact of Fiscal Expansion

amounting to 5 percent of GDP, or a little less than US\$500 million at the estimated 2010 level of GDP. Under perfect asset substitutability, this operation would have no effect. Under the model assumption about the degree of asset substitutability, however, this shock causes the exchange rate to appreciate as the stock of foreign assets in the private sector portfolio rises while that of domestic assets falls. The results are also sensitive to the characterization of monetary policy. If the policy interest rate responds to current-period inflation, interest rates will fall when the intervention occurs, as the exchange rate appreciation causes a downward shift in the price level. The lower interest rate moderates the upward movement in the exchange rate, in effect undoing some of the impact of the intervention. As this response of monetary policy seems inconsistent with the spirit of sterilized intervention, the model was run under an “inflation forecast targeting” regime, where the parameter in the Taylor Rule on future versus current-period inflation was set to 1. This moderated the downward impact on the policy interest rate, although it still responds to the current-period output gap and future inflation.

With this modification to monetary policy, the exchange rate jumps on impact by 5 percent, as shown in figure C.2. Output, inflation and the trade balance all fall in response to the exchange rate appreciation; the excess return on home assets also falls given the shift in portfolio composition. Output declines because the contractionary effect of the exchange rate appreciation more than offsets the modest drop in the real interest rate. Lower output reinforces the direct impact of the exchange rate appreciation in reducing absorption inflation. Of course, these effects are considered in isolation, whereas sterilized intervention is normally used to offset some other shock, for example an outflow of foreign capital. In this sense the shock-minus-control results should be considered relative to the outcome that would occur in the absence of intervention.

The impact effects of the shock are largely transitory given the assumption that the intervention is a one-time event, with the exchange rate and the other macro variables gradually returning toward their baseline values over the next few years. The impact on the steady-state values in the model is small.

Terms of trade shock

The third shock considered is a permanent improvement in Armenia’s terms of trade. In the model, the terms of trade equal the real exchange rate, and this variable cannot be shocked exogenously. A similar effect, however, can be generated by assuming a permanent increase in the “taste” of households in the ROW for Armenian output. Such a shock can be introduced in the model via an upward shift in the propensity of the ROW

to import Armenian goods, as reflected in the parameter im_f0 .¹²

The shock was calibrated to result in an increase in the real exchange rate of about 10 percent in the steady state, with the results shown in figure ???. The impact on output is expansionary, as the trade balance improves notwithstanding the appreciation of the real exchange rate. Absorption inflation falls on impact as a result of the direct effect of the exchange rate appreciation, but then remains close to its baseline level, as monetary policymakers pursue the inflation target as opposed to an exchange rate objective – i.e. they do not respond to the exchange rate appreciation by lowering interest rates.

The model can also be run under the assumption that policymakers attempt to stabilize the exchange rate as well as respond to inflation and output by imposing a positive value for the parameter $i4$. In a simulation where this value was set at 1, implying that a 1 percent appreciation in the exchange rate causes the policy interest rate to fall by 1 percentage point, the magnitude of the initial appreciation of the exchange rate is cut roughly in half relative to the above scenario. The output expansion is larger in this case, but at the expense of an increase in the inflation rate to almost 2 percentage points above target in the initial years. This rise in inflation eventually dominates the policy response - after 5 years, the real exchange rate appreciates by about the same amount in both cases. This comparison underscores the risks to the inflation objective if policymakers initially attempt to offset exchange rate movements that are the result of “real” factors, such as a terms of trade improvement.

The above three illustrative shocks show how the small GDE model can be used and the results it generates in terms of short- and long-run responses. Other shocks that have been performed on the model, but not described here, include: an increase in transfers to Armenia from abroad; a loss of confidence that causes a recession in Armenia’s trading partners; a portfolio shift toward Armenian assets (which could also be interpreted as foreign official lending); and changes in Armenian and ROW productivity growth. Of course, more complicated scenarios with multiple shocks can easily be constructed.

The sensitivity of the results to different calibrations of the policy response and other parameters can also be examined. Finally, it is possible to consider the effects of a sequence of “unanticipated” future shocks, i.e. those that are not incorporated in the rational expectations of agents at period t .

¹²Another way of introducing a terms of trade shock is to raise the price elasticity of demand for imports in the ROW to a high value, and then to introduce a “shift” factor that affects this relative price (i.e. the real exchange rate). Taking this approach yielded results that are similar to those obtained by the simpler method of shifting im_f0 .

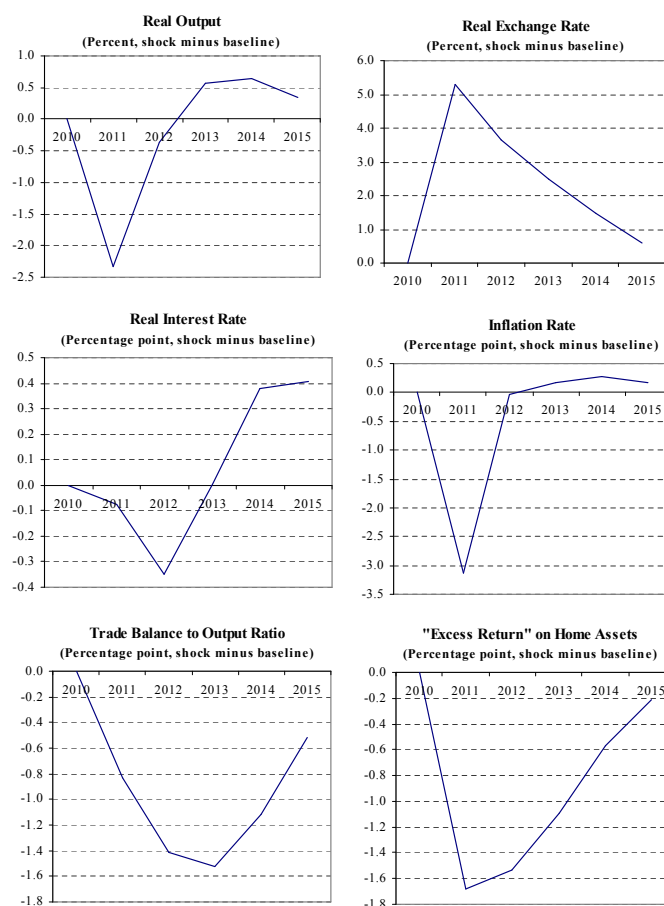


Figure C.2: Impact of Sterilized Intervention

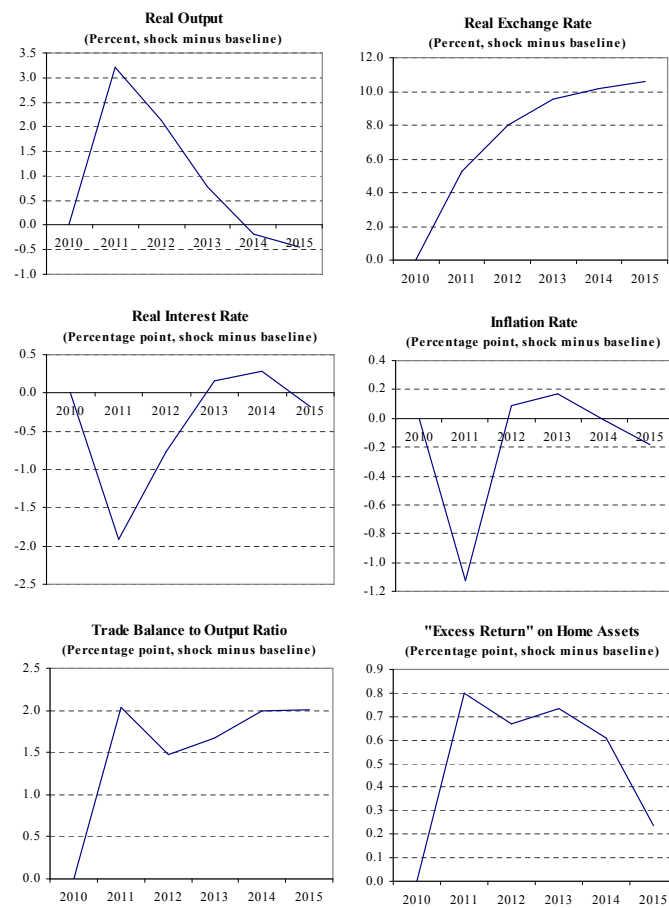


Figure C.3: Terms of Trade Shock

C.5 Concluding Remarks

The small DGE model in this paper is designed to be used as a simulation tool that provides a fully-consistent framework for evaluating the effects of policy and other shocks. As such, it complements rather than competes with models that are more suitable for baseline forecasting and the near-term assessment of monetary policy on economic outcomes. Such models have the strength of incorporating more high-frequency information and “realism” about the immediate situation of the economy, but on the other hand often assume that the equilibrium paths of the endogenous variables are exogenous. With the deeper structure afforded by DGE models, their simulation results can be fed into forecasting models to condition these equilibrium paths in analyzing alternative forecast scenarios.

The results obtained from DGE models of this type are, of course, driven by the assumptions about exogenous variables and parameter calibration, and users are encouraged to adjust these as appropriate to address specific needs. In the case of Armenia, the model has been calibrated on the basis of historical data, which may be inappropriate in the face of a dynamic and evolving economic structure. Informed judgment on the future prospects for the economy that are less tied to past outcomes would play an important role in refining the model properties.

Finally, some features of Armenia’s economic situation are not captured in the existing model. These include the large current account deficit and associated nonstationary path for net foreign assets. In addition, much of the current account deficit has been financed by foreign direct investment, which is not explicitly treated in the external financial flows in the model. Incorporating these phenomena would provide promising areas for future research.

Model Mnemonics

The following mnemonics apply to the home country; corresponding ROW values are indicated by $_F$ in the model code.

Variables are defined in terms of real levels (national accounts, external and fiscal data) or rates of change (e.g. inflation). The only exceptions are the nominal interest rate and current account balance. Stocks of financial and real assets are defined as beginning-of-period values.

Endogenous variables

AA	Capital share in CES production function
C	Total private consumption
CA	Current account balance (using nominal interest rates for investment income)
CBY	Consumption by “Blanchard-Yaari” (forward-looking) households
CH	Consumption of domestically-produced goods
EX	Exports of goods and non-factor services
GAM	Ratio of consumption to total wealth of Blanchard-Yaari households
GAX	Government debt
GB	Net general government debt
GBD	Government gross domestic debt
GBX	Government gross external debt
GP	Government primary spending
GTR	Ratio of government revenues to GDP (tax rate)
GTX	Government tax revenue
I	Nominal interest rate
IM	Imports of goods and non-factor services
IMC	Imports of consumption goods
IMI	Imports of private investment goods
INV	Private investment
INVH	Domestically-produced component of private investment
K	Private capital stock
KC	Total cost of adjustment of capital stock
KMC	Marginal cost of adjustment of the capital stock
NFA	Net foreign assets
PC	Relative price of absorption deflator to GDP deflator
PI	Inflation rate defined in terms of GDP deflator

PIC	Inflation rate defined in terms of absorption deflator
R	Real interest rate defined in terms of GDP deflator
RER	Real exchange rate (price of home country output relative to ROW output)
RERE	Expected level of RER in period T+1
RPC	Real interest rate defined in terms of absorption deflator
TB	Trade balance (goods and non-factor services)
TR	Net transfer from abroad
WF	Financial wealth of households
WH	Human wealth of households
XL	Gross liabilities of home country to ROW
XR	Expected excess return on home country relative to ROW financial assets
YH	Labor income in units of domestic absorption
YY	Real GDP (index)
YP	Potential real GDP (from production function)

Exogenous variables

BTH	Population birth rate
DPOP	Ratio of population in period t to period t-1 ($POP/POP(-1)$)
DTH	Population death rate
DYP	Ratio of steady-state potential output in period t to period t-1
G	Labor-augmenting productivity growth rate
PIT	Target inflation rate (defined in terms of absorption deflator)
POP	Population level
RELSIZE	Ratio of ROW GDP to home-country GDP
RISK	Risk premium on income from capital investment
THETA	Index of labor-augmenting productivity in production function

Parameters

a0	Parameter determining baseline share of capital in CES production function
beta0	Baseline ratio of external assets in home country financial wealth
bxr1	Parameter for response of excess return to changes in external liabilities
c1	Habit persistence in household consumption (weight on lagged consumption)
c2	Elasticity of substitution in consumption between home and imported goods
c3	Elasticity of intertemporal substitution for Blanchard-Yaari households
c5	Share of "rule of thumb" households (i.e. that consume out of labor income)
ces1	Elasticity of labor-capital substitution in CES production function
g0	Ratio of primary government spending to potential GDP
gax0	Ration of government (including central bank) holdings of assets to GDP
gax1	Response of government holdings of foreign assets to excess return

gd0	Target ratio of government debt to GDP in tax reaction function
gd1	Response of tax rate to gap between current and target government debt ratio
gd2	Response of tax rate to gap between lagged and target government debt ratio
gbx0	Ratio of government foreign debt to GDP
gbx1	Share of concessional debt to total government foreign debt
i2	Parameter on gap between actual and target inflation in Taylor Rule
i3	Parameter on gap between actual and potential output in Taylor Rule
i4	Parameter on change in real exchange rate in Taylor Rule
i5	Weight on period t+1 versus period t inflation in Taylor Rule
im0	Baseline ratio of imports to domestic absorption
k0	Capital depreciation rate
k1	Parameter determining cost of adjustment of capital stock
nfa0	Baseline ratio of home country NFA to GDP
pi1	Parameter on lagged versus future inflation in Phillips curve
pi2	Parameter on output gap in Phillips curve
pi3	Parameter on absorption inflation in Phillips curve
rer1	Weight on lagged exchange rate in expected exchange rate
rer2	Weight on current exchange rate in expected exchange rate
rho	Household rate of time preference
tr0	ratio of remittances to home country GDP
xl0	Baseline ratio of gross external liabilities to GDP
xl1	Share of gross external liabilities denominated in foreign currency
xr0	Baseline excess return on home-country external debt